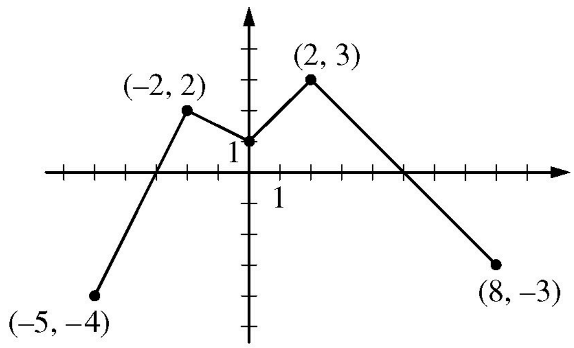
5-6

1.



Graph of f

The continuous function f is defined on the interval $-5 \le x \le 8$. The graph of f, which consists of four line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + \int_{-2}^{x} f(t)\,dt$.

(a) Find $g(0)$ and $g(-5)$.

(b) Find $g'(x)$ in terms of $f(x)$. For each of $g''(4)$ and $g''(-2)$, find the value or state that it does not exist.

(c) On what intervals, if any, is the graph of g concave down? Give a reason for your answer.

(d) The function h is given by $h(x) = g(x^3 + 1)$. Find $h'(1)$. Show the work that leads to your answer.

Part A

The response can earn up to 2 points:

1 point is earned for $g(0)$

$$g(0) = 2 \cdot 0 + \int_{-2}^{0} f(t)dt = 3$$

1 point is earned for g(−5)

$$g(-5) = 2 \cdot (-5) + \int_{-2}^{-5} f(t)dt = -10 + 3 = -7$$

0	1	2

The response earns all of the following points:

1 point is earned for $g(0)$

$$g(0) = 2 \cdot 0 + \int_{-2}^{0} f(t)dt = 3$$

1 point is earned for g(−5)

5-6

$$g(-5) = 2 \cdot (-5) + \int_{-2}^{-5} f(t)dt = -10 + 3 = -7$$

Part B

The response can earn up to 3 points:

1 point: $g'(x)$

$g'(x) = 2 + f(x)$

1 point: $g''(4)$

$g''(x) = f'(x)$

$g''(4) = f'(4) = -1$

1 point: Stating $g''(-2)$ does not exist

$g''(-2) = f'(-2)$ does not exist.

0	1	2	3

The response earns all of the following points:

1 point: $g'(x)$

$g'(x) = 2 + f(x)$

1 point: $g''(4)$

$g''(x) = f'(x)$

$g''(4) = f'(4) = -1$

1 point: Stating $g''(-2)$ does not exist

$g''(-2) = f'(-2)$ does not exist.

Part C

The response can earn up to 1 point:

1 point is earned for intervals and reason

The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.

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0	1

The response earns all of the following point:

1 point is earned for intervals and reason

The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.

Part D

The response can earn up to 3 points:

1 point is earned for chain rule

$h'(x) = g'(x^3 + 1) \cdot 3x^2$

1 point is earned for correct answer

$$\begin{aligned} h'(1) &= g'(2) \cdot 3 = (2 + f(2)) \cdot 3 \\ &= (2 + 3) \cdot 3 = 15 \end{aligned}$$

0	1	2	3

The response earns all of the following point:

1 point is earned for chain rule

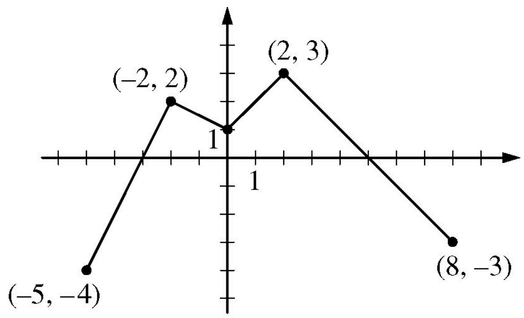
$h'(x) = g'(x^3 + 1) \cdot 3x^2$

1 point is earned for correct answer

$$\begin{aligned} h'(1) &= g'(2) \cdot 3 = (2 + f(2)) \cdot 3 \\ &= (2 + 3) \cdot 3 = 15 \end{aligned}$$

5-6

2.



Graph of f

The continuous function f is defined on the interval $-5 \leq x \leq 8$. The graph of f, which consists of four line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + \int_{-2}^{x} f(t)\, dt$.

On what intervals, if any, is the graph of g concave down? Give a reason for your answer.

Part C

The response can earn up to 1 point:

1 point: For intervals and reason

The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.

0	1

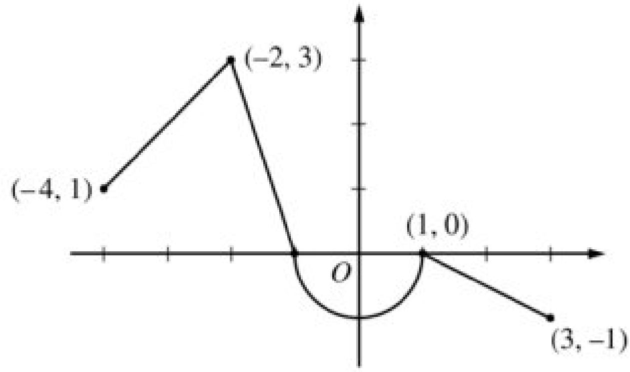
The response earn the following point:

1 point: For intervals and reason

The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.

5-6

3.



Graph of f

Let f be the continuous function defined on [-4, 3] whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t)dt$.

(a) Find the values of $g(2)$ and $g(-2)$.

(b) For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.

(c) Find the x-coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.

(d) For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

Part A

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t)dt = -\frac{1}{2}(1)\left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$g(-2) = \int_1^{-2} f(t)dt = -\int_{-2}^{1} f(t)dt$$

$$= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \frac{\pi}{2} - \frac{3}{2}$$

0	1	2

The student response earns all of the following points:

1 point is earned for $g(2)$

5-6

1 point is earned for g(-2)

$$g(2) = \int_1^2 f(t)dt = -\frac{1}{2}(1)\left(\tfrac{1}{2}\right) = -\tfrac{1}{4}$$

$$g(-2) = \int_1^{-2} f(t)dt = -\int_{-2}^1 f(t)dt$$

$$= -\left(\tfrac{3}{2} - \tfrac{\pi}{2}\right) = \tfrac{\pi}{2} - \tfrac{3}{2}$$

Part B

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$
$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

0	1	2

The student response earns all of the following points:

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$
$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

Part C

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

5-6

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

Part D

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

0	1	2 ✓

The student response earns all of the following points:

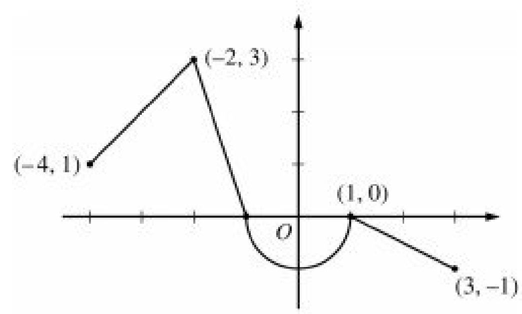
1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

5-6

4.



Graph of f

Let f be the continuous function defined on [-4, 3] whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t)dt.$

For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

Part D

One point is earned for the answer

One point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

0	1	2

The student earns all of the following points:

One point is earned for the answer

One point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

5-6

5. Let $g(x) = xe^{-x} + be^{-x}$, where b is a positive constant.

Find all values of b, if any, for which the graph of g has a point of inflection on the interval $0 < x < \infty$. Justify your answer.

Part C

2 points are earned for g''(x)

1 point is earned for interval for *b*

1 point is earned for justification

If $0 < b < 2$, then $g''(x)$ will change sign at $x = 2 - b > 0$.Therefore, the graph of g will have a point of inflection on the interval $0 < x < \infty$ when $0 < b < 2$.

0	1	2	3	4 ✓

The student response earns all of the following points:

2 points are earned for g''(x)

1 point is earned for interval for *b*

1 point is earned for justification

If $0 < b < 2$, then $g''(x)$ will change sign at $x = 2 - b > 0$.Therefore, the graph of g will have a point of inflection on the interval $0 < x < \infty$ when $0 < b < 2$.

5-6

6. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The function f is defined by $f(x) = \int_0^x \left(t^2 - 3t - 10\right) dt$.

(a) Write an equation for the line tangent to the graph of f at $x = 6$.

(b) Find all values of x at which f has a critical point. Determine whether f has a relative minimum, a relative maximum, or neither at each of these values. Justify your answers.

(c) On what open intervals, if any, is the graph of f concave down? Give a reason for your answer.

(d) Let g be the function defined by $g(x) = f(\tan x)$. Find $g'\left(\frac{\pi}{4}\right)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An unsupported value of -42 does not earn the first point; the minimal supporting work required is $72 - 54 - 60$.

The second point can be earned without presenting an explicit expression for $f'(x)$ by correctly finding $f'(6)$ with supporting work.

Eligibility for third point: The response has earned the second point and used $f(x) = \frac{1}{3}x^3 - \frac{3}{2}x^2 - 10x$ to find $f(6)$. In this case the third point is earned for any form of the equation $y = f(6) + 8(x - 6)$ for the declared value of $f(6)$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ $f(6)$
- ☐ $f'(x)$
- ☐ Equation for tangent line

5-6

Solution:

$f(6) = \int_0^6 \left(t^2 - 3t - 10\right) dt = \left(\frac{1}{3}t^3 - \frac{3}{2}t^2 - 10t\right)\Big|_0^6$

$= \frac{1}{3} \cdot 6^3 - \frac{3}{2} \cdot 6^2 - 10 \cdot 6 = 72 - 54 - 60 = -42$

$f'(x) = x^2 - 3x - 10$

$f'(6) = 6^2 - 3 \cdot 6 - 10 = 36 - 18 - 10 = 8$

An equation for the tangent line is $y = -42 + 8(x - 6)$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

If any additional critical points are listed or either critical point is missing, the first point is not earned.

If only one of $x = -2$ or $x = 5$ is listed as a critical point and that point is correctly justified as a relative maximum or relative minimum, the response earns one point.

“The slope changes from positive to negative” is not a sufficient justification for the second point. The response must be clear about what slope is changing.

Justifications that appeal only to $f(x)$ changing from increasing to decreasing (or vice versa) are not sufficient for the second point.

A response that imports from part (a) an incorrect $f'(x)$ of the form $ax^2 + bx + c$ (typically, $f'(x) = x^2 - 3x$) may earn the first point for finding the critical points associated with their $f'(x)$. Note: The incorrect $f'(x)$ must be imported from part (a) to be eligible for the first point. Such a response is not eligible for the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Critical points
- ☐ Answers with justification

Solution:

$f'(x) = x^2 - 3x - 10 = (x - 5)(x + 2) = 0$
$\Rightarrow x = -2$ and $x = 5$

f has a relative maximum at $x = -2$ because $f'(x)$ changes from positive to negative there.

5-6

f has a relative minimum at $x = 5$ because $f'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents $2x - 3 < 0$ (or $2x - 3 = 0$ or $2x - 3 > 0$) earns the first point.

To earn the second point, a response must explain that the graph of f is concave down on $\left(-\infty, \frac{3}{2}\right)$ (or $x < \frac{3}{2}$), because $f''(x) < 0$ on that interval.

The interval may include the right endpoint $x = \frac{3}{2}$.

A response that presents any interval other than $\left(-\infty, \frac{3}{2}\right)$ does not earn the second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Find $f''(x)$ and compare to zero
- ☐ Answer with reason

Solution:

$$f''(x) = 2x - 3 < 0 \Rightarrow x < \frac{3}{2}$$

The graph of f is concave down on the interval $\left(-\infty, \frac{3}{2}\right)$ because $f''(x) < 0$ there.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $g'(x) = f'(\tan x) \cdot \left(\sec^2 x\right)$.

The answer does not need to be simplified: $\left(\tan^2\left(\frac{\pi}{4}\right) - 3\tan\left(\frac{\pi}{4}\right) - 10\right)\left(\sec^2\left(\frac{\pi}{4}\right)\right)$ or equivalent earns the second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

5-6

- ☐ Chain rule
- ☐ Answer

Solution:

$g'(x) = f'(\tan x) \cdot \left(\sec^2 x\right) = \left(\tan^2 x - 3\tan x - 10\right)\left(\sec^2 x\right)$

$g'\left(\frac{\pi}{4}\right) = \left(\tan^2\left(\frac{\pi}{4}\right) - 3\tan\left(\frac{\pi}{4}\right) - 10\right)\left(\sec^2\left(\frac{\pi}{4}\right)\right)$
$= (1 - 3 - 10)(2) = -24$

5-6

7. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be a twice-differentiable function such that $f'(1) = 0$. The second derivative of f is given by $f''(x) = x^2 \cos\left(x^2 + \pi\right)$ for $-1 \le x \le 3$.

(a) On what open intervals contained in $-1 < x < 3$ is the graph of f concave up? Give a reason for your answer.

(b) Does f have a relative minimum, a relative maximum, or neither at $x = 1$? Justify your answer.

(c) Use the Mean Value Theorem on the closed interval $[-1, 1]$ to show that $f'(-1)$ cannot equal 2.5.

(d) Does the graph of f have a point of inflection at $x = 0$? Give a reason for your answer.

Part A

A maximum of 1 out of 2 points is earned for one correct interval with reason and no incorrect intervals.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ intervals
- ☐ reason

Solution:

5-6

The graph of f is concave up on $(1.253,\ 2.170\ \text{(or } 2.171))$ and $(2.802,\ 3)$ because $f''(x)$ is positive on those intervals.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ justification

Solution:

Because $f'(1) = 0$ and $f''(1) = -0.540302 < 0$, by the Second Derivative Test f has a relative maximum at $x = 1$.

Part C

The first point requires use of average rate of change of f' on $[-1,\ 1]$.

One point is earned for using MVT to conclude that there would be a number c in the interval $(-1,\ 1)$ such that $f''(c) = -1.25$. Two points are earned for showing that this is not possible.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ considers average rate of change
- ☐ conclusion using Mean Value Theorem

Solution:

5-6

If $f'(-1) = 2.5$, then the average rate of change of f' on the closed interval $[-1,\ 1]$ would be $\frac{f'(1)-f'(-1)}{1-(-1)} = \frac{0-2.5}{2} = -1.25$.

Since f' is continuous on $[-1,\ 1]$ and differentiable on $(-1,\ 1)$, by the Mean Value Theorem there would be a number c in the interval $(-1,\ 1)$ such that $f''(c) = -1.25$.

However, since $f''(x) = x^2 \cos\left(x^2 + \pi\right)$, it follows that $-1 \le f''(x) \le 1$ on the interval $(-1,\ 1)$. Therefore, there is no number c in the interval such that $f''(c) = -1.25$.

Thus, it is not possible that $f'(-1) = 2.5$.

Part D

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- [ ] answer
- [ ] reason

Solution:

Because $f''(x)$ does not change sign at $x = 0$, the graph of f does not have a point of inflection there.

5-6

8. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be a twice-differentiable function such that $f'(2) = 0$. The second derivative of f is given by $f''(x) = x^2e^{2-x} - 1$ for $0 \leq x \leq 6$.

(a) On what open intervals contained in $0 < x < 6$ is the graph of f concave down? Give a reason for your answer.

(b) Does f have a relative minimum, a relative maximum, or neither at $x = 2$? Justify your answer.

(c) Use the Mean Value Theorem on the closed interval $[2, 4]$ to show that $f'(4)$ cannot equal 8.5.

(d) Does the graph of f have a point of inflection at $x = 5$? Give a reason for your answer.

Part A

A maximum of 1 out of 2 points is earned for one correct interval with reason and no incorrect intervals.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ intervals
- ☐ reason

Solution:

5-6

The graph of f is concave down on $(0,\ 0.464)$ (or $(0,\ 0.463)$) and $(5.357,\ 6)$ (or $(5.356,\ 6)$) because $f''(x)$ is negative on those intervals.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ justification

Solution:

Because $f'(2) = 0$ and $f''(2) = 3 > 0$, by the Second Derivative Test f has a relative minimum at $x = 2$.

Part C

The first point requires use of average rate of change of f' on $[2,\ 4]$.

The second point is earned for using MVT to conclude that there would be a number c in the interval $(2,\ 4)$ such that $f''(c) = 4.25$. The third point is earned for showing that this is not possible.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ considers average rate of change
- ☐ uses Mean Value Theorem to determine $f''(c) = 4.25$
- ☐ conclusion of “not possible” using Mean Value Theorem

Solution:

5-6

If $f'(4) = 8.5$, then the average rate of change of f' on the closed interval $[2,\ 4]$ would be $\frac{f'(4)-f'(2)}{4-2} = \frac{8.5-0}{2} = 4.25.$

Since f' is continuous on $[2,\ 4]$ and differentiable on $(2,\ 4)$, by the Mean Value Theorem there would be a number c in the interval $(2,\ 4)$ such that $f''(c) = 4.25$.

However, since $f''(x) = x^2e^{2-x} - 1$, it follows that $f''(x) = 4.25$ has no solution on the interval $[2,\ 4]$. Therefore, there is no number c in the interval $(2,\ 4)$ such that $f''(c) = 4.25$.

Thus, it is not possible that $f'(4) = 8.5$.

Part D

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ reason

Solution:

Because $f''(x)$ does not change sign at $x = 5$, the graph of f does not have a point of inflection there. $f''(x) > 0$ on $(0.464,\ 5.357)$.

5-6

9. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

The functions f and g are twice differentiable. The table shown gives values of the functions and their first derivatives at selected values of x.

(a) Let h be the function defined by $h(x) = f(g(x))$. Find $h'(7)$. Show the work that leads to your answer.

(b) Let k be a differentiable function such that $k'(x) = (f(x))^2 \cdot g(x)$. Is the graph of k concave up or concave down at the point where $x = 4$? Give a reason for your answer.

(c) Let m be the function defined by $m(x) = 5x^3 + \int_0^x f'(t)\,dt$. Find $m(2)$. Show the work that leads to your answer.

(d) Is the function m defined in part (c) increasing, decreasing, or neither at $x = 2$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $h'(x) = f'(g(x)) \cdot g'(x)$ or $h'(7) = f'(g(7)) \cdot g'(7)$.

If the first point is earned, the second point is earned only for an answer of 12 (or equivalent).

If the first point is not earned, the second point can be earned only for a response of either $f'(0) \cdot 8 = 12$ or $\frac{3}{2} \cdot 8$.

A response of 12 with no supporting work does not earn either point.

5-6

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Chain rule
- [ ] Answer

Solution:

$h'(x) = f'(g(x)) \cdot g'(x)$

$h'(7) = f'(g(7)) \cdot g'(7) = f'(0) \cdot 8 = \frac{3}{2} \cdot 8 = 12$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $k''(x) = 2f(x) \cdot f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $k''(4) = 2f(4) \cdot f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$.

The first point is also earned by any of the following incorrect expressions, each of which has a single error in the application of the product rule or the chain rule:

- $2f(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $2f(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $2f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $2f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $2f(x) \cdot f'(x) \cdot g'(x)$ or $2f(4) \cdot f'(4) \cdot g'(4)$
- Note: A response that presents one of these expressions cannot earn the second point.

To earn the second point a response must correctly find $k''(4) = -40$ (or equivalent) with supporting work.

The third point is earned for an answer and reason that are consistent with any declared nonzero value of $k''(4)$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Product or chain rule

5-6

- ☐ $k''(4)$
- ☐ Answer with reason

Solution:

$$k''(x) = 2f(x) \cdot f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$$

$$k''(4) = 2f(4) \cdot f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$$
$$= 2 \cdot 4 \cdot 3 \cdot (-3) + 4^2 \cdot 2 = -72 + 32 = -40$$

The graph of k is concave down at the point where $x = 4$ because $k''(4) < 0$ and k'' is continuous.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned only for an answer of 37 (or equivalent) with supporting work equivalent to $5 \cdot 8 + (f(2) - f(0))$, $40 + (f(2) - f(0))$, $5 \cdot 8 + (7 - 10)$, $40 + (7 - 10)$.

An answer of 37 with no supporting work does not earn the point.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer with supporting work

Solution:

$$m(2) = 5 \cdot 8 + \int_0^2 f'(t)dt = 40 + (f(2) - f(0))$$
$$= 40 + (7 - 10) = 37$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for considering $m'(x)$, $m'(2)$, or m'. This consideration may appear in a justification statement.

The second point is earned for $m'(2) = 15 \cdot 2^2 + f'(2)$, $m'(2) = 60 + f'(2)$, or $m'(2) = 60 - 8$ but is not earned for an unsupported response of $m'(2) = 52$.

The third point is earned for an answer and justification consistent with any declared value of $m'(2)$.

5-6

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $m'(x)$
- ☐ $m'(2)$ with supporting work
- ☐ Answer with justification

Solution:

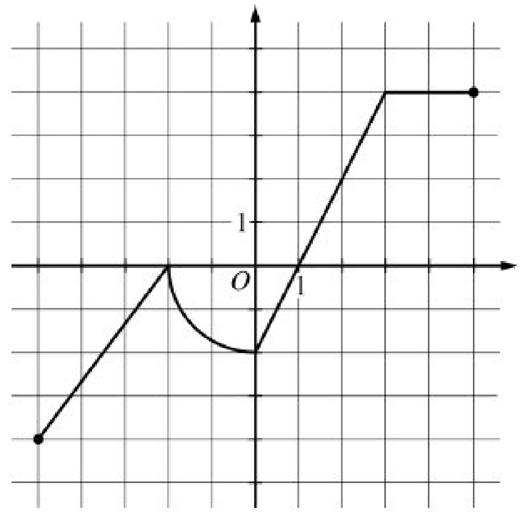
$m'(x) = 15x^2 + f'(x)$

$m'(2) = 15 \cdot 4 + f'(2) = 60 + (-8) = 52$

The graph of m is increasing at $x = 2$ because $m'(2) > 0$.

5-6

10. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.



Graph of f

The graph of the function f, consisting of three line segments and a quarter of a circle, is shown above. Let g be the function defined by $g(x) = \int_1^x f(t)dt$.

(a) Find the average rate of change of g from $x = -5$ to $x = 5$.

(b) Find the instantaneous rate of change of g with respect to x at $x = 3$, or state that it does not exist.

(c) On what open intervals, if any, is the graph of g concave up? Justify your answer.

(d) Find all x-values in the interval $-5 < x < 5$ at which g has a critical point. Classify each critical point as the location of a local minimum, a local maximum, or neither. Justify your answers.

Part A

1 point is earned for: difference quotient

2 points are earned for: answer

$$\frac{g(5)-g(-5)}{5-(-5)} = \frac{12-(\pi+7)}{10} = \frac{5-\pi}{10}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: difference quotient

2 points are earned for: answer

5-6

$$\frac{g(5)-g(-5)}{5-(-5)} = \frac{12-(\pi+7)}{10} = \frac{5-\pi}{10}$$

Part B

1 point is earned for: answer

$g'(x) = f(x)$
$g'(3) = f(3) = 4$

The instantaneous rate of change of g at $x = 3$ is 4.

0	1

The student response earns all of the following points:

1 point is earned for: answer

$g'(x) = f(x)$
$g'(3) = f(3) = 4$

The instantaneous rate of change of g at $x = 3$ is 4.

Part C

2 points are earned for: intervals with justification

The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$, because $g'(x) = f(x)$ is increasing on these intervals.

0	1	2

The student response earns all of the following points:

2 points are earned for: intervals with justification

The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$, because $g'(x) = f(x)$ is increasing on these intervals.

Part D

1 point is earned for: considers $f(x) = 0$

5-6

1 point is earned for: critical points at $x = -2$ and $x = 1$

1 point is earned for: answers with justifications

$g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$ because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to positive there.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: considers $f(x) = 0$

1 point is earned for: critical points at $x = -2$ and $x = 1$

1 point is earned for: answers with justifications

$g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$ because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to positive there.

5-6

11.

	$0 < x < 1$	$1 < x < 2$
$f(x)$	Positive	Negative
$f'(x)$	Negative	Negative
$f''(x)$	Negative	Positive

Let f be a function that is twice differentiable on $-2 < x < 2$ and satisfies the conditions in the table above. If $f(x) = f(-x)$ what are the x-coordinates of the points of inflection of the graph of f on $-2 < x < 2$?

(A) $x = 0$ only

(B) $x = 1$ only

(C) $x = 0$ and $x = 1$

(D) $x = -1$ and $x = 1$ ✓

(E) There are no points of inflection on $-2 < x < 2$.

12.

Let f be the function defined by $f(x) = \int_0^x \left(2t^3 - 15t^2 + 36t\right) dt$. On which of the following intervals is the graph of $y = f(x)$ concave down?

(A) $(-\infty, 0)$ only

(B) $(-\infty, 2)$

(C) $(0, \infty)$

(D) $(2, 3)$ only ✓

(E) $(3, \infty)$ only

13.

The function f is defined on the open interval $0.4 < x < 2.4$ and has first derivative f' given by $f'(x) = \sin(x^2)$. Which of the following statements are true?

i. f has a relative maximum on the interval $0.4 < x < 2.4$

ii. f has a relative minimum on the interval $0.4 < x < 2.4$

iii. The graph of f has two points of inflection on the interval $0.4 < x < 2.4$

(A) I only

(B) II only

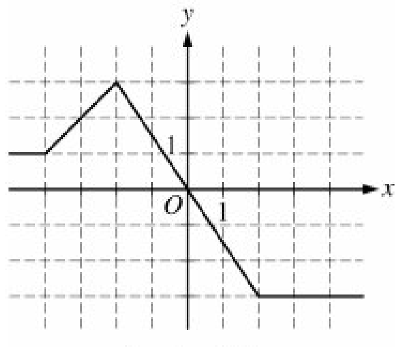
(C) III only

(D) I and III only ✓

(E) II and III only

5-6

14.



Graph of f'

The graph of f', the derivative of the function f, is shown in the figure above. Which of the following statements about f at $x = -2$ is true?

(A) f is not continuous at $x = -2$.

(B) f has an absolute maximum at $x = -2$.

(C) The derivative of f does not exist at $x = -2$.

(D) The graph of f has a point of inflection at $x = -2$.

(E) The graph of f has a vertical tangent line at $x = -2$.

5-6

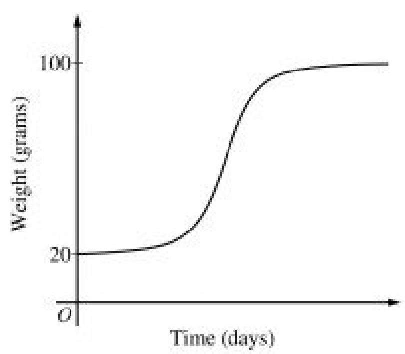
15. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

(a) Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.

(b) Find $\frac{d^2B}{dt^2}$ in terms of B. Use $\frac{d^2B}{dt^2}$ to explain why the graph of B cannot resemble the following graph.



(c) Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.

Part A

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for the answer with reason

$$\left.\frac{dB}{dt}\right|_{B=40} = \frac{1}{5}(60) = 12$$
$$\left.\frac{dB}{dt}\right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left.\frac{dB}{dt}\right|_{B=40} > \left.\frac{dB}{dt}\right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

0	1	2

The student earns all of the following points:

1 point is earned for using $\frac{dB}{dt}$

5-6

1 point is earned for the answer with reason

$$\frac{dB}{dt}\Big|_{B=40} = \tfrac{1}{5}(60) = 12$$
$$\frac{dB}{dt}\Big|_{B=70} = \tfrac{1}{5}(30) = 6$$

Because $\frac{dB}{dt}\Big|_{B=40} > \frac{dB}{dt}\Big|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

Part B

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5}\frac{dB}{dt} = -\frac{1}{5}\cdot\frac{1}{5}(100-B) = -\frac{1}{25}(100-B)$$

Therefore, the graph of B is concave down for $20 \le B < 100$. A portion of the given graph is concave up.

0	1	2

The student earns all of the following points:

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5}\frac{dB}{dt} = -\frac{1}{5}\cdot\frac{1}{5}(100-B) = -\frac{1}{25}(100-B)$$

Therefore, the graph of B is concave down for $20 \le B < 100$. A portion of the given graph is concave up.

Part C

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

5-6

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$
$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$
$$-\ln|100 - B| = \frac{1}{5}t + c$$
Because $20 \le B < 100,\ |100 - B| = 100 - B.$
$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$
$$100 - B = 80e^{-t/5}$$
$$B(t) = 100 - 80e^{-t/5},\ t \ge 0$$

0	1	2	3	4	5

The student earns all of the following points:

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

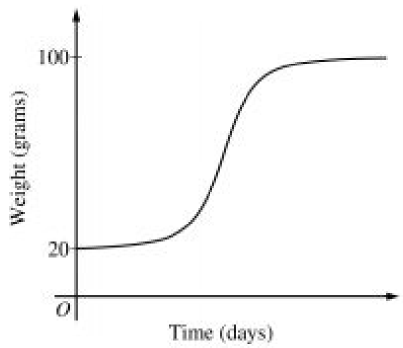
$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$
$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$
$$-\ln|100 - B| = \frac{1}{5}t + c$$
Because $20 \le B < 100,\ |100 - B| = 100 - B.$
$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$
$$100 - B = 80e^{-t/5}$$
$$B(t) = 100 - 80e^{-t/5},\ t \ge 0$$

5-6

16. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then $\frac{db}{dt} = \frac{1}{5}(100 - B)$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

Find $\frac{d^2B}{dt^2}$ in terms of B. Use $\frac{d^2B}{dt^2}$ to explain why the graph of B cannot resemble the following graph.



Part B

One point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

One point is earned for the explanation

$$\frac{d^2B}{dt^2}\Big| = -\frac{1}{5}\frac{dB}{dt} = -\frac{1}{5}\cdot\frac{1}{5}(100 - B) = -\frac{1}{25}(100 - B)$$

Therefore, the graph of B is concave down for $20 \leq B \leq 100$. A portion of the given graph is concave up.

0	1	2

The student earns all of the following points:

One point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

One point is earned for the explanation

$$\frac{d^2B}{dt^2}\Big| = -\frac{1}{5}\frac{dB}{dt} = -\frac{1}{5}\cdot\frac{1}{5}(100 - B) = -\frac{1}{25}(100 - B)$$

Therefore, the graph of B is concave down for $20 \leq B \leq 100$. A portion of the given graph is concave up.

5-6

5-6

17.

t (minutes)	0	4	8	12	16
$H(t)$ (°C)	65	68	73	80	91

The temperature, in degrees Celsius (C), of an oven being heated is modeled by an increasing differentiable function H of time t, where t is measured in minutes. The table above gives the temperature as recorded every 4 minutes over a 16-minute period.

Are the data in the table consistent with or do they contradict the claim that the temperature of the oven is increasing at an increasing rate? Give a reason for your answer.

Part D

1 point is earned for considers slopes of four secant lines

1 point is earned for the explanation

1 point is earned for the conclusion consistent with explanation

If a continuous function is increasing at an increasing rate, then the slopes of the secant lines of the graph of the function are increasing. The slopes of the secant lines for the four intervals in the table are $\frac{3}{4}, \frac{5}{4}, \frac{7}{4}, and \frac{10}{4}$, respectively.

Since the slopes are increasing, the data are consistent with the claim.

OR

By the Mean Value Theorem, the slopes are also the values of $H'(c_k)$ for some times $c_1 < c_2 < c_3 < c_4$, respectively. Since these derivative values are positive and increasing, the data are consistent with the claim.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for considers slopes of four secant lines

1 point is earned for the explanation

1 point is earned for the conclusion consistent with explanation

If a continuous function is increasing at an increasing rate, then the slopes of the secant lines of the graph of the function are increasing. The slopes of the secant lines for the four intervals in the table are $\frac{3}{4}, \frac{5}{4}, \frac{7}{4}, and \frac{10}{4}$, respectively.

Since the slopes are increasing, the data are consistent with the claim.

5-6

OR

By the Mean Value Theorem, the slopes are also the values of $H'(c_k)$ for some times $c_1 < c_2 < c_3 < c_4$, respectively. Since these derivative values are positive and increasing, the data are consistent with the claim.

18. For $-1.5 < x < 1.5$, let f be a function with first derivative given by $f'(x) = e^{(x^4-2x^2+1)} - 2$. Which of the following are all intervals on which the graph of f is concave down?

(A) (-0.418, 0.418) only

(B) (-1, 1)

(C) (-1.354, -0.409) and (0.409, 1.354)

(D) (-1.5, -1) and (0,1) ✓

(E) (-1.5, -1.354), (-0.409, 0), and (1.354, 1.5)

19. The first derivative of the function f is defined by $f'(x) = \left(x^2+1\right)\sin(3x-1)$ for $-1.5 < x < 1.5$. On which of the following intervals is the graph of f concave up?

(A) $(-1.5,\ -1.341)$ and $(-0.240,\ 0.964)$ ✓

(B) $(-1.341,\ -0.240)$ and $(0.964,\ 1.5)$

(C) $(-0.714,\ 0.333)$ and $(1.381,\ 1.5)$

(D) $(-1.5,\ -0.714)$ and $(0.333,\ 1.381)$

20. The function f has first derivative given by $f'(x) = x^4 - 6x^2 - 8x - 3$. On what intervals is the graph of f concave up?

(A) $(2, \infty)$ only ✓

(B) $(0, \infty)$

(C) $(-1, 2)$

(D) $(-\infty, -1)$ and $(3, \infty)$

21. Let f be the function given by $f(x) = x^3 - 6x^2$. The graph of f is concave up when

(A) $x > 2$ ✓

(B) $x < 2$

(C) $0 < x < 4$

(D) $x < 0$ or $x > 4$ only

(E) $x > 6$ only

5-6

22. If $f''(x) = x(x+2)^2$, then the graph of f is concave up for

(A) $x < 0$

(B) $x > 0$

(C) $-2 < x < 0$

(D) $x < -2$ and $x > 0$

(E) all real numbers

23. Let f be a function with derivative given by $f'(x) = x^3 - 5x^2 + e^x$. On which of the following intervals is the graph of f concave down?

(A) $(-\infty, 0.117)$ only

(B) $(-\infty, 0.144)$

(C) $(0.116, 2.062)$

(D) $(0.673, 2.863)$

(E) $(2.863, \infty)$

5-6

24. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = 10 - 2y$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(0) = 2$.

(a) Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 0$. Use the tangent line to approximate $f(0.5)$.

(b) Find the value of $\frac{d^2y}{dx^2}$ at the point $(0, 2)$. Is the graph of $y = f(x)$ concave up or concave down at the point $(0, 2)$? Give a reason for your answer.

(c) Find $y = f(x)$, the particular solution to the differential equation with the initial condition $f(0) = 2$.

(d) For the particular solution $y = f(x)$ found in part (c), find $\lim_{x \to \infty} f(x)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=(0,\ 2)} = 10 - 2 \cdot 2 = 6$$

An equation for the line tangent to the graph of $y = f(x)$ at $x = 0$ is $y = 6x + 2$.

$$f(0.5) \approx 6 \cdot 0.5 + 2 = 5$$

5-6

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $\left.\frac{d^2y}{dx^2}\right|_{(x,\ y)=(0,\ 2)}$
- [ ] concave down with reason

Solution:

$$\frac{d^2y}{dx^2} = -2\frac{dy}{dx} = -2\,(10 - 2y) = 4y - 20$$

$$\left.\frac{d^2y}{dx^2}\right|_{(x,\ y)=(0,\ 2)} = 4 \cdot 2 - 20 = -12$$

Because $\left.\frac{d^2y}{dx^2}\right|_{(x,\ y)=(0,\ 2)} < 0$ and $\frac{d^2y}{dx^2}$ is continuous, the graph of $y = f\,(x)$ is concave down at the point $(0,\ 2)$.

Part C

Zero out of 4 points earned if no separation of variables.

At most 2 out of 4 points earned [1-1-0-0] if no constant of integration.

The fourth point requires an expression for y.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3	4

The student response accurately includes all four of the criteria below.

- [ ] separation of variables
- [ ] antiderivative
- [ ] constant of integration and uses initial condition
- [ ] solves for y

5-6

Solution:

$$\frac{1}{10-2y}\,dy = dx$$
$$\int \frac{1}{10-2y}\,dy = \int 1\,dx$$
$$-\tfrac{1}{2}\ln|10-2y| = x + C$$

Since the solution curve includes the point $(0,\ 2)$,

$$-\tfrac{1}{2}\ln(10-2y) = x + C$$
$$-\tfrac{1}{2}\ln(10-2\cdot 2) = 0 + C \ \Rightarrow\ C = -\tfrac{1}{2}\ln 6$$
$$-\tfrac{1}{2}\ln(10-2y) = x - \tfrac{1}{2}\ln 6$$
$$\ln(10-2y) = -2x + \ln 6$$
$$10 - 2y = e^{-2x+\ln 6} = 6e^{-2x}$$
$$y = 5 - 3e^{-2x}$$

Part D

No supporting work is required to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The student response accurately includes a correct value.

Solution:

$$\lim_{x\to\infty}\left(5 - 3e^{-2x}\right) = 5 - 3\cdot 0 = 5$$

5-6

25. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Consider the curve given by the equation $2(x-y)=3+\cos y$. For all points on the curve, $\frac{2}{3} \leq \frac{dy}{dx} \leq 2$.

(a) Show that $\frac{dy}{dx}=\frac{2}{2-\sin y}$.

(b) For $-\frac{\pi}{2}<y<\frac{\pi}{2}$, there is a point P on the curve through which the line tangent to the curve has slope 1. Find the coordinates of the point P.

(c) Determine the concavity of the curve at points for which $-\frac{\pi}{2}<y<\frac{\pi}{2}$. Give a reason for your answer.

(d) Let $y=f(x)$ be a function, defined implicitly by $2(x-y)=3+\cos y$, that is continuous on the closed interval $[2,\ 2.1]$ and differentiable on the open interval $(2,\ 2.1)$. Use the Mean Value Theorem on the interval $[2,\ 2.1]$ to show that $\frac{1}{15} \leq f(2.1)-f(2) \leq \frac{1}{5}$.

Part A

1 point is earned for: implicit differentiation

1 point is earned for: verification

$$\frac{d}{dx}(2(x-y))=\frac{d}{dx}(3+\cos y)$$
$$2-2\frac{dy}{dx}=(-\sin y)\frac{dy}{dx}$$
$$2=(2-\sin y)\frac{dy}{dx}$$
$$\frac{dy}{dx}=\frac{2}{2-\sin y}$$

0	1	2

The student response earns all of the following points:

1 point is earned for: implicit differentiation

1 point is earned for: verification

$$\frac{d}{dx}(2(x-y))=\frac{d}{dx}(3+\cos y)$$
$$2-2\frac{dy}{dx}=(-\sin y)\frac{dy}{dx}$$
$$2=(2-\sin y)\frac{dy}{dx}$$
$$\frac{dy}{dx}=\frac{2}{2-\sin y}$$

Part B

5-6

1 point is earned for: $\frac{dy}{dx} = 1$

1 point is earned for: answer

$$\frac{dy}{dx} = \frac{2}{2 - \sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$$
$$2(x - 0) = 3 + \cos 0 \Rightarrow 2x = 4 \Rightarrow x = 2$$

Point P has coordinates $(2,\ 0)$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\frac{dy}{dx} = 1$

1 point is earned for: answer

$$\frac{dy}{dx} = \frac{2}{2 - \sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$$
$$2(x - 0) = 3 + \cos 0 \Rightarrow 2x = 4 \Rightarrow x = 2$$

Point P has coordinates $(2,\ 0)$.

Part C

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for: answer with reason

$$\frac{d^2y}{dx^2} = \frac{-2}{(2 - \sin y)^2}(-\cos y)\frac{dy}{dx} = \frac{4\cos y}{(2 - \sin y)^3}$$

$\frac{d^2y}{dx^2} > 0$ for all y in the interval $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

Therefore, the curve is concave up for $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

0	1	2	3

The student response earns all of the following points:

5-6

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for: answer with reason

$$\frac{d^2y}{dx^2} = \frac{-2}{(2-\sin y)^2}(-\cos y)\frac{dy}{dx} = \frac{4\cos y}{(2-\sin y)^3}$$

$\frac{d^2y}{dx^2} > 0$ for all y in the interval $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

Therefore, the curve is concave up for $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

Part D

1 point is earned for: applies Mean Value Theorem

1 point is earned for: verification

By the Mean Value Theorem, for some value c in the interval $(2,\ 2.1)$, $f'(c) = \frac{f(2.1)-f(2)}{0.1}$.

For all points on the curve, $\frac{2}{3} \leq f'(x) \leq 2$.

Thus, $\frac{2}{3} \leq \frac{f(2.1)-f(2)}{0.1} \leq 2 \Rightarrow \frac{1}{15} \leq f(2.1) - f(2) \leq \frac{1}{5}$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: applies Mean Value Theorem

1 point is earned for: verification

By the Mean Value Theorem, for some value c in the interval $(2,\ 2.1)$, $f'(c) = \frac{f(2.1)-f(2)}{0.1}$.

For all points on the curve, $\frac{2}{3} \leq f'(x) \leq 2$.

Thus, $\frac{2}{3} \leq \frac{f(2.1)-f(2)}{0.1} \leq 2 \Rightarrow \frac{1}{15} \leq f(2.1) - f(2) \leq \frac{1}{5}$.

5-6

26. Let f be the function defined by $f(x) = -3 + 6x^2 - 2x^3$. What is the largest open interval on which the graph of f is both concave up and increasing?

(A) (0, 1) ✓

(B) (1, 2)

(C) (0, 2)

(D) $(2, \infty)$

27. The first derivative of the function f is given by $f'(x) = x - 4e^{-\sin(2x)}$. How many points of inflection does the graph of f have on the interval $0 < x < 2\pi$?

(A) Three

(B) Four ✓

(C) Five

(D) Six

(E) Seven

28. For what values of x does the graph of $y = 3x^5 + 10x^4$ have a point of inflection?

(A) $x = -\frac{8}{3}$ only

(B) $x = -2$ only ✓

(C) $x = 0$ only

(D) $x = 0$ and $x = -\frac{8}{3}$

(E) $x = 0$ and $x = -2$

29. The function f is defined by $f(x) = \sin x + \cos x$ for $0 \leq x \leq 2\pi$. What is the x-coordinate of the point of inflection where the graph of f changes from concave down to concave up?

(A) $\frac{\pi}{4}$

(B) $\frac{3\pi}{4}$ ✓

(C) $\frac{5\pi}{4}$

(D) $\frac{7\pi}{4}$

(E) $\frac{9\pi}{4}$

5-6

30. The second derivative of a function f is given by $f''(x) = \sin(3x) - \cos(x^2)$. How many points of inflection does the graph of f have on the interval $0 < x < 3$?

(A) One

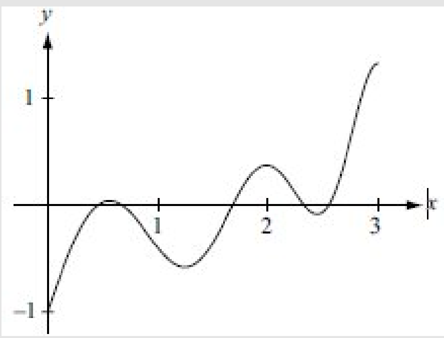
(B) Three

(C) Four

(D) Five

Answer D

This option is correct. Graph f ¢¢(x) on the interval $0 \le x \le 3$.



A point of inflection occurs where the second derivative f ''(x) changes sign. This happens five times on the interval: twice between 0 and 1, once between 1 and 2, and twice between 2 and 3.

31. Let f be the function with derivative defined by $f'(x) = x^3 - 4x$. At which of the following values of x does the graph of f have a point of inflection?

(A) 0

(B) $\frac{2}{3}$

(C) $\frac{2}{\sqrt{3}}$

(D) $\frac{4}{3}$

(E) 2

5-6

32. $f''(x) = x(x-1)^2(x+2)^3$

$g''(x) = x(x-1)^2(x+2)^3 + 1$

$h''(x) = x(x-1)^2(x+2)^3 - 1$

The twice-differentiable functions *f, g,* and *h* have second derivatives given above. Which of the functions *f, g*, and *h* have a graph with exactly two points of inflection?

(A) *g* only

(B) *h* only

(C) *f* and *g* only ✓

(D) *f, g*, and *h*

33. The first derivative of the function *f* is given by $f'(x) = 3x^4 - 12x^3$. What are the *x*-coordinates of the points of inflection of the graph of *f*?

(A) $x = 3$ only ✓

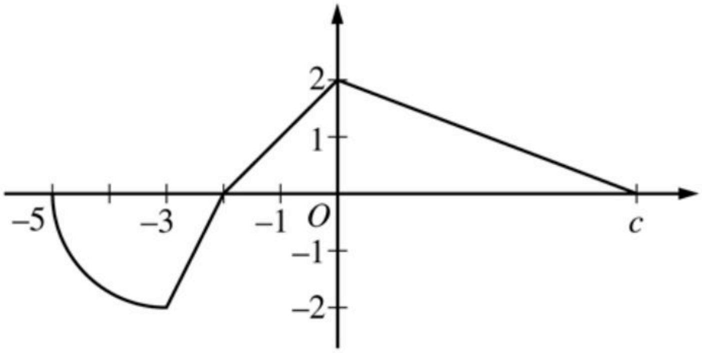
(B) $x = 4$ only

(C) $x = 0$ and $x = 2$

(D) $x = 0$ and $x = 3$

(E) $x = 0$ and $x = 4$

5-6



Graph of f

The function f is defined on the interval $-5 \leq x \leq c$, where $c > 0$ and $f(c) = 0$. The graph of f, which consists of three line segments and a quarter of a circle with center (-3, 0) and radius 2, is shown in the figure above.

34. For $-5 \leq x \leq c$, let g be the function defined by $g(x) = \int_{-1}^{x} f(t)dt$. Find the x-coordinate of each point of inflection of the graph of g. Justify your answer.

Part B

1 point(s) earned for: g'(x) = f(x)

g'(x) = f(x)

1 point(s) earned for: identifies x = -3 and x =0

The graph of g has a point of inflection at x = −3 because g′ = f changes from decreasing to increasing at this point

1 point(s) earned for: justification

The graph of g has a point of inflection at x = 0 because g′ = f changes from increasing to decreasing at this point

0	1	2	3

Student response earns 3 of the following 3 point(s)

1 point(s) earned for: g'(x) = f(x)

g'(x) = f(x)

1 point(s) earned for: identifies x = -3 and x =0

5-6

The graph of g has a point of inflection at $x = -3$ because $g' = f$ changes from decreasing to increasing at this point

1 point(s) earned for: justification

The graph of g has a point of inflection at $x = 0$ because $g' = f$ changes from increasing to decreasing at this point

5-6

35. For $-5 \leq x \leq c$, let g be the function defined by $g(x) = \int_{-1}^{x} f(t)dt$. Find the x-coordinate of each point of inflection of the graph of g. Justify your answer.

Part B

3 points can be earned;

1 point for $g'(x) = f(x)$

1 point for x=-3 and x=0

1 point for justification

$g'(x) = f(x)$

The graph of g has a point of inflection at x = −3 because

g′ = f changes from decreasing to increasing at this point.

The graph of g has a point of inflection at x = 0 because

g′ = f changes from increasing to decreasing at this point.

0	1	2	3

The student response earns none of the following points:

3 points can be earned;

1 point for $g'(x) = f(x)$

1 point for x=-3 and x=0

1 point for justification

$g'(x) = f(x)$

The graph of g has a point of inflection at x = −3 because

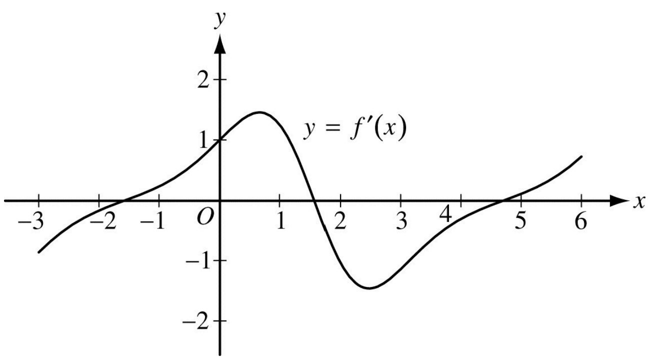
5-6

$g' = f$ changes from decreasing to increasing at this point.

The graph of g has a point of inflection at $x = 0$ because

$g' = f$ changes from increasing to decreasing at this point.

36.



The figure above shows the graph of f', the derivative of the function f, on the interval $[-3,6]$. If the derivative of the function h is given by $h'(x)=2f'(x)$, how many points of inflection does the graph of h have on the interval $[-3,6]$?

(A) One

(B) Two ✓

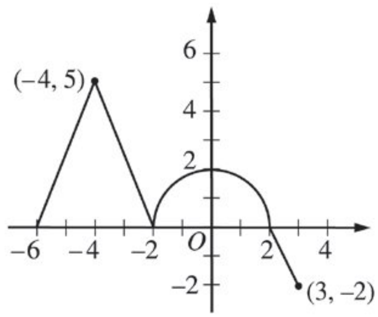
(C) Three

(D) Four

(E) Five

5-6

37.



Graph of f

The graph of the continuous function f, consisting of three line segments and a semicircle, is shown above. Let g be the function given by $g(x) = \int_{-2}^{x} f(t)dt$.

Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.

Part D

The response can earn up to 3 points:

2 points: For the correct values of x

1 point: For the explanation

The graph of g has a point inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

0	1	2	3

The response earns all three of the following points:

2 points: For the correct values of x

1 point: For the explanation

The graph of g has a point inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

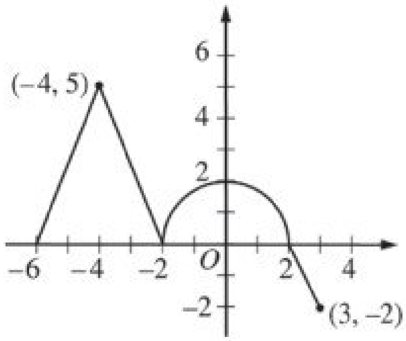
5-6

OR

$g''(x) = f(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

5-6

38.



Graph of f

The graph of the continuous function f, consisting of three line segments and a semicircle, is shown above. Let g be the function given by $g(x) = \int_{-2}^{x} f(t)dt$.

Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.

Part D

Two points are earned for the value of x

One point is earned the explanation

The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

0	1	2	3

The student earns all of the following points:

Two points are earned for the value of x

One point is earned the explanation

The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

5-6

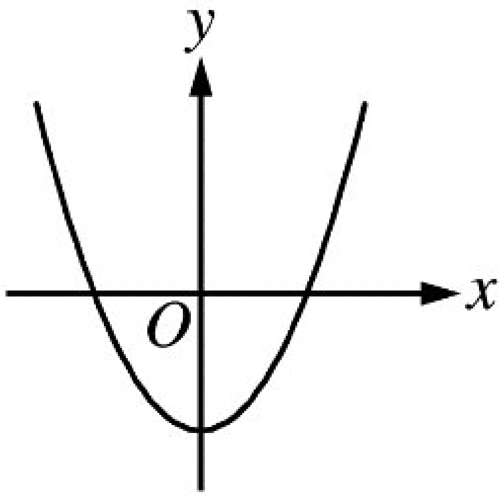
$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

5-6

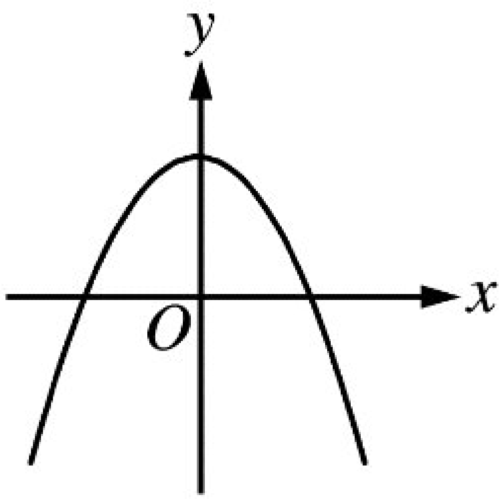
39. The derivative of a function f is increasing for $x < 0$ and decreasing for $x > 0$. Which of the following could be the graph of f?

5-6

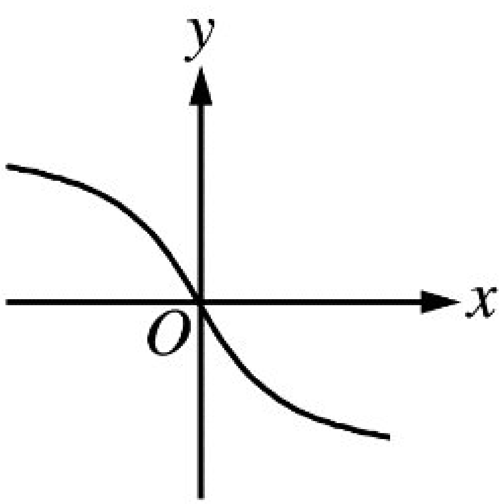
(A)



(B)

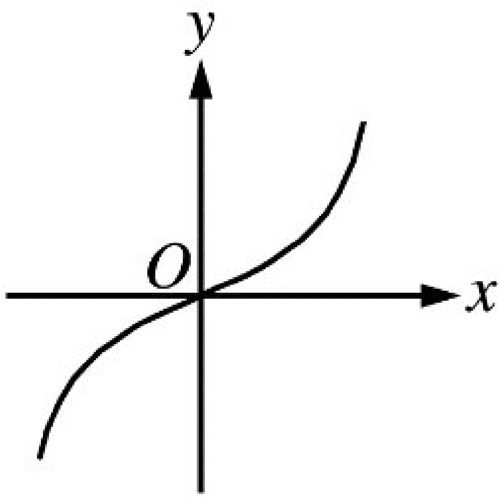


(C)

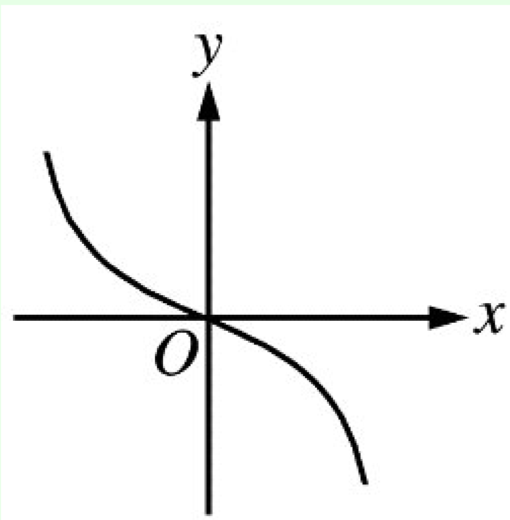


5-6

(D)



(E)



40. Let $y = f(x)$ define a twice-differentiable function and let $y = t(x)$ be the line tangent to the graph of f at $x = 2$. If $t(x) \geq f(x)$ for all real x, which of the following must be true?

(A) $f(2) \geq 0$

(B) $f'(2) \geq 0$

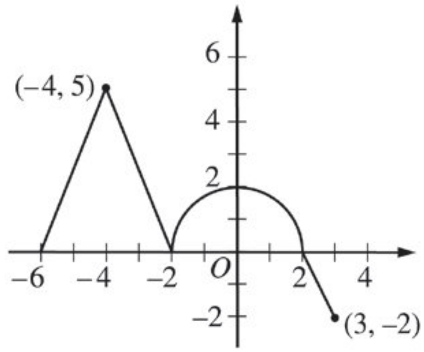
(C) $f'(2) \leq 0$

(D) $f''(2) \geq 0$

(E) $f''(2) \leq 0$

5-6

41.



Graph of f

The graph of the continuous function f, consisting of three line segments and a semicircle, is shown above. Let g be the function given by $g(x) = \int_{-2}^{x} f(t)dt$.

(a) Find $g(-6)$ and $g(3)$.

(b) Find $g'(0)$.

(c) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.

(d) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.

Part A

The response can earn up to 2 points:

1 point is earned for $g(-6)$

$$g(-6) = \int_{-2}^{-6} f(t)dt = -\int_{-6}^{-2} f(t)dt = -\frac{1}{2} \cdot 4 \cdot 5 = -10$$

1 point is earned for $g(3)$

$$g(3) = \int_{-2}^{3} f(t)dt = \frac{1}{2}\pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$

0	1	2

The response earns all of the following points:

1 point is earned for $g(-6)$

5-6

$$g(-6) = \int_{-2}^{-6} f(t)dt = -\int_{-6}^{-2} f(t)dt = -\frac{1}{2} \cdot 4 \cdot 5 = -10$$

1 point is earned for $g(3)$

$$g(3) = \int_{-2}^{3} f(t)dt = \frac{1}{2}\pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$

Part B

The response can earn up to 1 point:

1 point is earned for the correct evaluation of $g'(0)$

$g'(0) = f(0) = 2$

0	1 ✓

The response earns all of the following point:

1 point is earned for the correct evaluation of $g'(0)$

$g'(0) = f(0) = 2$

Part C

The response can earn up to 3 points:

1 point is earned for the horizontal tangent at x = -2 and x = 2

The graph of g has a horizontal tangent at x = -2 and x = 2 where $g'(x) = f(x) = 0$.

2 points are earned for correct answers with justifications

The graph of g has neither a local maximum nor a local minimum at x = -2 because $g'(x) = f(x)$ does not change sign at x = -2.

The graph of g has a local maximum at x = 2 because $g'(x) = f(x)$ changes sign from positive to negative at x = 2.

0	1	2	3 ✓

The response earns all of the following points:

5-6

1 point is earned for the horizontal tangent at $x = -2$ and $x = 2$

The graph of g has a horizontal tangent at $x = -2$ and $x = 2$ where $g'(x) = f(x) = 0$.

2 points are earned for correct answers with justifications

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

The graph of g has a local maximum at $x = 2$ because $g'(x) = f(x)$ changes sign from positive to negative at $x = 2$.

Part D

The response can earn up to 3 points:

2 points are earned for the correct values of x

1 point is earned for the explanation

The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

0	1	2	3

The response earns all of the following points:

2 points are earned for the correct values of x

1 point is earned for the explanation

The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

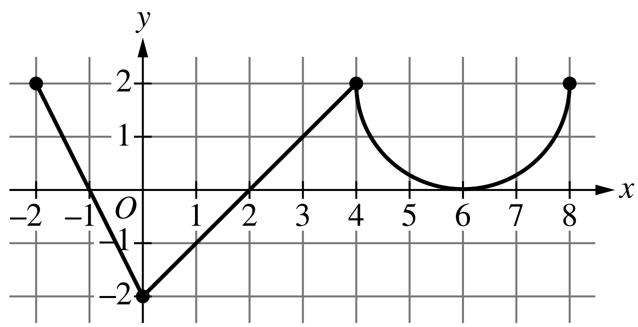
5-6

42. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f'

The function f is defined on the closed interval $[-2, 8]$ and satisfies $f(2) = 1$. The graph of f', the derivative of f, consists of two line segments and a semicircle, as shown in the figure.

(a) Does f have a relative minimum, a relative maximum, or neither at $x = 6$? Give a reason for your answer.

(b) On what open intervals, if any, is the graph of f concave down? Give a reason for your answer.

(c) Find the value of $\lim\limits_{x \to 2} \frac{6f(x) - 3x}{x^2 - 5x + 6}$, or show that it does not exist. Justify your answer.

(d) Find the absolute minimum value of f on the closed interval $[-2, 8]$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that declares $f'(x)$ does not change sign at $x = 6$, so neither, is sufficient to earn the point.

A response does not have to present intervals on which $f'(x)$ is positive or negative, but if any are given, they must be correct. Any presented intervals may include none, one, or both endpoints.

A response that declares $f'(x) > 0$ before and after $x = 6$ does not earn the point.

0	1

5-6

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

$f'(x) > 0$ on $(2, 6)$ and $f'(x) > 0$ on $(6, 8)$.

$f'(x)$ does not change sign at $x = 6$, so there is neither a relative maximum nor a relative minimum at this location.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only by an answer of $(-2, 0)$ and $(4, 6)$, or these intervals including one or both endpoints.

A response must earn the first point to be eligible for second point.

To earn the second point a response must correctly discuss the behavior of f' or the slopes of f'.

Special case: A response that presents exactly one of the two correct intervals with a correct reason earns 1 out of 2 points.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Intervals
- [ ] Reason

Solution:

The graph of f is concave down on $(-2, 0)$ and $(4, 6)$ because f' is decreasing on these intervals.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the presentation of two separate limits for the numerator and denominator.

A response that presents a limit explicitly equal to $\frac{0}{0}$ does not earn the first point.

The second point is earned by applying L'Hospital's Rule, that is, by presenting at least one correct derivative in the limit of a ratio of derivatives.

5-6

The third point is earned for the correct answer with supporting work.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator
- ☐ Uses L'Hospital's Rule
- ☐ Answer

Solution:

Because f is differentiable at $x=2$, f is continuous at $x=2$, so $\lim_{x\to 2} f(x)=f(2)=1$.

$\lim_{x\to 2}(6f(x)-3x)=6\cdot 1-3\cdot 2=0$

$\lim_{x\to 2}\left(x^2-5x+6\right)=0$

Because $\lim_{x\to 2}\frac{6f(x)-3x}{x^2-5x+6}$ is of indeterminate form $\frac{0}{0}$, L'Hospital's Rule can be applied.

Using L'Hospital's Rule,
$\lim_{x\to 2}\frac{6f(x)-3x}{x^2-5x+6}=\lim_{x\to 2}\frac{6f'(x)-3}{2x-5}=\frac{6\cdot 0-3}{2\cdot 2-5}=3.$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must state $f'=0$ or equivalent. Listing the zeros of f' is not sufficient.

A response that presents any error in evaluating f at any critical point or endpoint will not earn the justification point.

A response need not present the value of $f(-1)$ provided $x=-1$ is eliminated because it is the location of a local maximum. A response need not present the value of $f(6)$ provided $x=6$ is eliminated by reference to part (a) or eliminated through analysis.

A response need not present the value of $f(8)$ provided there is a presentation that argues $f'(x)\geq 0$ for $x>2$ and, therefore, $f(8)>f(2)$.

A response that does not consider both endpoints does not earn the justification point.

5-6

The answer point is earned only for indicating that the minimum value is 1. It is not earned for noting that the minimum occurs at $x = 2$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $f'(x) = 0$
- ☐ Justification
- ☐ Answer

Solution:

$f'(x) = 0 \Rightarrow x = -1,\ x = 2,\ x = 6$

The function f is continuous on $[-2, 8]$, so the candidates for the location of an absolute minimum for f are $x = -2$, $x = -1$, $x = 2$, $x = 6$, and $x = 8$.

x	$f(x)$
-2	3
-1	4
2	1
6	$7 - \pi$
8	$11 - 2\pi$

The absolute minimum value of f is $f(2) = 1$.

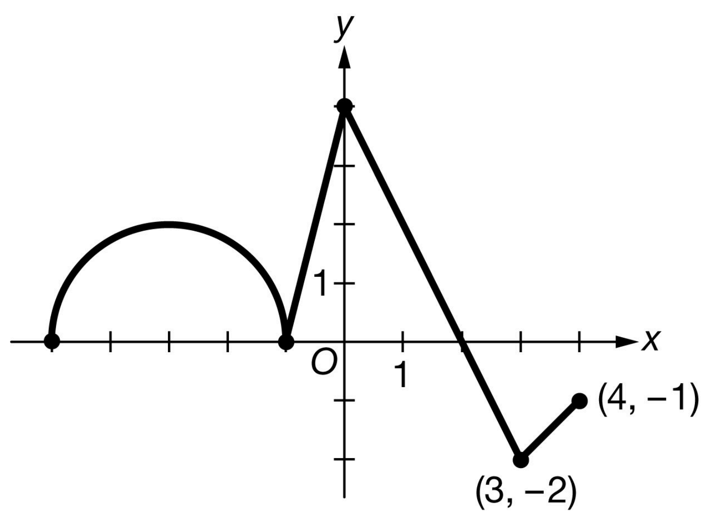
5-6

43. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g'

Let g be a function that is defined on the closed interval $-5 \leq x \leq 4$ with $g(0) = 5$. The graph of g', the derivative of g, consists of a semicircle and three line segments, as shown above.

(a) On what intervals, if any, is g decreasing? Give a reason for your answer.

(b) Find the average rate of change of g on the interval $[-3, 3]$. Is there a value c, $-3 < c < 3$, for which $g'(c)$ is equal to the average rate of change of g on the interval $[-3, 3]$? Justify your answer.

(c) Find the x-coordinate of each point of inflection of the graph of g on the open interval $(-5, 4)$. Justify your answer.

(d) Let $H(x) = e^{3x} g(x)$. Find the value of $H'(0)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response does not have to say that g is continuous to earn the point.

A response with correct endpoints can use open, closed, or half-open interval notation to earn the point.

5-6

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

g is decreasing on the interval $[2,\ 4]$ because g is continuous and $g'\ (x) < 0$ on $(2,\ 4]$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for declaring that the average rate of change is either $\frac{g(3)-g(-3)}{3-(-3)}$ or $\frac{\int_{-3}^{3} g'(x)\ dx}{6}$. The difference quotient and the definite integral must contain $\frac{1}{6}$ to earn the point.

The second point is earned for finding the value of the average rate of change, $\frac{\pi+5}{6}$ or equivalent.

The third point is for stating that g is continuous because g is differentiable (on $[-3,\ 3]$). Stating that g is both differentiable and continuous on the interval is not enough to earn this point.

The fourth point is for applying either the Mean Value Theorem (MVT) and concluding "yes," or for applying the Intermediate Value Theorem (IVT) and concluding "yes." Neither theorem needs to be named if the appropriate conditions are checked and the correct conclusion is reached.

If a response does not appropriately cite/apply the MVT or IVT, the response does not earn the fourth point.

0	1	2	3	4

The student response accurately includes **all four** of the criteria below.

- [ ] Difference quotient or definite integral
- [ ] Average rate of change
- [ ] Differentiable implies continuous
- [ ] Conclusion using Mean Value Theorem
  – OR –
  Conclusion using Intermediate Value Theorem

5-6

Solution:

Average rate of change $= \frac{g(3)-g(-3)}{3-(-3)} = \frac{\int_{-3}^{3} g'(x)\,dx}{6}$

$= \frac{\pi+6-1}{6} = \frac{\pi+5}{6}$

g is differentiable on $[-3,\ 3]$. $\Rightarrow$ g is continuous on $[-3,\ 3]$.

The Mean Value Theorem guarantees that there exists a value of c in the interval $(-3,\ 3)$ such that $g'(c)$ is equal to the average rate of change of g on the interval $[-3,\ 3]$.

– OR –

g' is continuous on $[-3,\ 3]$.

The range of g' on the interval $[-3,\ 3]$ is $-2 \leq g'(x) \leq 4$. Because $-2 < \frac{\pi+5}{6} < 4$, there exists a value of c in the interval $(-3,\ 3)$ such that $g'(c)$ is equal to the average rate of change of g on the interval $[-3,\ 3]$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must identify all four answers.

A response that provides a correct identification and justification for just two (or three) of the four answers earns 1 of the 2 points.

The second point can be earned by correctly discussing the signs of the slopes of the graph of g'.

The second point cannot be earned by the use of vague or undefined terms such as "it" or "the graph" or "the derivative."

The second point cannot be earned by only discussing the concavity of the graph of g.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Answers
- ☐ Justification

Solution:

5-6

The graph of g will have a point of inflection where the graph of g' changes from increasing to decreasing or from decreasing to increasing. This occurs at $x = -3,\ \ x = -1,\ \ x = 0$, and $x = 3$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for using the product rule and chain rule to correctly find $H'(x)$ or $H'(0)$.

The second point is earned for correctly using the values of $g(0)$ and $g'(0)$ in the calculation of $H'(0)$. This point can be earned without the first point if there is a single error in finding $H'(x)$.

A response of $3 \cdot 5 + 4$ is the minimal response that earns both points.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $H'(x)$
- ☐ Answer

Solution:

$H'(x) = 3e^{3x} \cdot g(x) + e^{3x} \cdot g'(x)$

$H'(0) = 3 \cdot g(0) + g'(0) = 3 \cdot 5 + 4 = 19$

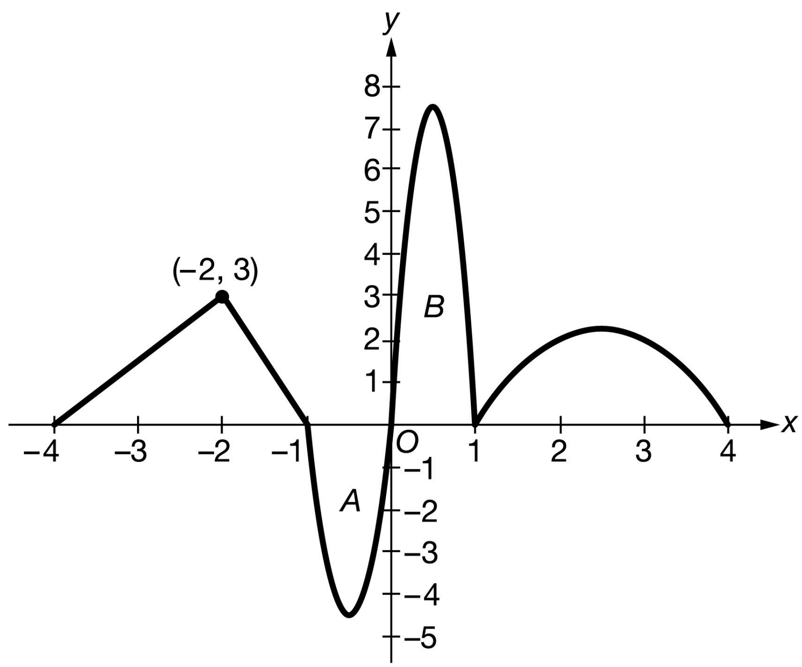
5-6

44. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The continuous function f is defined for $-4 \le x \le 4$. The graph of f, shown above, consists of two line segments and portions of three parabolas. The graph has horizontal tangents at $x = -\frac{1}{2}$, $x = \frac{1}{2}$, and $x = \frac{5}{2}$. It is known that $f(x) = -x^2 + 5x - 4$ for $1 \le x \le 4$. The areas of regions A and B bounded by the graph of f and the x-axis are 3 and 5, respectively. Let g be the function defined by $g(x) = \int_{-4}^{x} f(t)\,dt$.

(a) Find $g(0)$ and $g(4)$.

(b) Find the absolute minimum value of g on the closed interval $[-4, 4]$. Justify your answer.

(c) Find all intervals on which the graph of g is concave down. Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Simplification is not required for the first point.

5-6

The second point is earned with a correct integral or with work that shows that the integral has been taken. For example, $\int_1^4 f(t)\, dt$, or $\int_1^4 \left(-t^2+5t-4\right)\, dt$, or $\left[-\frac{1}{3}t^3+\frac{5}{2}t^2-4t\right]_1^4$ would earn the second point.

The third point is earned with $-\frac{1}{3}t^3+\frac{5}{2}t^2-4t$.

Simplification is not required for the fourth point.

0	1	2	3	4

The student response accurately includes **all four** of the criteria below.

- ☐ $g(0)$
- ☐ integral of f over $1 \le t \le 4$
- ☐ antiderivative
- ☐ $g(4)$

Solution:

$$g(0)=\int_{-4}^{0} f(t)\, dt=\frac{9}{2}-3=\frac{3}{2}$$

$$\begin{aligned}
g(4)&=\int_{-4}^{4} f(t)\, dt\\
&=\int_{-4}^{0} f(t)\, dt+\int_{0}^{1} f(t)\, dt+\int_{1}^{4} f(t)\, dt\\
&=\frac{3}{2}+5+\int_1^4 \left(-t^2+5t-4\right)\, dt\\
&=\frac{3}{2}+5+\left[-\frac{1}{3}t^3+\frac{5}{2}t^2-4t\right]_1^4\\
&=\frac{3}{2}+5+\left[\left(-\frac{1}{3}\cdot 4^3+\frac{5}{2}\cdot 4^2-4\cdot 4\right)-\left(-\frac{1}{3}\cdot 1^3+\frac{5}{2}\cdot 1^2-4\cdot 1\right)\right]\\
&=\frac{3}{2}+5+\left(\frac{8}{3}-\left(-\frac{11}{6}\right)\right)=11
\end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an explicit statement connecting $g' = f$. Note that it is possible that this connection is made in an earlier part and will earn the point here.

The second point is earned for values appearing in an equation, table, or paragraph.

5-6

The third point is earned by ruling out all other candidates and committing to the minimum value of 0.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ $g'(x) = f(x)$
- ☐ identifies $x = -4$ and $x = 0$ as candidates
- ☐ answer with justification

Solution:

$g'(x) = f(x)$ is negative for $-1 < x < 0$, and nonnegative elsewhere. Thus, the absolute minimum value of g on $[-4,\ 4]$ can only occur at $x = -4$ or $x = 0$.

$g(-4) = 0$

$g(0) = \frac{3}{2}$

The absolute minimum value of g on $[-4,\ 4]$ is $g(-4) = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with all three intervals, with or without the endpoints. Verbal expressions rather than intervals also earn the point.

The second point is earned with reasoning clearly tied to the given graph of f.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ intervals
- ☐ reason

Solution:

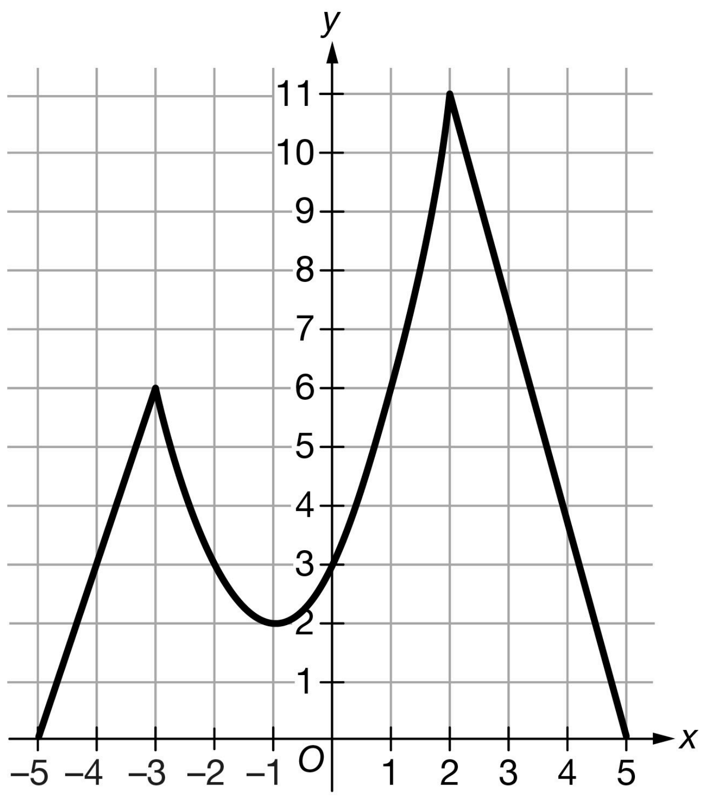
The graph of g is concave down on the intervals $-2 < x < -\frac{1}{2}$, $\frac{1}{2} < x < 1$, and $\frac{5}{2} < x < 4$ because $g'(x) = f(x)$ is decreasing on these intervals.

45. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The continuous function f is defined on the closed interval $[-5, 5]$. The graph of f consists of a parabola and two line segments, as shown in the figure above. Let g be a function such that $g'(x) = f(x)$.

(a) Fill in the missing entries in the table below to describe the behavior of f' and f''. Indicate Positive, Negative, or 0. Give reasons for your answers.

x	$-5 < x < -3$	$-3 < x < -1$	$-1 < x < 2$	$2 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$				
$f''(x)$				

(b) There is no value of x in the open interval $(-1, 5)$ at which $f'(x) = \frac{f(5)-f(-1)}{5-(-1)}$. Explain why this does not violate the Mean Value Theorem.

5-6

(c) Find all values of x in the open interval $(-5, 5)$ at which the graph of g has a point of inflection. Explain your reasoning.

(d) At what value of x does g attain its absolute maximum on the closed interval $[-5, 5]$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] behavior for $f'(x)$
- [ ] behavior for $f''(x)$
- [ ] reasons

Solution:

x	$-5<x<-3$	$-3<x<-1$	$-1<x<2$	$2<x<5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$	**Positive**	**Negative**	**Positive**	**Negative**
$f''(x)$	**0**	**Positive**	**Positive**	**0**

$f'(x)$ is positive for $-5 < x < -3$ and $-1 < x < 2$ because f is increasing there.

$f'(x)$ is negative for $-3 < x < -1$ and $2 < x < 5$ because f is decreasing there.

$f''(x)$ is positive for $-3 < x < -1$ and $-1 < x < 2$ because the graph of f is concave up there.

$f''(x) = 0$ for $-5 < x < -3$ and $2 < x < 5$ because the graph of f is linear there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The student response accurately includes a correct explanation.

Solution:

5-6

This does not violate the Mean Value Theorem because f is not differentiable at $x = 2$ due to a sharp corner. The Mean Value Theorem cannot be applied on the interval $-1 < x < 5$.

Part C

The response is not eligible for any points if incorrect values of x are included.

A response may earn the first point for two correct values of x without explanations or with incomplete, incorrect explanations.

A maximum of 1 point may be earned for three correct values of x without explanations.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- [ ] One correct value of x with explanation OR Two correct values of x without explanations
- [ ] A second correct value of x with explanation
- [ ] A third correct value of x with explanation

Solution:

The graph of g has a point of inflection at $x = -3$, $x = -1$, and $x = 2$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -3$ and $x = 2$, and $g'(x) = f(x)$ changes from decreasing to increasing at $x = -1$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- [ ] answer
- [ ] reason

Solution:

5-6

Because $g'(x) = f(x) > 0$ on the interval $(-5,\ 5)$, g is increasing on the interval $(-5,\ 5)$. Therefore, the absolute maximum value of g on the closed interval $[-5,\ 5]$ occurs at the right endpoint $x = 5$.

5-6

46. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

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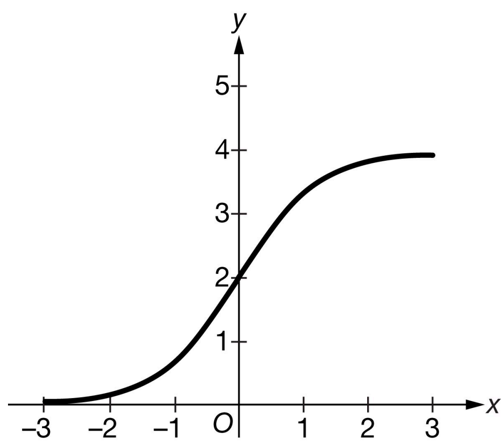


Graph of g

The graph of the continuous function g is shown above for $-3 \leq x \leq 3$. The function g is twice differentiable, except at $x = 0$.

Let f be the function with $f(0) = -1$ and derivative given by $f'(x) = \left(x^2 - \frac{5}{4}\right)e^x$.

(a) Find the x-coordinate of each critical point of f. Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.

(b) Find all values of x at which the graph of f has a point of inflection. Give reasons for your answers.

(c) Fill in the missing entries in the table below to describe the behavior of g' and g'' on the interval $-3 \leq x \leq 3$. Indicate Positive or Negative. Give reasons for your answers.

x	$x = -3$	$-3 < x < 0$	$x = 0$	$0 < x < 3$	$x = 3$
$g(x)$	0	Positive	2	Positive	4
$g'(x)$	0		2		0
$g''(x)$	0		Undefined		0

(d) Let h be the function defined by $h(x) = f(x)g(x)$. Is h increasing or decreasing at $x = 0$? Give a reason for your answer.

Part A

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

5-6

The first point requires reference to $f'(x) = 0$.

A maximum of 1 out of 3 points is earned for only one correct critical point with correct identification and justification, and no incorrect critical points are included.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] critical points
- [ ] relative maximum at $x = -\frac{\sqrt{5}}{2}$ with justification
- [ ] relative minimum at $x = \frac{\sqrt{5}}{2}$ with justification

Solution:

$$f'(x) = \left(x^2 - \tfrac{5}{4}\right)e^x = 0 \;\Rightarrow\; x^2 = \tfrac{5}{4} \;\Rightarrow\; x = -\frac{\sqrt{5}}{2},\; x = \frac{\sqrt{5}}{2}$$

f has a relative maximum at $x = -\frac{\sqrt{5}}{2}$ because $f'(x)$ changes from positive to negative there.

f has a relative minimum at $x = \frac{\sqrt{5}}{2}$ because $f'(x)$ changes from negative to positive there.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

A maximum of 1 out of 2 points is earned if only one point of inflection with reason and no incorrect points of inflection.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $f''(x)$
- [ ] answers with reasons

5-6

Solution:

$$f''(x) = 2xe^x + \left(x^2 - \frac{5}{4}\right)e^x = e^x\left(x^2 + 2x - \frac{5}{4}\right) = 0$$

$$\Rightarrow\ x^2 + 2x - \frac{5}{4} = 0$$

$$\Rightarrow\ x = \frac{-2 \pm \sqrt{4 - 4(1)\left(-\frac{5}{4}\right)}}{2} = \frac{-2 \pm \sqrt{9}}{2} = \frac{-2 \pm 3}{2}$$

$$\Rightarrow\ x = -\frac{5}{2} \text{ and } x = \frac{1}{2}$$

Because $f''(x)$ changes sign at $x = -\frac{5}{2}$ and $x = \frac{1}{2}$, the graph of f has points of inflection there.

Part C

A maximum of 1 out of 2 points is earned if only behavior for $g'(x)$ with reason OR behavior for $g''(x)$ with reason.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ behavior for $g'(x)$ and $g''(x)$
- ☐ reasons

Solution:

x	$x=-3$	$-3<x<0$	$x=0$	$0<x<3$	$x=3$
$g(x)$	0	Positive	2	Positive	4
$g'(x)$	0	**Positive**	2	**Positive**	0
$g''(x)$	0	**Positive**	Undefined	**Negative**	0

$g'(x)$ is positive for $-3 < x < 0$ and $0 < x < 3$ because g is increasing there.

$g''(x)$ is positive for $-3 < x < 0$ because the graph of g is concave up there, and $g''(x)$ is negative for $0 < x < 3$ because the graph of g is concave down there.

Part D

The response is not eligible for the second point with an incorrect product rule.

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] product rule
- [ ] answer with reason

Solution:

$h'(x) = f'(x)g(x) + f(x)g'(x)$

$h'(0) = f'(0)g(0) + f(0)g'(0)$

Because $f(0) < 0,\ f'(0) < 0,\ g(0) > 0$, and $g'(0) > 0$, it follows that $h'(0) < 0$ and h is decreasing at $x = 0$.

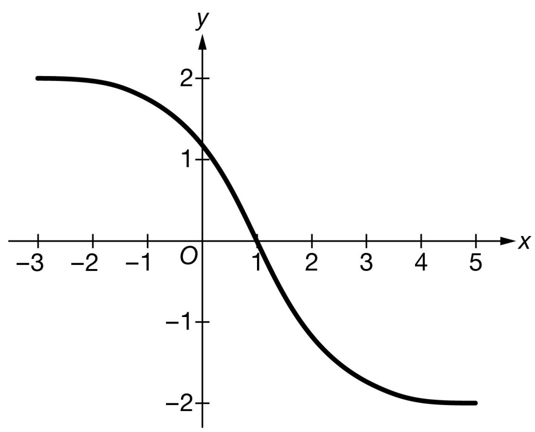
5-6

47. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the continuous function g is shown above for $-3 \leq x \leq 5$. The function g is twice differentiable except at $x = 1$.

Let f be the function with $f(1) = 3$ and derivative given by $f'(x) = \left(6x^2 - 5x\right)e^x$.

(a) Find the x-coordinate of each critical point of f. Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.

(b) Find all values of x at which the graph of f has a point of inflection. Give reasons for your answers.

Fill in the missing entries in the table below to describe the behavior of g' and g'' on the interval $-3 \leq x \leq 5$. Indicate Positive or Negative. Give reasons for your answers.

x	$x = -3$	$-3 < x < 1$	$x = 1$	$1 < x < 5$	$x = 5$
$g(x)$	2	Positive	0	Negative	-2
$g'(x)$	0		$-\frac{3}{2}$		0
$g''(x)$	0		Undefined		0

(d) Let h be the function defined by $h(x) = f(x)g(x)$. Is h increasing or decreasing on the interval $1 < x < 5$? Give a reason for your answer.

Part A

5-6

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The first point requires reference to $f'(x) = 0$.

A maximum of 1 out of 3 points is earned for only one correct critical point with correct identification and justification, and no incorrect critical points are included.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ critical points
- ☐ relative maximum at $x = 0$ with justification.
- ☐ relative minimum at $x = \frac{5}{6}$ with justification.

Solution:

$$f'(x) = \left(6x^2 - 5x\right)e^x = 0 \;\Rightarrow\; 6x^2 - 5x = 0 \Rightarrow\; x(6x - 5) = 0 \Rightarrow\; x = 0,\; x = \frac{5}{6}$$

f has a relative maximum at $x = 0$ because $f'(x)$ changes from positive to negative there.

f has a relative minimum at $x = \frac{5}{6}$ because $f'(x)$ changes from negative to positive there.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

A maximum of 1 out of 2 points is earned if only one point of inflection with reason and no incorrect points of inflection.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ $f''(x)$

5-6

- ☐ answers with reasons

Solution:

$f''(x) = (12x - 5)e^x + \left(6x^2 - 5x\right)e^x = e^x\left(6x^2 + 7x - 5\right) = 0$
$\Rightarrow\ 6x^2 + 7x - 5 = (3x + 5)(2x - 1) = 0$
$\Rightarrow\ x = -\frac{5}{3}$ and $x = \frac{1}{2}$

Because $f''(x)$ changes sign at $x = -\frac{5}{3}$ and $x = \frac{1}{2}$, the graph of f has points of inflection there.

Part C

A maximum of 1 out of 2 points is earned if only behavior for $g'(x)$ with reason OR behavior for $g''(x)$ with reason.

Select a point value to view scoring criteria, solutions, and/or examples to score the response

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ behavior for $g'(x)$ and $g''(x)$
- ☐ reasons

Solution:

x	$x=-3$	$-3<x<1$	$x=1$	$1<x<5$	$x=5$
$g(x)$	2	Positive	0	Negative	-2
$g'(x)$	0	**Negative**	$-\frac{3}{2}$	**Negative**	0
$g''(x)$	0	**Negative**	Undefined	**Positive**	0

$g'(x)$ is negative for $-3 < x < 1$ and $1 < x < 5$ because g is decreasing there.

$g''(x)$ is negative for $-3 < x < 1$ because the graph of g is concave down there, and $g''(x)$ is positive for $1 < x < 5$ because the graph of g is concave up there.

Part D

The response is not eligible for the second point with an incorrect product rule.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-6

0	1	2 ✓

The student response accurately includes both of the criteria below.

- [ ] product rule
- [ ] answer with reason

Solution:

$h'(x) = f'(x)g(x) + f(x)g'(x)$

Because $f(1) = 3$ and $f'(x) > 0$ on $1 < x < 5$, it follows that $f(x) > 0$ on $1 < x < 5$. Because $g(x) < 0$ and $g'(x) < 0$ on $1 < x < 5$, it follows that $h'(x) < 0$. Thus h is decreasing on the interval $1 < x < 5$.

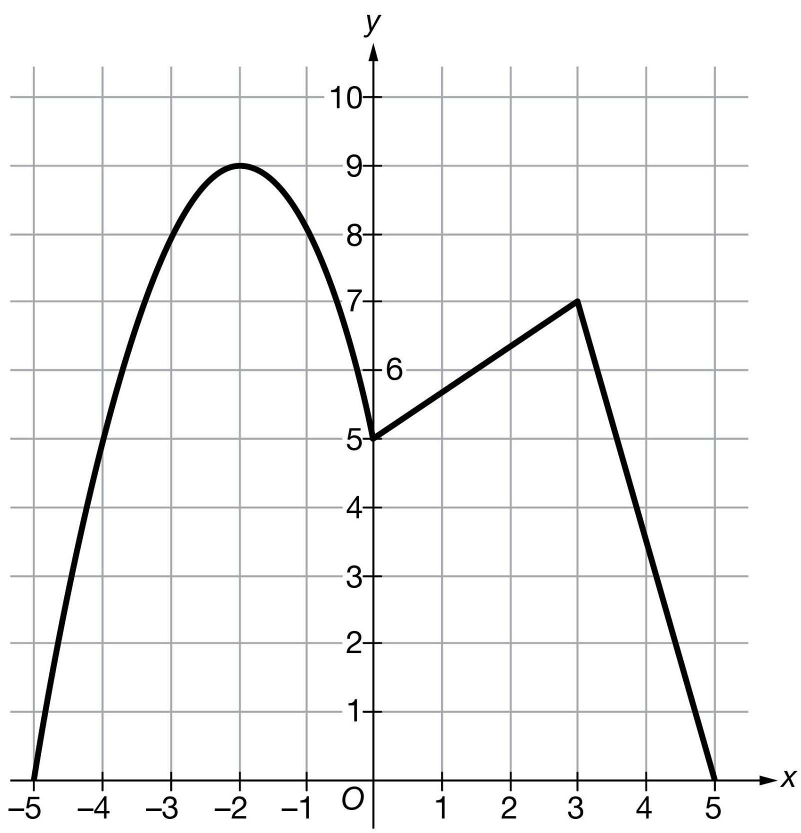
5-6

48. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The continuous function f is defined on the closed interval $[-5, 5]$. The graph of f consists of a parabola and two line segments, as shown in the figure above. Let g be a function such that $g'(x) = f(x)$.

(a) Fill in the missing entries in the table below to describe the behavior of f' and f''. Indicate Positive, Negative, or 0. Give reasons for your answers.

x	$-5 < x < -2$	$-2 < x < 0$	$0 < x < 3$	$3 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$				
$f''(x)$				

5-6

(b) There is no value of x in the open interval $(-1, 3)$ at which $f'(x) = \frac{f(3)-f(-1)}{3-(-1)}$. Explain why this does not violate the Mean Value Theorem.

(c) Find all values of x in the open interval $(-5, 5)$ at which the graph of g has a point of inflection. Explain your reasoning.

(d) At what value of x does g attain its absolute maximum on the closed interval $[-5, 5]$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ behavior for $f'(x)$
- ☐ behavior for $f''(x)$
- ☐ reasons

Solution:

x	$-5<x<-2$	$-2<x<0$	$0<x<3$	$3<x<5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$	**Positive**	**Negative**	**Positive**	**Negative**
$f''(x)$	**Negative**	**Negative**	**0**	**0**

$f'(x)$ is positive for $-5 < x < -2$ and $0 < x < 3$ because f is increasing there.

$f'(x)$ is negative for $-2 < x < 0$ and $3 < x < 5$ because f is decreasing there.

$f''(x)$ is negative for $-5 < x < -2$ and $-2 < x < 0$ because the graph of f is concave down there.

$f''(x) = 0$ for $0 < x < 3$ and $3 < x < 5$ because the graph of f is linear there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

5-6

The student response accurately includes a correct explanation.

Solution:

This does not violate the Mean Value Theorem because f is not differentiable at $x = 0$ due to a sharp corner. The Mean Value Theorem cannot be applied on the interval $-1 < x < 3$.

Part C

Two values of x must be correct without explanation to earn one point, and no incorrect values included.

A maximum of 1 point may be earned for one correct value with explanation.

A maximum of 2 points may be earned for two correct values with explanation.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ values of x (2 points full credit)
- ☐ explanation

Solution:

The graph of g has a point of inflection at $x = -2$, $x = 0$, and $x = 3$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -2$ and $x = 3$, and $g'(x) = f(x)$ changes from decreasing to increasing at $x = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ reason

5-6

Solution:

Because $g'(x) = f(x) > 0$ on the interval $(-5,\ 5)$, g is increasing on the interval $(-5,\ 5)$. Therefore, the absolute maximum value of g on the closed interval $[-5,\ 5]$ occurs at the right endpoint $x = 5$.

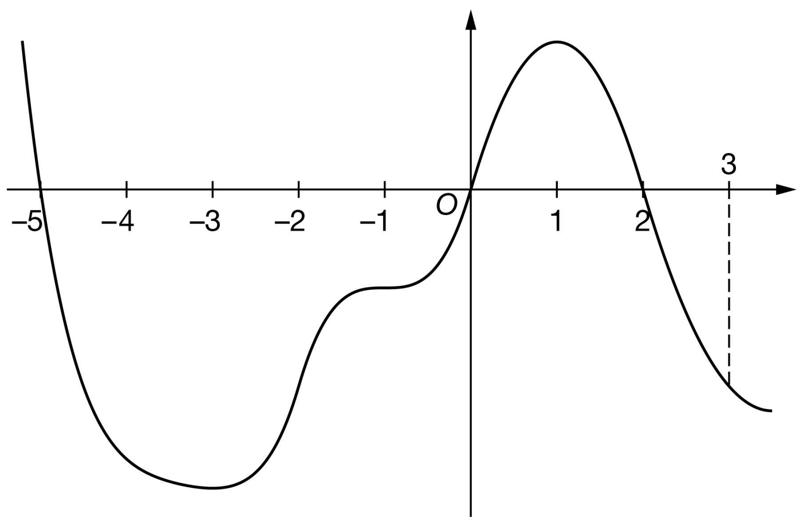
5-6

49. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-5.5 \leq x \leq 3.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of f on the intervals $[-5, 0]$, $[0, 2]$, and $[2, 3]$ are 57, 12, and 7, respectively. The graph of f has horizontal tangents at $x = -3$, $x = -1$, and $x = 1$. Let g be the function defined by $g(x) = \int_2^x f(t)\,dt$ for $-5.5 \leq x \leq 3.5$.

(a) Find the x-coordinate of each critical point of g on the interval $-5.5 < x < 3.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of g on the interval $-5.5 < x < 3.5$. Justify your answer.

(d) Find the value of $g(3)$.

(e) Find the value of $g(-5)$. Show the computations that lead to your answer.

(f) Must there exist a value of k, for $-5 < k < 0$, such that $g'(k)$ is equal to the average rate of change of g over the interval $-5 \leq x \leq 0$? Justify your answer.

(g) Find $\lim_{x\to 3} \frac{5g(x)}{2x}$. Show the computations that lead to your answer.

(h) The function r is defined by $r(x) = x^2 \cdot g(x)$. It is known that $f(3) = -12$. Find $r'(3)$. Show the computations that lead to your answer.

5-6

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -5,\quad x = 0$, and $x = 2$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5.5,\ 3.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5.5, 3.5)$. The point is still earned provided the three critical points, $x = -5,\quad x = 0$, and $x = 2$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-5,\ 0),\quad (0,\ 0),\quad (2,\ 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -5$, $x = 0$, and $x = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $f(x)$ (that is, $g'(x) = f(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -5$, g has a relative maximum) is eligible for at most one of the three points.

5-6

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-5)$ changes from positive to negative”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $h(-5.5) < 45$

· $h(-5) = 45$

· $h(0) = -12$

· $h(2) = 0$

· $h(3) = -7$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -5$, g has a relative maximum because $g' = f$ is positive for $x < -5$ and $g' = f$ is negative for $x > -5$.

· At $x = 0$, g has a relative minimum because $g' = f$ is negative for $x < 0$ and $g'(x) = f(x)$ is positive for $x > 0$.

· At $x = 2$, g has a relative maximum because $g' = f$ is positive for $x < 2$ and $g' = f$ is negative for $x > 2$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Relative maximum with justification
- [ ] Relative minimum with justification

5-6

- [ ] Relative maximum with justification

Solution:

At $x = -5$, g Alt text: g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = 0$, g Alt text: g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

At $x = 2$, g Alt text: g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only g' is eligible for all three points.

If a response gives justifications based only on f, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $f(x)$ (that is, $g'(x) = f(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = f(x)$.

Justification for the first two points should reference one of the following:

· $g'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $g'(x)$

· $g''(x)$ changes sign

· $f'(x)$ changes sign

“ $g'(x) = f(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to f as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn both points.

A response that discusses g' at a particular point (i.e., $g'(-3)$) when the discussion should have been about $g'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

5-6

If $x = -3$ and $x = 1$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ $x = -3$ with justification
- ☐ $x = 1$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -3$, the graph of g has a point of inflection because $g'(x) = f(x)$ changes from decreasing to increasing there.

At $x = 1$, the graph of g has a point of inflection because $g'(x) = f(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -7 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

- ☐ ·Answer

Solution:

$$g(3) = \int_{2}^{3} f(t)\,dt = -7$$

Part E

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 57 and ± 12 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 57) \pm (\pm 12)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 57 and ± 12 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-57)-(-12)$ earns both points because numerical expressions need not be simplified.

· The expression $(-57)-12$ earns the first point, but not the second point.

· The expression $-57-(-12)-7$ earns neither point.

· The expression $\int_2^{-5} f(t)\, dt = 45$ earns neither point.

· The expression $-\int_{-5}^{0} f(t)dt - \int_0^2 f(t)dt = 45$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 57 and ± 12, but it does earn the second point.

· $-57, -12$ earns neither point

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ $\left|\int_{-5}^{0} f(t)\, dt\right| = 57$ and $\left|\int_{2}^{0} f(t)\, dt\right| = 12$
- ☐ Answer

Solution:

$$g(-5) = \int_2^{-5} f(t)\, dt = -\int_{-5}^{2} f(t)\, dt$$
$$= -\int_{-5}^{0} f(t)\, dt - \int_0^2 f(t)\, dt = -(-57) - 12 = 45$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-6

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-5,\ 0)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $g\ (x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $g\ (x)$ is continuous and differentiable, therefore there must be a value where $g'\ (k)$ equals the average rate of change of $g\ (x)$ " earns both points.

· "Yes, $g'\ (x)$ (the derivative of $g\,(x)$) is continuous and differentiable, therefore there must be a value where $g'\ (k)$ equals the average rate of change of $g\ (x)$ " does not earn the first point, but does the earn second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $g\ (x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $g\ (x)$ is continuous" earns neither point.

· "Yes, because $g\ (x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = f \Rightarrow g$ is differentiable on $-5 \leq x \leq 0. \Rightarrow g$ is continuous on $-5 \leq x \leq 0.$

Therefore, the Mean Value Theorem can be applied to g on the interval $-5 \leq x \leq 0$ to guarantee that there exists a value of k , for $-5 < k < 0$, such that $g'\ (k)$ equals the average rate of change of g over the interval $-5 \leq x \leq 0$.

Part G

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the first point include:

- $\dfrac{5(-7)}{2 \cdot 3}$
- $-\dfrac{35}{6}$

Responses that do not earn the first point include:

- $\dfrac{5 \cdot g(3)}{6}$
- $\dfrac{5 \cdot g(3)}{2 \cdot 3}$

Note: an incorrect value for $g\,(3)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that g is continuous. If the response does state that g is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g' = f \Rightarrow g$ is continuous. $\Rightarrow \lim\limits_{x \to 3} g\,(x) = g\,(3)$

$$\lim_{x \to 3} \frac{5g(x)}{2x} = \frac{5g(3)}{2 \cdot (3)} = \frac{5(-7)}{6} = -\frac{35}{6}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of r that contains a sum of two terms, with at least one term written as a product, and one term involving g and the other term involving g' (or f).

5-6

Fundamental Theorem of Calculus is evidenced by linking g' and f either directly or indirectly in the presented derivative of $r(x)$ or in $r'(3)$. Since $f(3) = -12$ is given in the question, replacing $g'(3)$ directly with -12 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $g(3)$ that is brought in from part (d) can earn the third point.

$9(-12) + 6(-7)$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a, b, c, or d is incorrect, will earn zero points.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Product rule
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$r'(x) = x^2 \cdot g'(x) + 2x \cdot g(x)$

$= x^2 \cdot f(x) + 2x \cdot g(x)$

$r'(3) = 3^2 \cdot f(3) + 2 \cdot 3 \cdot g(3) = 9 \cdot (-12) + 6 \cdot (-7) = -150$

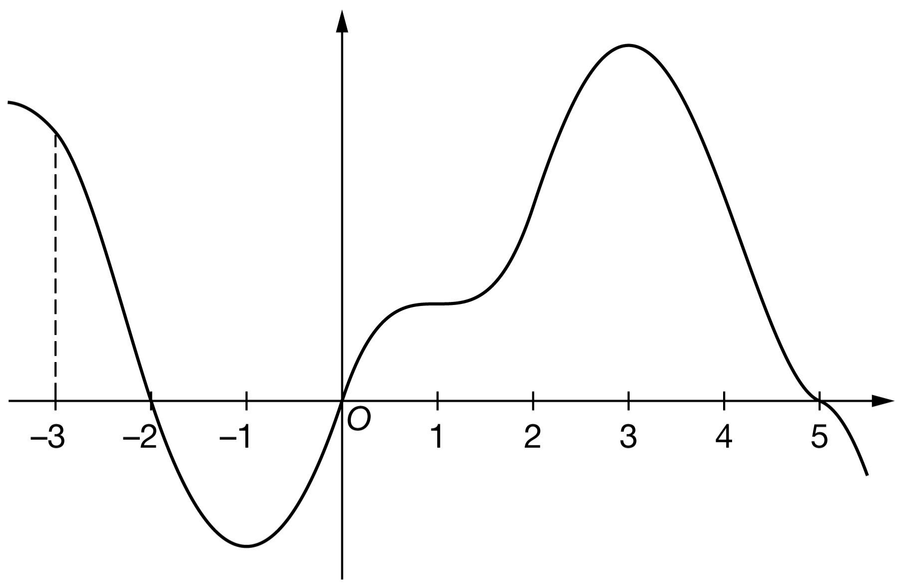
5-6

50. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the differentiable function g with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of g on the intervals $[-3,-2]$, $[-2,0]$, and $[0,5]$ are 10, 13, and 59, respectively. The graph of g has horizontal tangents at $x=-1$, $x=1$, and $x=3$. Let h be the function defined by $h(x)=\displaystyle\int_0^x g(t)\,dt$ for $-3.5 \leq x \leq 5.5$.

(a) Find the x-coordinate of each critical point of h on the interval $-3.5 < x < 5.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of h on the interval $-3.5 < x < 5.5$. Justify your answer.

(d) Find the value of $h(5)$.

(e) Find the value of $h(-3)$. Show the computations that lead to your answer.

(f) Must there exist a value of p, for $-3 < p < 2$, such that $h'(p)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 2$? Justify your answer.

(g) Find $\lim\limits_{x \to -3} \frac{h(x)}{2x}$. Show the computations that lead to your answer.

5-6

(h) The function f is defined by $f(x) = x \cdot h(x)$. It is known that $g(-3) = 18$. Find $f'(-3)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -2$, $x = 0$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5,\ 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5,\ 5.5)$. The point is still earned provided the three critical points, $x = -2$, $x = 0$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-2,\ 0)$, $(0, 0)$, $(5, 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -2$, $x = 0$, and $x = 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -2$, h has a relative maximum) is

5-6

eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-2)$ changes from positive to negative”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $h(-3) = 3$

· $h(-2) = 13$

· $h(0) = 0$

· $h(5) = 59$

· $h(5.5) < 59$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -2$, h has a relative maximum because $h' = g$ is positive for $x < -2$ and $h' = g$ is negative for $x > -2$.

· At $x = 0$, h has a relative minimum because $h' = g$ is negative for $x < 0$ and $h' = g$ is positive for $x > 0$.

· At $x = 5$, h has a relative maximum because $h' = g$ is positive for $x < 5$ and $h' = g$ is negative for $x > 5$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

☐ Relative maximum with justification

5-6

- ☐ Relative minimum with justification
- ☐ Relative maximum with justification

Solution:

At $x = -2$, h has a relative maximum because $h'\ (x) = g\ (x)$ changes from positive to negative there.

At $x = 0$, h has a relative minimum because $h'\ (x) = g\ (x)$ changes from negative to positive there.

At $x = 5$, h has a relative maximum because $h'\ (x) = g\ (x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on g, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $h'\ (x)$ and $g\ (x)$ (that is, $h'(x) = g\ (x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of g or h' as $g\ (x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g\ (x)$.

Justification for the first two points should reference one of the following:

· $h'\ (x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $h'\ (x)$

· $h''(x)$ changes sign

· $g'\ (x)$ changes sign

"$h'(x) = g\ (x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to g as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn both points.

A response that discusses h' at a particular point (i.e., $h'\ (-1)$) when the discussion should have been about $h'\ (x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

5-6

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from decreasing to increasing there.

At $x = 3$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 59 is all that is needed to earn the point. No supporting work is required.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$h(5) = \int_0^5 g(t)\,dt = 59$$

Part E

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 10 and ± 13 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 10) \pm (\pm 13)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 10 and ± 13 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-10 - (-13)$ earns both points because numerical expressions need not be simplified.

· The expression $(-10) - 13$ earns the first point, but not the second point.

· The expression $-10 - (-13) - 3$ earns neither point.

· The expression $\int_0^{-3} g\,(t)\ dt = 3$ earns neither point.

· The expression $-\int_{-3}^{-2} g\,(t)\ dt - \int_{-2}^{0} g\,(t)\ dt = 3$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 10 and ± 13, but it does earn the second point.

· $-10, 13$ earns neither point

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\left|\int_{-2}^{-3} g\,(t)\ dt\right| = 10$ and $\left|\int_{-2}^{0} g\,(t)\ dt\right| = 13$
- ☐ Answer

Solution:

$$h\,(-3) = \int_0^{-3} g\,(t)\ dt = -\int_{-3}^{0} g\,(t)\ dt$$

$$= -\int_{-3}^{-2} g\,(t)\ dt - \int_{-2}^{0} g\,(t)\ dt = -10 - (-13) = 3$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-6

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3,\ 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· “Yes, $h\ (x)$ is continuous and differentiable so MVT applies” earns both points.

· “Yes, $h\ (x)$ is continuous and differentiable, therefore there must be a value where $h'\ (p)$ equals the average rate of change of $h\ (x)$ ” earns both points.

· “Yes, $h'\ (x)$ (the derivative of $h\ (x)$) is continuous and differentiable, therefore there must be a value where $h'\ (p)$ equals the average rate of change of $h\ (x)$ ” does not earn the first point, but does the earn second point.

· “Yes because of MVT” does not earn the first point, but does earn the second point.

· “Yes, because $h\ (x)$ is continuous and differentiable” earns the first point, but not the second point.

· “Yes, because $h\ (x)$ is continuous” earns neither point.

· “Yes, because $h\ (x)$ is differentiable” earns neither point.

· “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 2. \Rightarrow h$ is continuous on $-3 \leq x \leq 2$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of p, for $-3 < p < 2$, such that $h'\ (p)$ equals the average rate of change of h over the interval $-3 \leq x \leq 2$.

Part G

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

- $\frac{h(-3)}{-6} = \frac{3}{-6}$
- $\frac{3}{-6}$

Responses that do not earn the point are:

- $\frac{h(-3)}{-6}$
- $\frac{h(-3)}{2 \cdot (-3)}$

Note: an incorrect value for $h(-3)$ can be brought in from part (e) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h' = g \Rightarrow h$ is continuous. $\Rightarrow \lim_{x \to -3} h(x) = h(-3)$

$$\lim_{x \to -3} \frac{h(x)}{2x} = \frac{h(-3)}{2 \cdot (-3)} = \frac{3}{-6} = -\frac{1}{2}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of f that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or g).

5-6

Fundamental Theorem of Calculus is evidenced by linking h' and g either directly or indirectly in the presented derivative of $f(x)$ or in $f'(-3)$. Since $g(-3) = 18$ is given in the question, replacing $h'(-3)$ directly with 18 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(-3)$ that is brought in from part (e) can earn the third point.

$-3 \cdot 18 + 3$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c$ where any of $a,\ b$, or c is incorrect, will earn zero points.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$f'(x) = x \cdot h'(x) + 1 \cdot h(x)$$

$$= x \cdot g(x) + h(x)$$

$$f'(-3) = -3 \cdot g(-3) + h(-3) = -3 \cdot (18) + 3 = -51$$

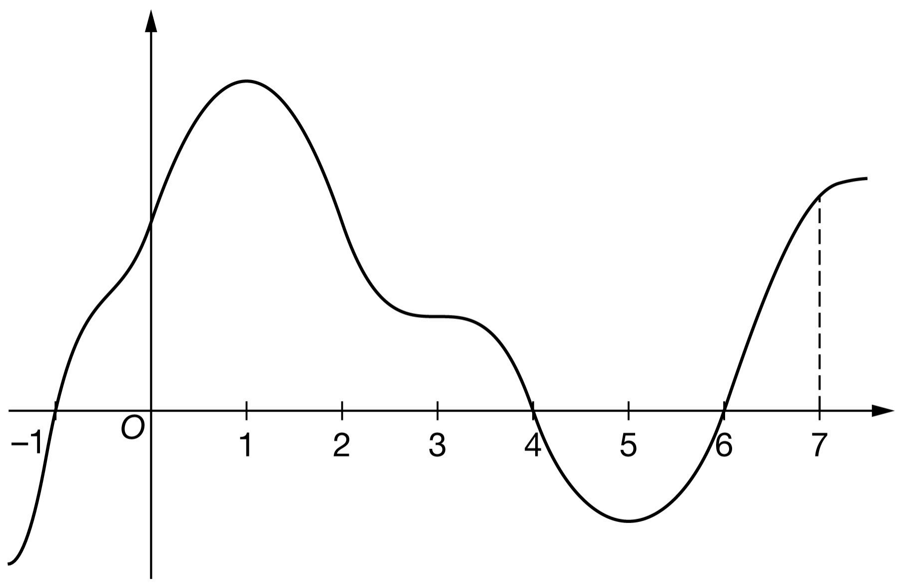
5-6

51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-1.5 \leq x \leq 7.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of f on the intervals $[-1, 4]$, $[4, 6]$, and $[6, 7]$ are 56, 10, and 8, respectively. The graph of f has horizontal tangents at $x = 1$, $x = 3$, and $x = 5$. Let h be the function defined by $h(x) = \int_6^x f(t)\,dt$ for $-1.5 \leq x \leq 7.5$.

(a) Find the x-coordinate of each critical point of h on the interval $-1.5 < x < 7.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of h on the interval $-1.5 < x < 7.5$. Justify your answer.

(d) Find the value of $h(7)$.

(e) Find the value of $h(-1)$. Show the computations that lead to your answer.

(f) Must there exist a value of d, for $-1 < d < 4$, such that $h'(d)$ is equal to the average rate of change of h over the interval $-1 \leq x \leq 4$? Justify your answer.

(g) Find $\lim_{x \to 7} \frac{h(x)}{4x}$. Show the computations that lead to your answer.

5-6

(h) The function q is defined by $q(x) = 3x \cdot h(x)$. It is known that $f(7) = 14$. Find $q'(7)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -1$, $x = 4$, and $x = 6$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1.5, 7.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1.5, 7.5)$. The point is still earned provided the three critical points, $x = -1$, $x = 4$, and$x = 6$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(4, 0)$, $(6, 0)$, read the x-coordinates and disregard the y-coordinates.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h\prime(x) = f(x) = 0$ at $x = -1$, $x = 4$, and $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $f(x)$ (that is, $h'(x) = f(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question. m

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = f(x)$. Any Any additional error (for example, not justifying that at $x = -1$, h has a relative minimum) is

5-6

eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., "$h'\ (-1)$ changes from negative to positive”) when the discussion should have been about $h'\ (x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $h(-1.5) > -46$

· $h(-1) > -46$

· $h(4) = 10$

· $h(6) = 0$

· $h(7) = 8$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -1$, h has a relative minimum because $h' = f$ is negative for $x < -1$ and $h' = f$ is positive for $x > -1$.

· At $x = 4$, h has a relative maximum because $h' = f$ is positive for $x < 4$ and $h' = f$ is negative for $x > 4$.

· At $x = 6$, h has a relative minimum because $h' = f$ is negative for $x < 6$ and $h' = f$ is positive for $x > 6$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

☐ Relative minimum with justification

5-6

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification

Solution:

At $x = -1$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = 4$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

At $x = 6$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on f, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $f(x)$ (that is, $h'(x) = f(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = f(x)$.

Justification for the first two points should reference one of the following:

· $h'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $h'(x) = f(x)$

· h'' changes sign

· $f'(x)$ changes sign

" $h'(x) = f(x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to f as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn both points.

A response that discusses h' at a particular point (i.e., $h'(-1)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

5-6

If $x = 1$ and $x = 5$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] $x = 1$ with justification
- [ ] $x = 5$ with justification
- [ ] Response includes no other points of inflection

Solution:

At $x = 1$, the graph of h has a point of inflection because $h'(x) = f(x)$ changes from increasing to decreasing there.

At $x = 5$, the graph of h has a point of inflection because $h'(x) = f(x)$ changes from decreasing to increasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 8 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$$h(7) = \int_6^7 f(t)\,dt = 8$$

Part E

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 56 and ± 10 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 56) \pm (\pm 10)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 56 and ± 10 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-56-(-10)$ earns both points because numerical expressions need not be simplified.

· The expression $(-56)-10$ earns the first point, but not the second point.

· The expression $-56-(-10)-8$ earns neither point.

· The expression $\int_{6}^{-1} g(t)\,dt = -46$ earns neither point.

· The expression $-\int_{-1}^{4} f(t)\,dt - \int_{4}^{6} f(t)\,dt = -46$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 56 and ± 10, but it does earn the second point.

· $-56,\ 10$ earns neither point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ $\left|\int_{4}^{-1} f(t)\,dt\right| = 56$ and $\left|\int_{4}^{6} f(t)\,dt\right| = 10$
- ☐ Answer

Solution:

$$h(-1) = \int_{6}^{-1} f(t)\,dt = -\int_{-1}^{6} f(t)\,dt$$

$$= -\int_{-1}^{4} f(t)\,dt - \int_{4}^{6} f(t)\,dt = -56-(-10) = -46$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-6

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-1,\ 4)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· “Yes, $h\ (x)$ is continuous and differentiable so MVT applies” earns both points.

· “Yes, $h\ (x)$ is continuous and differentiable, therefore there must be a value where $h'\ (d)$ equals the average rate of change of $h\ (x)$“ earns both points.

· “Yes, $h'\ (x)$ (the derivative of $h\ (x)$) is continuous and differentiable, therefore there must be a value where $h'\ (d)$ equals the average rate of change of $h\ (x)$“ does not earn the first point, but does the earn second point.

· “Yes because of MVT” does not earn the first point but does earn the second point.

· “Yes, because $h\ (x)$ is continuous and differentiable” earns the first point, but not the second point.

· “Yes, because $h\ (x)$ is continuous” earns neither point.

· “Yes, because $h\ (x)$ is differentiable” earns neither point.

· “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = f \Rightarrow h$ is differentiable on $-1 \leq x \leq 4. \Rightarrow h$ is continuous on $-1 \leq x \leq 4$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-1 \leq x \leq 4$ to guarantee that there exists a value of d, for $-1 < d < 4$, such that $h'\ (d)$ equals the average rate of change of h over the interval $-1 \leq x \leq 4$.

Part G

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

- $\frac{h(7)}{28} = \frac{2}{7}$
- $\frac{8}{28}$

Responses that do not earn the point include:

- $\frac{h(7)}{4 \cdot 7}$
- $\frac{h(7)}{28}$

Note: an incorrect value for $h(7)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h' = f \Rightarrow h$ is continuous. $\Rightarrow \lim_{x \to 7} h(x) = h(7)$

$$\lim_{x \to 7} \frac{h(x)}{4x} = \frac{h(7)}{4 \cdot (7)} = \frac{8}{28} = \frac{2}{7}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of q that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or f).

5-6

Fundamental Theorem of Calculus is evidenced by linking h' and f either directly or indirectly in the presented derivative of $q(x)$ or in $q'(7)$. Since $f(7) = 15$ is given in the question, replacing $h'(7)$ directly with 14 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(7)$ that is brought in from part (d) can earn the third point.

$21 \cdot 14 + 3 \cdot 8$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a, b, c, or d is incorrect, will earn zero points.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$q'(x) = 3x \cdot h'(x) + 3 \cdot h(x)$

$= 3x \cdot f(x) + 3 \cdot h(x)$

$q'(7) = 3 \cdot 7 \cdot f(7) + 3h(7) = 21 \cdot (14) + 3 \cdot (8) = 318$

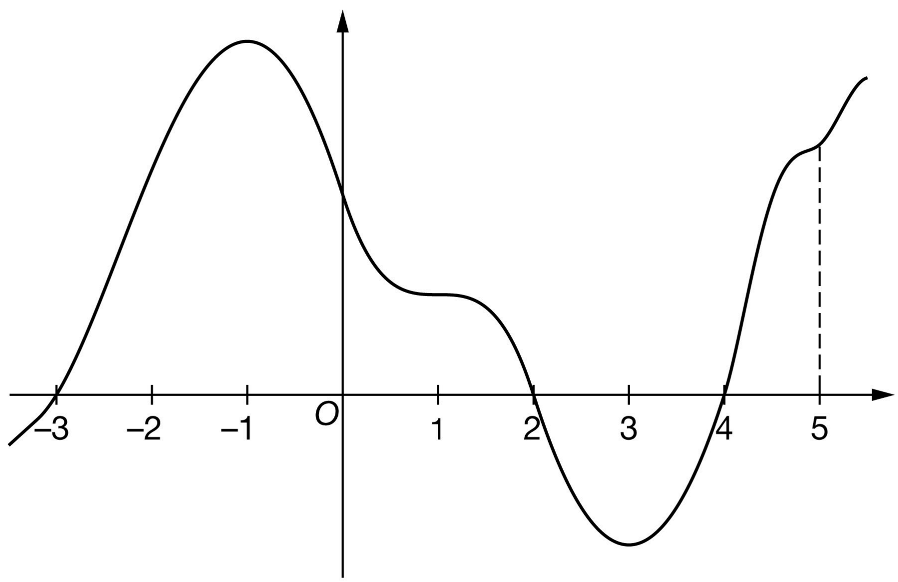
5-6

52. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the differentiable function g with domain $-3.5 \le x \le 5.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of g on the intervals $[-3, 2]$, $[2, 4]$, and $[4, 5]$ are 55, 12, and 10, respectively. The graph of g has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let h be the function defined by $h(x) = \int_4^x g(t)\,dt$ for $-3.5 \le x \le 5.5$.

(a) Find the x-coordinate of each critical point of h on the interval $-3.5 < x < 5.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of h on the interval $-3.5 < x < 5.5$. Justify your answer.

(d) Find the value of $h(5)$.

(e) Find the value of $h(-3)$. Show the computations that lead to your answer.

(f) Must there exist a value of r, for $-3 < r < 2$, such that $h'(r)$ is equal to the average rate of change of h over the interval $-3 \le x \le 2$? Justify your answer.

(g) Find $\lim_{x \to 5} \frac{h(x)}{2x}$. Show the computations that lead to your answer.

5-6

(h) The function k is defined by $k(x) = x \cdot h(x)$. It is known that $g(5) = 15$. Find $k'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5, 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5,\ 5.5)$. The point is still earned provided the three critical points $x = -3$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g.,$(-3, 0)$, $(2, 0)$, $(4, 0)$, read the x -coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative minimum) is

5-6

eligible for at most one of the three points.

Read references to "the graph" as references to $g(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., "$h'(3)$ changes from negative to positive") when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $h(-3.5) > -43$

· $h(-3) = -43$

· $h(2) = 12$

· $h(4) = 0$

· $h(5) = 10$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative minimum because $h' = g$ is negative for $x < -3$ and $h' = g$ is positive for $x > -3$.

· At $x = 2$, h has a relative maximum because $h' = g$ h prime equals g is positive for $x < 2$ and $h' = g$ is negative for $x > 2$.

· At $x = 4$, h has a relative minimum because $h' = g$ is negative for $x < 4$ and $h' = g$ is positive for $x > 4$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

☐ Relative minimum with justification

5-6

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 2$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = 4$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on g, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$.

Justification for the first two points should reference one of the following:

· $h'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $h'(x) = g(x)$

· $h''(x)$ changes sign

· $g'(x)$ changes sign

"$h'(x) = g(x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to g as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn both points.

A response that discusses h' at a particular point (i.e., $h'(-3)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

5-6

If $x=-1$ and $x=3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ $x=-1$ with justification
- ☐ $x=3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x=-1$, the graph of h has a point of inflection because $h'(x)=g(x)$ changes from increasing to decreasing there.

At $x=3$, the graph of h has a point of inflection because $h'(x)=g(x)$ changes from decreasing to increasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 10 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$h(5)=\int_4^5 g(t)\,dt=10$$

Part E

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 55 and ± 12 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 55) \pm (\pm 12)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 55 and ± 12 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-55-(-12)$ earns both points because numerical expressions need not be simplified.

· The expression $(-55)-12$ earns the first point, but not the second point.

· The expression $-55-(-12)-8$ earns neither point.

· The expression $\int_{4}^{-3} g(t)\, dt = -43$ earns neither point.

· The expression $-\int_{-3}^{2} g(t)\, dt - \int_{2}^{4} g(t)\, dt = -43$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 55 and ± 12, but it does earn the second point.

· $-55, 12$ earns neither point

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_{2}^{-3} g(t)\; dt\right| = 55$ and $\left|\int_{2}^{4} g(t)\; dt\right| = 12$
- [ ] Answer

Solution:

$$h(-3) = \int_{4}^{-3} g(t)\; dt = -\int_{-3}^{4} g(t)\; dt$$
$$= -\int_{-3}^{2} g(t)\; dt - \int_{2}^{4} g(t)\; dt = -55-(-12) = -43$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-6

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3, 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.

· “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ ” earns both points.

· “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does the earn second point.

· “Yes because of MVT” does not earn the first point, but does earn the second point.

· “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.

· “Yes, because $h(x)$ is continuous” earns neither point.

· “Yes, because $h(x)$ is differentiable” earns neither point.

· “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 2$. $\Rightarrow h$ is continuous on $-3 \leq x \leq 2$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of r, for $-3 < r < 2$, such that $h'(r)$ equals the average rate of change of h over the interval $-3 \leq x \leq 2$.

Part G

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

- $\dfrac{h(5)}{10} = 1$
- $\dfrac{10}{10}$

Responses that do not earn the point include:

- $\dfrac{h(5)}{10}$
- 1

Note: an incorrect value for $h(5)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h' = g \Rightarrow h$, is continuous. $\Rightarrow \lim\limits_{x \to 5} h(x) = h(5)$

$$\lim_{x \to 5} \frac{h(x)}{2x} = \frac{h(5)}{2 \cdot 5} = \frac{10}{10} = 1$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of k that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or g).

5-6

Fundamental Theorem of Calculus is evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(5)$. Since $g(5) = 15$ is given in the question, replacing $h'(5)$ directly with 15 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(5)$ that is brought in from part (d) can earn the third point.

$5 \cdot 15 + 10$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c$ where any of a, b, or c is incorrect, will earn zero points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$k'(x) = x \cdot h'(x) + 1 \cdot h(x)$

$= x \cdot g(x) + h(x)$

$k'(5) = 5g(5) + h(5) = 5 \cdot 15 + 10 = 85$

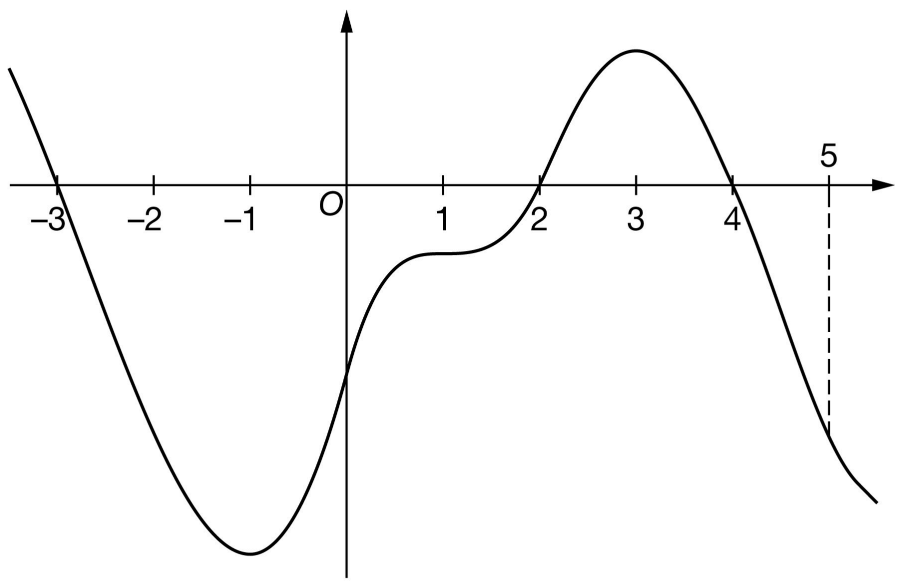
5-6

53. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h

The graph of the differentiable function h with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of h on the intervals $[-3, 2]$, $[2, 4]$, and $[4, 5]$ are 58, 11, and 8, respectively. The graph of h has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let g be the function defined by $g(x) = \int_4^x h(t)\,dt$ for $-3.5 \leq x \leq 5.5$.

(a) Find the x-coordinate of each critical point of g on the interval $-3.5 < x < 5.5$.

(b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(c) Find the x-coordinate of each point of inflection for the graph of g on the interval $-3.5 < x < 5.5$. Justify your answer.

(d) Find the value of $g(5)$.

(e) Find the value of $g(-3)$. Show the computations that lead to your answer.

(f) Must there exist a value of b, for $-3 < b < 2$, such that $g'(b)$ is equal to the average rate of change of g over the interval $-3 \leq x \leq 2$? Justify your answer.

(g) Find $\lim_{x \to 5} \frac{g(x)}{3x}$. Show the computations that lead to your answer.

5-6

(h) The function a is defined by $a(x) = x^2 \cdot g(x)$. It is known that $h(5) = -16$. Find $a'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3, \quad x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5, \quad 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5, \quad 5.5)$. The point is still earned provided the three critical points, $x = -3, \quad x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, \quad 0), \quad (2, \quad 0), \quad (4, \quad 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g'(x) = h(x) = 0$ at $x = -3$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on h, then:

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $h(x)$ (that is, $g'(x) = h(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = h(x)$ Any additional error (for example, not justifying that at $x = 2$, h has a relative minimum) is eligible

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for at most one of the three points.

Read references to “the graph” as references to $h\ (x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'\ (-3)$ changes from negative to positive”) when the discussion should have been about $g'\ (x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g\ (x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

· $g\ (-3.5) < 47$

· $g\ (-3) = 47$

· $g\ (2) = -11$

· $g\ (4) = 0$

· $g\ (5) = -8$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, g has a relative maximum because $g' = h$ is positive for $x < -3$ and $g' = h$ is negative for $x > -3$.

· At $x = 2$, g has a relative minimum because $g' = h$ is negative for $x < 2$ and $g' = h$ is positive for $x > 2$.

· At $x = 4$, g has a relative maximum because $g' = h$ is positive for $x < 4$ and $g' = h$ is negative for $x > 4$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Relative maximum with justification

Relative minimum with justification

5-6

- [ ]
- [ ] Relative maximum with justification

Solution:

At $x=-3$, g has a relative maximum because $g'(x)=h(x)$ changes from positive to negative there.

At $x=2$, g has a relative minimum because $g'(x)=h(x)$ changes from negative to positive there.

At $x=4$, g has a relative maximum because $g'(x)=h(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only g is eligible for all three points.

If a response gives justifications based only on h, then (in parallel with the approach followed in part (b)):

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $h(x)$ (that is, $g'(x)=h(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of h or g' as $h(x)=g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x)=h(x)$.

Justification for the first two points should reference one of the following:

· $g'(x)$ changing from increasing to decreasing, or vice versa

· relative extrema of $g'(x)$

· $g''(x)$ changes sign

· $h'(x)$ changes sign

" $g'(x)=h(x)$ changes direction" earns neither of the first two points.

Read references to "the graph" as references to h as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn both points.

A response that discusses g' at a particular point (i.e., $g'(-3)$) when the discussion should have been about $g'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

5-6

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] $x = -1$ with justification
- [ ] $x = 3$ with justification
- [ ] Response includes no other points of inflection

Solution:

At $x = -1$, the graph of g has a point of inflection because $g'(x) = h(x)$ changes from decreasing to increasing there.

At $x = 3$, the graph of g has a point of inflection because $g'(x) = h(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -8 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$$g(5) = \int_4^5 h(t)\,dt = -8$$

Part E

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 58 and ± 11 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 58) \pm (\pm 11)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values. The ± 58 and ± 11 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-58) - 11$ earns both points because numerical expressions need not be simplified.

· The expression $(-58) - 11$ earns the first point, but not the second point.

· The expression $-58 - 11 - 8$ earns neither point.

· The expression $\int_4^{-3} h(t)\, dt = 47$ earns neither point.

· The expression $-\int_{-3}^{2} h(t)\, dt - \int_2^4 h(t)\, dt = 47$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 58 and ± 11, but it does earn the second point.

· $-58, \quad 11$ earns neither point

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\left| \int_{-3}^{2} h(t)\, dt \right| = 58$ and $\left| \int_4^2 h(t) dt \right| = 11$
- ☐ Answer

Solution:

$$g(-3) = \int_4^{-3} h(t)\, dt = -\int_{-3}^{4} h(t)\, dt$$
$$= -\int_{-3}^{2} h(t)\, dt - \int_2^4 h(t)\, dt = -(-58) - 11 = 47$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-6

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., "the function is continuous and differentiable") does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3,\ 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

· "Yes, $g\ (x)$ is continuous and differentiable so MVT applies" earns both points.

· "Yes, $g\ (x)$ is continuous and differentiable, therefore there must be a value where $g'\ (b)$ equals the average rate of change of $g\ (x)$" earns both points.

· "Yes, $g'\ (x)$ (the derivative of $g\ (x)$) is continuous and differentiable, therefore there must be a value where $g'\ (b)$ equals the average rate of change of $g\ (x)$ " does not earn the first point, but does the earn second point.

· "Yes because of MVT" does not earn the first point, but does earn the second point.

· "Yes, because $g\ (x)$ is continuous and differentiable" earns the first point, but not the second point.

· "Yes, because $g\ (x)$ is continuous" earns neither point.

· "Yes, because $g\ (x)$ is differentiable" earns neither point.

· "Yes" earns neither point.

A response of "No" earns neither point, regardless of any other explanation including any reference to MVT.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = h \Rightarrow g$ is differentiable on $-3 \leq x \leq 2. \Rightarrow g$ is continuous on $-3 \leq x \leq 2.$

Therefore, the Mean Value Theorem can be applied to g on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of b, for $-3 < b < 2$, such that $g'\ (b)$ equals the average rate of change of g over the interval $-3 \leq x \leq 2.$

Part G

5-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

- $\frac{g(5)}{15} = \frac{-8}{15}$
- $\frac{-8}{15}$

Responses that do not earn the point are:

- $\frac{g(5)}{15}$
- $\frac{g(5)}{3 \cdot 5}$

Note: an incorrect value for $g(5)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that g is continuous. If the response does state that g is continuous, an explanation is not required.

The response does not need to use limit notation.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g' = h \Rightarrow g$ is continuous. $\Rightarrow \lim\limits_{x \to 5} g(x) = g(5)$

$$\lim_{x \to 5} \frac{g(x)}{3x} = \frac{g(5)}{3 \cdot (5)} = \frac{-8}{15}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of a that contains a sum of two terms, with at least one term written as a product, and one term involving g and the other term involving g' (or h).

5-6

Fundamental Theorem of Calculus is evidenced by linking g' and h either directly or indirectly in the presented derivative of $g(x)$ or in $g'(5)$. Since $h(5) = -16$ is given in the question, replacing $g'(5)$ directly with -16 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $g(5)$ that is brought in from part (d) can earn the third point.

$25 \cdot (-16) + 10 \cdot (-8)$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a, b, c, or d is incorrect, will earn zero points.

✓

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$a'(x) = x^2 \cdot g'(x) + 2x \cdot g(x)$$

$$= x^2 \cdot h(x) + 2x \cdot g(x)$$

$$a'(5) = 25 \cdot h(5) + 2 \cdot 5 \cdot g(5) = 25 \cdot (-16) + 10 \cdot (-8) = -480$$

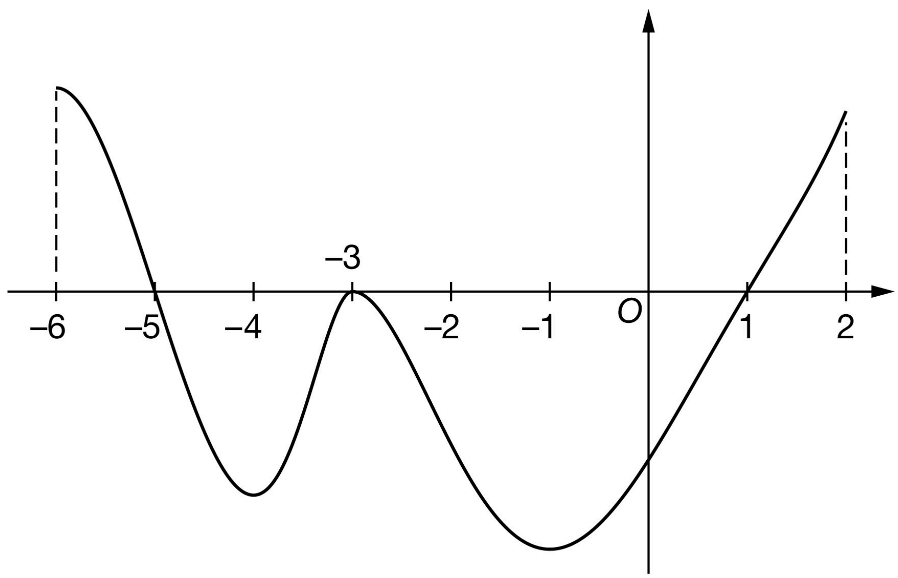
5-6

54. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the differentiable function g with domain $-6 \leq x \leq 2$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of g on the intervals $[-6,-5]$, $[-5,-3]$, $[-3,1]$, and $[1,2]$ are 9, 17, 42, and 6, respectively. The graph of g has horizontal tangents at $x=-4$, $x=-3$, and $x=-1$. Let h be the function defined by $h(x)=\int_{-3}^{x} g(t)\,dt$ for $-6 \leq x \leq 2$.

(a) Find the value of $h(1)$.

(b) Find the value of $h(-6)$. Show the computations that lead to your answer.

(c) Find the x-coordinate of each critical point of h on the interval $-6 \leq x \leq 2$.

(d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(e) For $-6 \leq x \leq 2$, find the x-coordinate of each point of inflection of the graph of h. Justify your answers.

(f) Explain why there must be a value of k, for $-3<k<1$, such that $h(k)=-33$.

(g) Find $\lim\limits_{x \to 1} \frac{3x^2-3}{h(x)+42x}$. Give a reason for your answer.

(h) The function p is defined by $p(x)=h(2x)$. It is known that $g(-2)=-10.8$. Find $p'(-1)$. Show the computations that lead to your answer.

5-6

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -42 is all that is needed to earn the point. No supporting work is required.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h\left(1\right)=\int_{-3}^{1}g\left(t\right)\,dt=-42$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ±9 and ±17 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ±9 and ±17 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-9-\left(-17\right)$ earns both points because numerical expressions need not be simplified.

· The expression $-9-17$ earns the first point, but not the second point.

· The expression $-9-\left(-17\right)+42$ earns neither point.

Note: Just the list $-9,\quad-17$ earns neither point. However, if the values -9 and -17 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals equal to 8 earns just the second point because not enough work has been shown.

5-6

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\left|\int_{-6}^{-5} f(t)\,dt\right| = 9$ and $\left|\int_{-5}^{-3} f(t)\,dt\right| = 17$
- ☐ Answer

Solution:

$$h(-6) = \int_{-3}^{-6} g(t)\,dt = -\int_{-6}^{-3} g(t)\,dt$$

$$= -\int_{-6}^{-5} g(t)\,dt - \int_{-5}^{-3} g(t)\,dt = -9 - (-17) = 8$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -5$, $x = -3$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-6,\ 2)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-6,\ 2)$. The point is still earned provided the three critical points, $x = -5$, $x = -3$, and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-5,\ 0)$, $(-3,\ 0)$, and $(1,\ 0)$, read the x -coordinates, and disregard the y -coordinates.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -5$, $x = -3$, and $x = 1$.

5-6

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on $h\prime$ is eligible for all three points.

If a response gives justifications based only on g , then:

· the response is eligible for all three points, if the response presents the connection between $g\left(x\right)$ and $h'\left(x\right)$ (that is, $g\left(x\right)=h'\left(x\right)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of g or h' as $g\left(x\right)=h'\left(x\right)$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $g\left(x\right)=h'\left(x\right)$. Any additional error (for example, not justifying that at $x=-5,\ h$ has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g\left(x\right)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'\left(-5\right)$ changes from positive to negative”) when the discussion should have been about $h'\left(x\right)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h\left(x\right)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x=-5,\ x=-3,\ x=1$) must be included in the table or list. The correct values of relevant inputs are:

· $h\left(-6\right)=8$

· $h\left(-5\right)=17$

· $h\left(-3\right)=0$

· $h\left(1\right)=-42$

· $h\left(2\right)=-36$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that

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presents all three of the following statements in part (d) earns two points:

· At $x = -5$, h has a relative maximum because $h' = g$ is positive for $x < -5$ and $h' = g$ is negative for $x > -5$.

· At $x = -3$, h has neither because $h\prime = g$ is negative for $x < -3$ and $h' = g$ is negative for $x > -3$.

· At $x = 1$, h has a relative minimum because $h' = g$ is negative for $x < 1$ and $h' = g$ is positive for $x > 1$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Relative maximum with justification
- [ ] Neither with justification
- [ ] Relative minimum with justification

Solution:

At $x = -5$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = -3$, h has neither a relative minimum nor a relative maximum, because $h'(x) = g(x)$ does not change sign there.

At $x = 1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-4, \quad -3, \quad -1$, and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $g(x)$ is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

· h' changes from increasing to decreasing or vice versa

· h'' changes from positive to negative or vice versa

· h' has a local extrema

5-6

Any incorrect statement (such as h' changes from increasing to decreasing at $x = -4$) fails to earn the second point.

If the response addresses at least one correct value of x, it is eligible to earn the second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = g$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = -4$, $x = -3$, and $x = -1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that h is differentiable $\Rightarrow h$ is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $-3 < x < 1$, or the point is not earned.

A response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, in the presence of the required function evaluation, the statement "since h is continuous, and $-42 < -33 < 0$, such a point exists" would earn the point.

The value of -33 must be bounded by -42 and 0 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(-3)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. We need to see A$h(-3) = 0$.

A response may reference a value of $h(1)$ determined in part (a). If the value of $h(1)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is less than -33.

0	1 ✓

5-6

The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable on $-3 \leq x \leq 1$. $\Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

$h\,(1) = -42 < -33 < 0 = h\,(-3)$

By the Intermediate Value Theorem, there must be a value of k, for $-3 < k < 1$, such that $h\,(k) = -33$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3 \cdot 1^2 - 3 = 0$ and $h\,(1) + 42 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3 \cdot 1^2 = 3$ and $h\,(1) = -42$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim\limits_{x \to 1} \frac{3x^2 - 3}{h(x) + 42x} = \frac{0}{0}$ does not earn the first point. The statement $\lim\limits_{x \to 1} \frac{3x^2 - 3}{h(x) + 42x} \to \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim\limits_{x \to 1} \frac{3x^2 - 3}{h(x) + 42x} = \frac{6x}{h'(x) + 42}$, will earn the second point, but not the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim\limits_{x \to 1} \frac{3x^2 - 3}{h(x) + 42x} = \frac{0}{0}$ and $\lim\limits_{x \to 1} \frac{3x^2 - 3}{h(x) + 42x} = \frac{6x}{h'(x) + 42}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

5-6

The third point is earned only for the answer of $\frac{1}{7}$, or its numeric equivalent.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.
$\Rightarrow h$ is continuous. $\Rightarrow \lim\limits_{x\to 1} h\,(x) = h\,(1)$ and $\lim\limits_{x\to 1} h'\,(x) = h'\,(1)$.

$$\lim_{x\to 1}\left(3x^2-3\right) = 3\cdot 1^2 - 3 = 0$$

$$\lim_{x\to 1}\left(h\,(x)+42x\right) = h\,(1) + 42\cdot 1 = -42+42 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x\to 1}\frac{3x^2-3}{h(x)+42x} = \lim_{x\to 1}\frac{6x}{h'(x)+42}$$

$$=\frac{6\cdot 1}{h'(1)+42} = \frac{6}{g(1)+42} = \frac{6}{0+42} = \frac{1}{7}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x=-1$ is substituted for x, then $x=-1$ must appear unsimplified to earn the first point. Responses that earn the point include:

- $h'\,(2x)\cdot 2$
- $g\,(2x)\cdot 2$
- $h'\,(2\cdot(-1))\cdot 2$
- $g\,(2\cdot(-1))\cdot 2$

Responses that do not earn the point include:

- $h'\,(x)\cdot 2$

5-6

$\cdot\, g(x) \cdot 2$

$\cdot\, h'(-2) \cdot 2$

$\cdot\, g(-2) \cdot 2$

$\cdot\, h(2x) \cdot 2$

The second point is earned for the application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $p(x)$ or in $p'(-1)$. For example, the second point is demonstrated by $g(2x)$ or by $g(2 \cdot 2)$.

The second point may be earned without earning the first point. For example, $h'(2x) = g(2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(-2)$ to -10.8. The function g does not have to be stated explicitly. For example, $h'(2 \cdot (-1)) \cdot 2 = -10.8 \cdot 2$ earns the first two points. Recall, if $x = -1$ is substituted for x, then $x = -1$ must appear unsimplified to earn the first point.

The first two points can be earned by a single expression. For example, both $g(2x) \cdot 2$ and $g(2 \cdot (-1)) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -21.6, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$p'(x) = h'(2x) \cdot 2$

$= g(2x) \cdot 2$

$p'(-1) = g(2 \cdot (-1)) \cdot 2 = g(-2) \cdot 2 = (-10.8) \cdot 2 = -21.6$

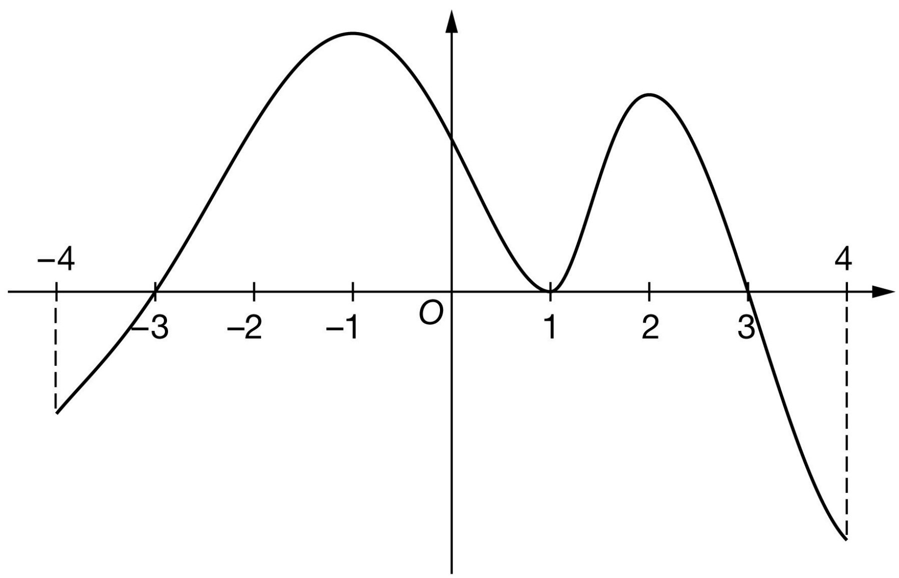
5-6

55. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-4 \le x \le 4$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of f on the intervals $[-4, -3]$, $[-3, 1]$, $[1, 3]$, and $[3, 4]$ are 5, 46, 18, and 11, respectively. The graph of f has horizontal tangents at $x = -1$, $x = 1$, and $x = 2$. Let h be the function defined by $h(x) = \int_1^x f(t)\, dt$ for $-4 \le x \le 4$.

(a) Find the value of $h(3)$.

(b) Find the value of $h(-4)$. Show the computations that lead to your answer.

(c) Find the x-coordinate of each critical point of h on the interval $-4 < x < 4$.

(d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(e) For $-4 < x < 4$, find the x-coordinate of each point of inflection of the graph of h. Justify your answers.

(f) Explain why there must be a value of c, for $1 < c < 3$, such that $h(c) = 13$.

(g) Find $\lim\limits_{x\to 3} \frac{x^2-9}{h(x)-6x}$. Give a reason for your answer.

(h) The function m is defined by $m(x) = h\left(x^2\right)$. It is known that $f(4) = -19.3$. Find $m'(2)$. Show the computations that lead to your answer.

5-6

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 18 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$$h(3) = \int_1^3 f(t)\, dt = 18$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 5 and ± 46 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 5 and ± 46 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-5) - 46$ earns both points because numerical expressions need not be simplified.

· The expression $-5 - 46$ earns the first point, but not the second point.

· The expression $-(-5) - 46 + 8$ earns neither point.

Note: Just the list $-5,\ 46$ earns neither point. However, if the values -5 and 46 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals -41 earns just the second point because not enough work has been shown.

5-6

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_{-4}^{-3} f(t)\, dt\right| = 5$ and $\left|\int_{-3}^{1} f(t)\, dt\right| = 46$
- [ ] Answer

Solution:

$$h(-4) = \int_{1}^{-4} f(t)\, dt = -\int_{-4}^{1} f(t)\, dt$$

$$= -\int_{-4}^{-3} f(t)\, dt - \int_{-3}^{1} f(t)\, dt = -(-5) - 46 = -41$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(1, 0)$, and $(3, 0)$, read the x-coordinates, and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = f(x) = 0$ at $x = -3$, $x = 1$, and $x = 3$.

5-6

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points, if the response presents the connection between $f(x)$ and $h'(x)$ (that is, $f(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $f(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $f(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., "$h'(-3)$ changes from positive to negative") when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = 1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

· $h(-4) = -41$

· $h(-3) = -46$

· $h(1) = 0$

· $h(3) = 18$

· $h(4) = 7$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that

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presents all three of the following statements in part (d) earns two points:

· At $x = -3$, h has a relative minimum because $h' = f$ is negative for $x < -3$ and $h' = f$ is positive for $x > -3$.

· At $x = 1$, h has neither because $h' = f$ is positive for $x < 1$ and $h' = f$ is positive for $x > 1$.

· At $x = -3$, h has a relative maximum because $h' = f$ is positive for $x < 3$ and $h' = f$ is negative for $x > 3$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = 1$, h has neither a relative minimum nor a relative maximum because $h'(x) = f(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-1, 1, 2$ and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $f(x)$ is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

· h' changes from increasing to decreasing or vice versa

· h'' changes from positive to negative or vice versa

· h' has a local extrema

5-6

Any incorrect statement (such as h' changes from decreasing to increasing at $x = -1$) fails to earn the second point.

If the response addresses at least one correct value of x it is eligible to earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Location of inflection points
- [ ] Justification

Solution:

The points of inflection for the graph of h occur where $h' = f$ changes from increasing to decreasing or vice versa. Therefore, the points of inflection occur at $x = -1$, $x = 1$, and $x = 2$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that h is differentiable $\Rightarrow$ that h is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, in the presence of the required function evaluation, the statement "since h is continuous, and $0 < 13 < 18$, such a point exists" would earn the point.

The value of 13 must be bounded by 0 and 18 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $h(1) = 0$ must be present.

A response may reference a value of $h(3)$ determined in part (a). If the value of $h(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 13.

0	1

5-6

The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

$h' = f$ is differentiable on $1 \leq x \leq 3$. $\Rightarrow h$ is continuous on $1 \leq x \leq 3$.

$h(1) = 0 < 13 < 18 = h(3)$

By the Intermediate Value Theorem, there must be a value of c, for $1 < c < 3$, such that $h(c) = 13$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3^2 - 9 = 0$ and $h(3) - 18 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $h(3) = 18$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{0}{0}$ does not earn the first point. The statement $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 6x} \to \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{2x}{h'(x) - 6}$, will earn the second point, but not the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{0}{0}$ and $\lim\limits_{x \to 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{2x}{h'(x) - 6}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

5-6

The third point is earned only for the answer of -1, or its numeric equivalent.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Limits of numerator and denominator equal 0
- [ ] Application of L'Hospital's Rule
- [ ] Answer

$h' = f$ is differentiable. $\Rightarrow h' = f$ is continuous.
$\Rightarrow h$ is continuous. $\Rightarrow \lim\limits_{x \to 3} h(x) = h(3)$ and $\lim\limits_{x \to 3} h'(x) = h'(3)$.

$$\lim_{x \to 3} \left(x^2 - 9\right) = 3^2 - 9 = 0$$

$$\lim_{x \to 3} \left(h(x) - 6x\right) = h(3) - 6 \cdot 3 = 18 - 18 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to 3} \frac{x^2 - 9}{h(x) - 6x} = \lim_{x \to 3} \frac{2x}{h'(x) - 6}$$

$$= \frac{2 \cdot 3}{h'(3) - 6} = \frac{6}{f(3) - 6} = \frac{6}{0 - 6} = -1$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 2$ is substituted for x, then $x = 2$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$\cdot\ h'(x^2) \cdot 2x$

$\cdot\ f(x^2) \cdot 2x$

$\cdot\ h'(2^2) \cdot 2 \cdot 2$

$\cdot\ f(2^2) \cdot 2 \cdot 2$

$\cdot\ h'(4) \cdot 2 \cdot 2$

$\cdot\ f(4) \cdot 2 \cdot 2$

$\cdot\ h'(2^2) \cdot 4$

5-6

· $f(2^2) \cdot 4$

Responses that do not earn the point include:

· $h'(x^2) \cdot 2$

· $f(x^2) \cdot 2$

· $h'(x) \cdot 2x$

· $f(x) \cdot 2x$

· $h'(4) \cdot 4$

· $f(4) \cdot 4$

· $h(x^2) \cdot 2x$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $m(x)$ or in $m'(2)$. For example, the second point is earned by $f(x^2)$ or by $f(2^2)$.

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = f(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and f are linked.

The second point may be earned by going directly from $h'(1)$ to -19.3. The function f does not have to be stated explicitly. Responses that earn the first two points include:

· $h'(2^2) \cdot 2 \cdot 2 = -19.3 \cdot 2 \cdot 2$

· $h'(2^2) \cdot 4 = -19.3 \cdot 4$

Recall, if $x = 2$ is substituted for x, then $x = 2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^2) \cdot 2x$ and $f(2^2) \cdot 2 \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -77.2, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $h'(x^2) \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

5-6

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$m'(x) = h'(x^2) \cdot 2x = f(x^2) \cdot 2x$

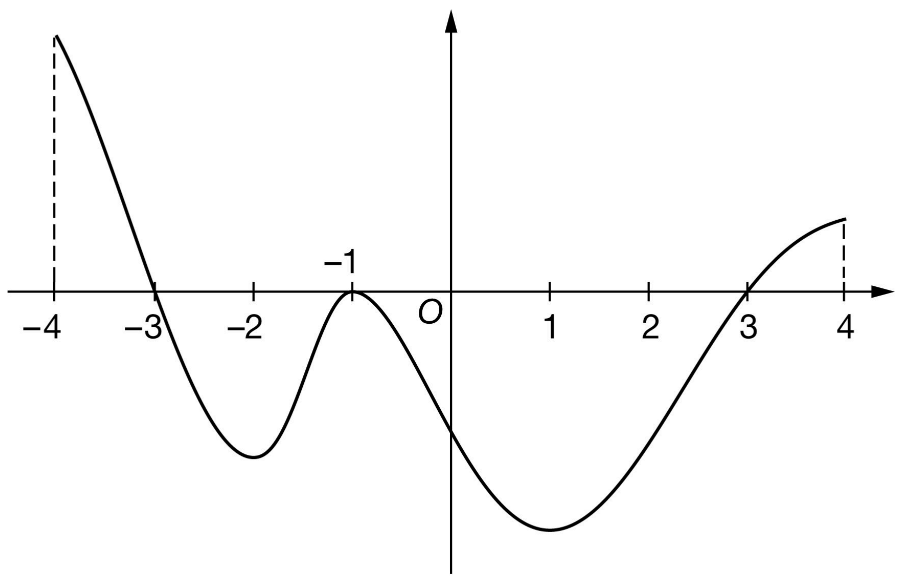
$m'(2) = f(2^2) \cdot 2 \cdot 2 = f(4) \cdot 4 = -19.3 \cdot 4 = -77.2$

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the differentiable function f with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of f on the intervals $[-4, -3]$, $[-3, -1]$, $[-1, 3]$, and $[3, 4]$ are 12, 17, 48, and 4, respectively. The graph of f has horizontal tangents at $x = -2$, $x = -1$, and $x = 1$. Let g be the function defined by $g(x) = \int_{-1}^{x} f(t)dt$ for $-4 \leq x \leq 4$.

(a) Find the value of $g(3)$.

(b) Find the value of $g(-4)$. Show the computations that lead to your answer.

(c) Find the x-coordinate of each critical point of g on the interval $-4 < x < 4$.

(d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(e) For $-4 < x < 4$, find the x-coordinate of each point of inflection of the graph of g. Justify your answers.

(f) Explain why there must be a value of d, for $-1 < d < 3$, such that $g(d) = -25$.

(g) Find $\lim\limits_{x \to 3} \frac{x^2-9}{g(x)+16x}$. Give a reason for your answer.

(h) The function a is defined by $a(x) = g(x^3)$. It is known that $f(1) = -21.1$. Find $a'(1)$. Show the computations that lead to your answer.

5-6

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -48 is all that is needed to earn the point. No supporting work is required.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$$g(3) = \int_{-1}^{3} f(t)\ dt = -48$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 12 and ± 17 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 12 and ± 17 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-12-(-17)$ earns both points because numerical expressions need not be simplified.

· The expression $-12-(-17)$ earns the first point, but not the second point.

· The expression $-12-(-17)+48$ earns neither point.

Note: Just the list $-12,\ -17$ earns neither point. However, if the values -12 and -17 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals 5 earns just the second point because not enough work has been shown.

5-6

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_{-3}^{-4} f(t)\,dt\right| = 12$ and $\left|\int_{-3}^{-1} f(t)\,dt\right| = 17$
- [ ] Answer

Solution:

$$g(-4) = \int_{-1}^{-4} f(t)\,dt = -\int_{-4}^{-1} f(t)\,dt$$

$$= -\int_{-4}^{-3} f(t)\,dt - \int_{-3}^{-1} f(t)\,dt = -12 - (-17) = 5$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = -1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4,\ 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4,\ 4)$. The point is still earned provided the three critical points, $x = -3$, $x = -1$, and $x = 3$, are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3,\ 0)$, $(-1,\ 0)$, and $(3,\ 0)$ read the x -coordinates, and disregard the y -coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -3$, $x = -1$, and $x = 3$.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f, then:

· the response is eligible for all three points, if the response presents the connection between $f(x)$ and $g'(x)$ (that is, $f(x) = g'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $f(x) = g'(x)$. Any additional error (for example, not justifying that at $x = -3$, g has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $f(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., "$g'(-3)$ changes from positive to negative") when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = -1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

· $g(-4) = 5$

· $g(-3) = 17$

· $g(-1) = 0$

· $g(3) = -48$

· $g(4) = -44$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that

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presents all three of the following statements in part (d) earns two points:

· At $x = -3$, g has a relative maximum because $g' = f$ is positive for $x < -3$ and $g' = f$ is negative for $x > -3$.

· At $x = -1$, g has neither because $g' = f$ is negative for $x < -1$ and $g' = f$ is negative for $x > -1$.

· At $x = 3$, g has a relative minimum because $g' = f$ is negative for $x < 3$ and $g' = f$ is positive for $x > 3$.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = -1$, g has neither a relative minimum nor a relative maximum because $g'(x) = f(x)$ does not change sign there.

At $x = 3$, g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-2, \ -1, \ 1$, and only this list, earns the first point.

A response referring only to $g'(x)$ is eligible to earn the second point.

A response referring only to $f(x)$ is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $g'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

· g' changes from increasing to decreasing or vice versa

· g'' changes from positive to negative or vice versa

· g' has a local extrema

5-6

Any incorrect statement (such as g' changes from increasing to decreasing at $x = -2$) fails to earn the second point.

If the response addresses at least one correct value of x, it is eligible to earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of g occur where $g' = f$ changes from increasing to decreasing or vice versa. Therefore, the points of inflection occur at $x = -2$, $x = -1$, and $x = 1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that g is continuous in order to be eligible to earn the point. The response is not required to state that g is differentiable $\Rightarrow g$ is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $-1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, in the presence of the required function evaluation, the statement "since g is continuous, and $-48 < -25 < 0$, such a point exists" would earn the point.

The value of -25 must be bounded by -48 and 0 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $g(-1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $g(-1) = 0$ must be present.

A response may reference a value of $g(3)$ determined in part (a). If the value of $g(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is less than -25.

0	1

5-6

The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

$g' = f$ is differentiable on $-1 \le x \le 3$. $\Rightarrow$ g is continuous on $-1 \le x \le 3$.

$g(3) = -48 < -25 < 0 = g(-1)$

By the Intermediate Value Theorem, there must be a value of d, for $-1 < d < 3$, such that $g(d) = -25$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that g and g' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3^2 - 9 = 0$ and $g(3) + 48 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $g(3) = -48$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim\limits_{x\to 3}\frac{x^2-9}{g(x)+16x} = \frac{0}{0}$ does not earn the first point. The statement $\lim\limits_{x\to 3}\frac{x^2-9}{g(x)+16x} \to \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim\limits_{x\to 3}\frac{x^2-9}{g(x)+16x} = \frac{2x}{g'(x)+16}$, will earn the second point, but not the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim\limits_{x\to 3}\frac{x^2-9}{g(x)+16x} = \frac{0}{0}$ and $\lim\limits_{x\to 3}\frac{x^2-9}{g(x)+16x} = \frac{2x}{g'(x)+16}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

5-6

The third point is earned only for the answer of $\frac{3}{8}$, or its numeric equivalent.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$g' = f$ is differentiable. $\Rightarrow g' = f$ is continuous.
$\Rightarrow g$ is continuous. $\Rightarrow \lim\limits_{x \to 3} g(x) = g(3)$ and $\lim\limits_{x \to 3} g'(x) = g'(3)$.

$$\lim_{x \to 3} \left(x^2 - 9\right) = 3^2 - 9 = 0$$

$$\lim_{x \to 3} \left(g(x) + 16x\right) = g(3) + 16 \cdot 3 = -48 + 48 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to 3} \frac{x^2 - 9}{g(x) + 16x} = \lim_{x \to 3} \frac{2x}{g'(x) + 16}$$

$$= \frac{2 \cdot 3}{g'(3) + 16} = \frac{6}{f(3) + 16} = \frac{6}{0 + 16} = \frac{3}{8}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 1$ is substituted for x, then $x = 1$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$\cdot\, g'\left(x^3\right) \cdot 3x^2$

$\cdot\, f\left(x^3\right) \cdot 3x^2$

$\cdot\, g'\left(1^3\right) \cdot 3 \cdot 1^2$

$\cdot\, f\left(1^3\right) \cdot 3 \cdot 1^2$

$\cdot\, g'(1) \cdot 3 \cdot 1^2$

$\cdot\, f\left(1^3\right) \cdot 3 \cdot 1$

5-6

Responses that do not earn the point include:

· $g'\left(x^3\right) \cdot 3x$

· $f\left(x^3\right) \cdot 3x$

· $g'(x) \cdot 3x^2$

· $f(x) \cdot 3x^2$

· $g'(1) \cdot 3$

· $f(1) \cdot 3$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking g' and f either directly or indirectly in the presented derivative of $a(x)$ or in $a'(1)$. For example, the second point is earned by $f\left(x^3\right)$ or by $f\left(1^3\right)$.

The second point may be earned without earning the first point. For example, $g'\left(x^3\right) \cdot 3x = f\left(x^3\right) \cdot 3x$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and f are linked.

The second point may be earned by going directly from $g'(1)$ to -21.1. The function f does not have to be stated explicitly. For example, $g'\left(1^3\right) \cdot 3 \cdot 1^2 = -21.1 \cdot 3 \cdot 1$ earns the first two points. Recall, if $x = 1$ is substituted for x, then $x = 1$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f\left(x^3\right) \cdot 3x^2$ and $f\left(1^3\right) \cdot 3 \cdot 1^2$ earn the first two points.

The third point is earned for the correct answer of -63.3, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $g'(x)^3 \cdot 3x^2$ or $\left(g'(x)\right)^3 \cdot 3x^2$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

5-6

$a'(x) = g'\left(x^3\right) \cdot 3x^2$

$= f\left(x^3\right) \cdot 3x^2$

$a'(1) = g'\left(1^3\right) \cdot 3 \cdot 1^2 = g'(1) \cdot 3 = f(1) \cdot 3 = (-21.1) \cdot 3 = -63.3$

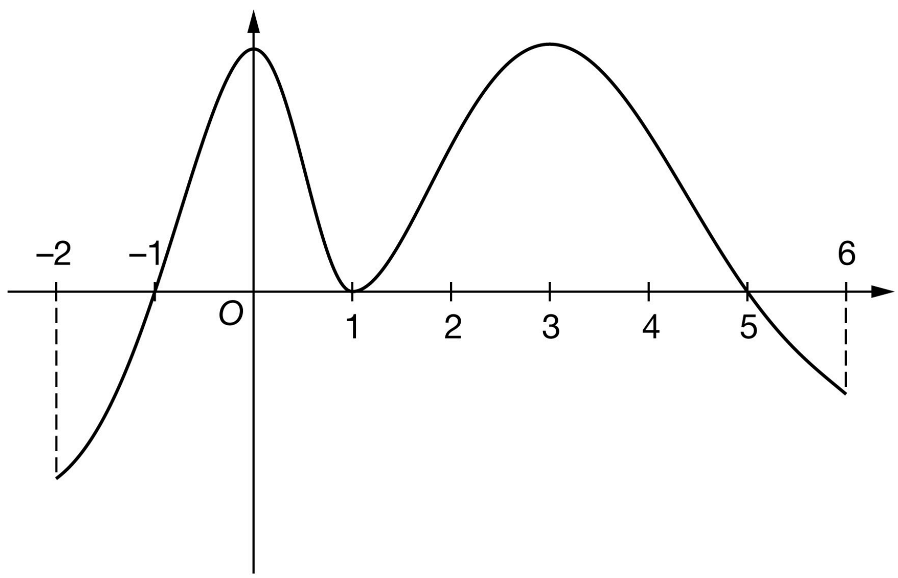
5-6

57. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h

The graph of the differentiable function h with domain $-2 \le x \le 6$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of h on the intervals $[-2,-1]$, $[-1,1]$, $[1,5]$, and $[5,6]$ are 8, 19, 40, and 4, respectively. The graph of h has horizontal tangents at $x=0$, $x=1$, and $x=3$. Let g be the function defined by $g(x)=\int_1^x h(t)\,dt$ for $-2 \le x \le 6$.

(a) Find the value of $g(5)$.

(b) Find the value of $g(-2)$. Show the computations that lead to your answer.

(c) Find the x-coordinate of each critical point of g on the interval $-2 < x < 6$.

(d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for g. Justify each of your answers.

(e) For $-2 < x < 6$, find the x-coordinate of each point of inflection of the graph of g. Justify your answers.

(f) Explain why there must be a value of w, for $1 < w < 5$, such that $g(w)=5$.

(g) Find $\lim\limits_{x \to 5} \frac{x^2-25}{g(x)-8x}$. Give a reason for your answer.

(h) The function z is defined by $z(x)=g(2x)$. It is known that $h(4)=11.25$. Find $z'(2)$. Show the computations that lead to your answer.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 40 is all that is needed to earn the point. No supporting work is required.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$g(5) = \int_1^5 h(t)\, dt = 40$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 8 and ± 19 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 8 and ± 19 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-8) - 19$ earns both points because numerical expressions need not be simplified.

· The expression $(-8) - 19$ earns the first point, but not the second point.

· The expression $-(-8) - 19 - 40$ earns neither point.

Note: Just the list $-8,\ 19$ earns neither point. However, if the values -8 and 19 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals -11 earns just the second point because not enough work has been shown.

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0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_{-2}^{-1} h(t)\, dt\right| = 8$ and $\left|\int_{1}^{-1} h(t)\, dt\right| = 19$
- [ ] Answer

Solution:

$$g(-2) = \int_{1}^{-2} h(t)\, dt = -\int_{-2}^{1} h(t)\, dt$$
$$= -\int_{-2}^{-1} h(t)\, dt - \int_{-1}^{1} h(t)\, dt = -(-8) - 19 = -11$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1,$, $x = 1,$,and $x = 5.$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-2,\ 6)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-2,\ 6)$. The point is still earned provided the three critical points, $x = -1$, $x = 1$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1,\ 0)$, $(1,\ 0)$, and $(5,\ 0)$ read the x -coordinates, and disregard the y -coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g'(x) = h(x) = 0$ at $x = -1$, $x = 1$, and $x = 5$.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justification based on g' is eligible for all three points.

If a response gives justifications based only on h, then:

· the response is eligible for all three points, if the response presents the connection between $h(x)$ and $g'(x)$ (that is, $h(x) = g'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $h(x) = g'(x)$. Any additional error (for example, not justifying that at $x = -1$, g has a relative minimum) is eligible for at most one of the three points.

Read references to "the graph" as references to $h(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., " $g'(-1)$ changes from negative to positive") when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -1$, $x = 1$, $x = 5$) must be included in the table or list. The correct values of relevant inputs are:

· $g(-2) = -11$

· $g(-1) = -19$

· $g(1) = 0$

· $g(5) = 40$

· $g(6) = 36$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that

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presents all three of the following statements in part (d) earns two points:

· At $x=-1$, g has a relative minimum because $g'=h$ is negative for $x<-1$ and $g'=h$ is positive for $x>-1$.

· At $x=1$, g has neither because $g'=h$ is positive for $x<1$ and $g'=h$ is positive for $x>1$.

· At $x=5$, g has a relative maximum because $g'=h$ is positive for $x<5$ and $g'=h$ is negative for $x>5$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Relative minimum with justification
- [ ] Neither with justification
- [ ] Relative maximum with justification

Solution:

At $x=-1$, g has a relative minimum because $g'(x)=h(x)$ changes from negative to positive there.

At $x=1$, g has neither a relative minimum nor a relative maximum because $g'(x)=h(x)$ does not change sign there.

At $x=5$, g has a relative maximum because $g'(x)=h(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $0,\ 1,\ 3$, and only this list, earns the first point.

A response referring only to $g'(x)$ is eligible to earn the second point.

A response referring only to $h(x)$ is not eligible to earn the second point, unless the response has established the link between $h(x)$ and $g'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

· g' changes from increasing to decreasing or vice versa

· g'' changes from positive to negative or vice versa

· g' has a local extrema

Any incorrect statement (such as g' changes from decreasing to increasing at $x=0$) fails to earn the second point.

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If the response addresses at least one correct value of x, it is eligible to earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of g occur where $g' = h$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = 0$, $x = 1$, and $x = 3$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that g is continuous in order to be eligible to earn the point. The response is not required to state that " g is differentiable $\Rightarrow$ g is continuous."

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $1 < x < 5$, or the point is not earned.

A response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, in the presence of the required function evaluation, the statement "since g is continuous, and $0 < 5 < 40$, such a point exists" would earn the point.

The value of 5 must be bounded by 0 and 40 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $g\left(1\right)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $g\left(1\right) = 0$ must be present.

A response may reference a value of $g\left(5\right)$ determined in part (a). If the value of $g\left(5\right)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 5.

0	1

The student response accurately includes the criteria below.

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- [ ] Explanation using Intermediate Value Theorem

Solution:

$g' = h$ is differentiable on $1 \leq x \leq 5$. $\Rightarrow g$ is continuous on $1 \leq x \leq 5$.

$g(1) = 0 < 5 < 40 = g(5)$

By the Intermediate Value Theorem, there must be a value of w, for $1 < w < 5$, such that $g(w) = 5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that g and g' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $5^2 - 25 = 0$ and $g(5) - 40 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $5^2 = 25$ and $g(5) = 40$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim\limits_{x \to 5} \frac{x^2-25}{g(x)-8x} = \frac{0}{0}$ does not earn the first point. The statement $\lim\limits_{x \to 5} \frac{x^2-25}{g(x)-8x} \to \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim\limits_{x \to 5} \frac{x^2-25}{g(x)-8x} = \frac{2x}{g'(x)-8}$, will earn the second point, but not the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim\limits_{x \to 5} \frac{x^2-25}{g(x)-8x} = \frac{0}{0}$ and $\lim\limits_{x \to 5} \frac{x^2-25}{g(x)-8x} = \frac{2x}{g'(x)-8}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-5/4$, or its numeric equivalent.

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0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$g' = h$ is differentiable. $\Rightarrow g' = h$ is continuous.
$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \to 5} g(x) = g(5)$ and $\lim_{x \to 5} g'(x) = g'(5)$.

$$\lim_{x \to 5} \left(x^2 - 25\right) = 5^2 - 25 = 0$$

$$\lim_{x \to 5} \left(g(x) - 8x\right) = g(5) - 8 \cdot 5 = 40 - 40 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to 5} \frac{x^2 - 25}{g(x) - 8x} = \lim_{x \to 5} \frac{2x}{g'(x) - 8}$$

$$= \frac{2 \cdot 5}{g'(5) - 8} = \frac{10}{h(5) - 8} = \frac{10}{0 - 8} = -\frac{5}{4}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 2$ is substituted for x, then $x = 2$ must appear unsimplified to earn the first point. Responses that earn the point include:

- $g'(2x) \cdot 2$
- $h(2x) \cdot 2$
- $g'(2 \cdot 2) \cdot 2$
- $h(2 \cdot 2) \cdot 2$

Responses that do not earn the point include:

- $g'(x) \cdot 2$
- $h(x) \cdot 2$

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$\cdot\, g'(2) \cdot 2$

$\cdot\, h(2) \cdot 2$

$\cdot\, g(2x) \cdot 2$

The second point is earned for the application of the Fundamental Theorem of Calculus as evidenced by linking g' prime and h either directly or indirectly in the presented derivative of $z(x)$ or in $z'(2)$. For example, the second point is demonstrated by $h(2x)$ or by $h(2 \cdot 2)$.

The second point may be earned without earning the first point. For example, $g'(2x) = h(2x)$ does not earn the first point because the Chain rule is applied incorrectly, but does earn the second point because g' and h are linked.

The second point may be earned by going directly from $g'(4)$ to 11.25. The function h does not have to be stated explicitly. For example, $g'(2 \cdot 2) \cdot 2 = 11.25 \cdot 2$ earns the first two points. Recall, if $x = 2$ is substituted for x, then $x = 2$ must appear unsimplified to earn the first point.

The first two points can be earned by a single expression. For example, both $h(2x) \cdot 2$ and $h(2 \cdot 2) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of 22.5, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$z'(x) = g'(2x) \cdot 2$

$= h(2x) \cdot 2$

$z'(2) = h(2 \cdot 2) \cdot 2 = h(4) \cdot 2 = 11.25 \cdot 2 = 22.5$

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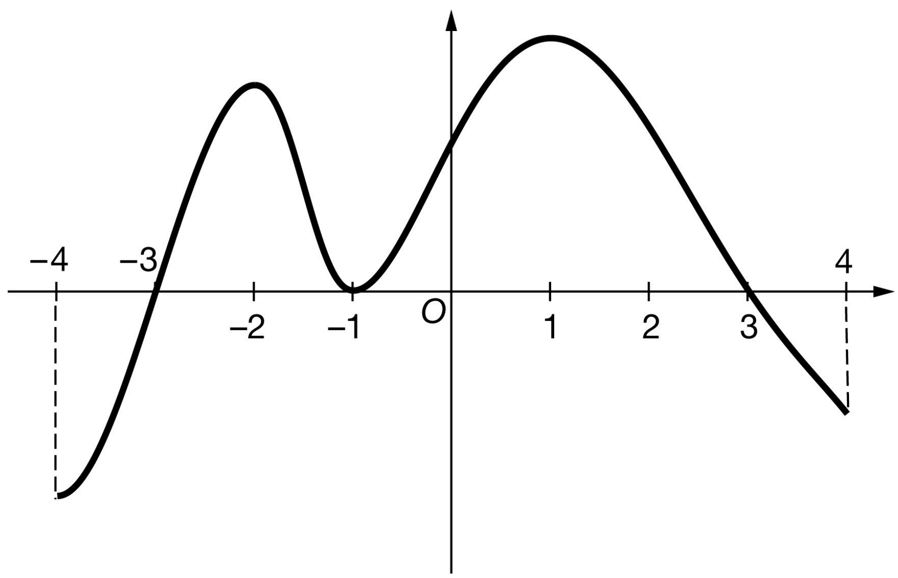
58. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the differentiable function g with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x-axis and the graph of g on the intervals $[-4, -3]$, $[-3, -1]$, $[-1, 3]$, and $[3, 4]$ are 10, 18, 45, and 5, respectively. The graph of g has horizontal tangents at $x = -2$, $x = -1$, and $x = 1$. Let h be the function defined by $h(x) = \int_{-1}^{x} g(t)\, dt$ for $-4 \leq x \leq 4$.

(a) Find the value of $h(3)$.

(b) Find the value of $h(-4)$. Show the computations that lead to your answer.

(c) Find the x-coordinate of each critical point of h on the interval $-4 < x < 4$.

(d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h. Justify each of your answers.

(e) For $-4 < x < 4$, find the x-coordinate of each point of inflection of the graph of h. Justify your answers.

(f) Explain why there must be a value of r, for $-1 < r < 3$, such that $h(r) = 20$.

(g) Find $\lim_{x \to 3} \frac{x^2 - 9}{h(x) - 15x}$. Give a reason for your answer.

(h) The function k is defined by $k(x) = h(x^2)$. It is known that $g(1) = 19.7$. Find $k'(-1)$. Show the computations that lead to your answer.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 45 is all that is needed to earn the point. No supporting work is required.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$$h(3) = \int_{-1}^{3} g(t)\, dt = 45$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 10 and ± 18 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 10 and ± 18 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

· The expression $-(-10) - 18$ earns both points because numerical expressions need not be simplified.

· The $(-10) - 18$ expression earns the first point, but not the second point.

· The expression $-(-10) - 18 - 45$ earns neither point.

Note: Just the list $-10,\ 18$ earns neither point. However, if the values -10 and 18 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals $= -8$ earns just the second point because not enough work has been shown.

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0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $\left|\int_{-4}^{-3} g(t)\,dt\right| = 10$ and $\left|\int_{-3}^{-1} g(t)\,dt\right| = 18$
- [ ] Answer

Solution:

$$h(-4) = \int_{-1}^{-4} g(t)\,dt = -\int_{-4}^{-1} g(t)\,dt$$
$$= -\int_{-4}^{-3} g(t)\,dt - \int_{-3}^{-1} g(t)\,dt = -(-10) - 18 = -8$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = -1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4,\ 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4,\ 4)$. The point is still earned provided the three critical points, $x = -3$, $x = -1$, and $x = 3$, are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3,\ 0)$, $(-1,\ 0)$, and $(3,\ 0)$, read the x -coordinates and disregard the y-coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = -1$, and $x = 3$.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g, then:

· the response is eligible for all three points, if the response presents the connection between $g(x)$ and $h'(x)$ (that is, $g(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

· the response is eligible for at most two of the three points, if the response does not present the connection $g(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to "the graph" as references to $g(x)$ as indicated by the stem.

A response referencing "the function" or "it" or "the slope" is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of "the function" or "it" or "the slope" with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., " $h'(-3)$ changes from negative to positive") when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = -1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

· $h(-4) = -8$

· $h(-3) = -18$

· $h(-1) = 0$

· $h(3) = 45$

· $h(4) = 40$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a "global" rather than a "local" argument. These earn two of three points. For example, a response that

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presents all three of the following statements in part (d) earns two points:

· At $x = -3$, h has a relative minimum because $h' = g$ is negative for $x < -3$ and $h' = g$ is positive for $x > -3$.

· At $x = -1$, h has neither because $h' = g$ is positive for $x < -1$ and $h' = g$ is positive for $x > -1$.

· At $x = 3$, h has a relative maximum because $h' = g$ is positive for $x < 3$ and $h' = g$ is negative for $x > 3$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Relative minimum with justification
- [ ] Neither with justification
- [ ] Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = -1$, h has neither a relative minimum nor a relative maximum, because $h'(x) = g(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-2, \ -1, \ 1$, and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $g(x)$ is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

· "h' changes from increasing to decreasing or vice versa"

· "h'' changes from positive to negative or vice versa"

· "h' has a local extrema"

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Any incorrect statement (such as h' changes from decreasing to increasing at $x = -2$) fails to earn the second point.

If the response addresses at least one correct value of x, it is eligible to earn the second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = g$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = -2$, $x = -1$, and $x = 1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that " h is differentiable $\Rightarrow h$ is continuous."

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval stated must contain the interval $-1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, in the presence of the required function evaluation, the statement "since h is continuous, and $0 < 20 < 45$, such a point exists" would earn the point.

The value of 20 must be bounded by 0 and 45 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h\,(-1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. We need to see $h\,(-1) = 0$.

A response may reference a value of $h\,(3)$ determined in part (a). If the value of $h\,(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 20.

0	1 ✓

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The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable on $-1 \leq x \leq 3$. $\Rightarrow h$ is continuous on $-1 \leq x \leq 3$.

$h(-1) = 0 < 20 < 45 = h(3)$

By the Intermediate Value Theorem, there must be a value of r, for $-1 < r < 3$, such that $h(r) = 20$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3^2 - 9 = 0$ and $h(3) - 45 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $h(3) = 45$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $0/0$ earns the first point.

· Indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $0/0$. For example, the statement $\lim\limits_{x\to3}\frac{x^2-9}{h(x)-15x} = \frac{0}{0}$ does not earn the first point, but the statement $\lim\limits_{x\to3}\frac{x^2-9}{h(x)-15x} \to \frac{0}{0}$ earns the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim\limits_{x\to3}\frac{x^2-9}{h(x)-15x} = \frac{2x}{h'(x)-15}$, will earn the second point, but not the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim\limits_{x\to3}\frac{x^2-9}{h(x)-15x} = \frac{0}{0}$ and $\lim\limits_{x\to3}\frac{x^2-9}{h(x)-15x} = \frac{2x}{h'(x)-15}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

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The third point is earned only for the answer of $-\frac{2}{5}$, or its numeric equivalent.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.
$\Rightarrow h$ is continuous $\Rightarrow \lim\limits_{x \to 3} h\left(x\right) = h\left(3\right)$ and $\lim\limits_{x \to 3} h'\left(x\right) = h'\left(3\right)$.

$$\lim_{x \to 3}\left(x^2 - 9\right) = 3^2 - 9 = 0$$

$$\lim_{x \to 3}\left(h\left(x\right) - 15x\right) = h\left(3\right) - 15 \cdot 3 = 45 - 45 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to 3}\frac{x^2 - 9}{h(x) - 15x} = \lim_{x \to 3}\frac{2x}{h'(x) - 15}$$

$$= \frac{2 \cdot 3}{h'(3) - 15} = \frac{6}{g(3) - 15} = \frac{6}{0 - 15} = -\frac{2}{5}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$\cdot\, h'\left(x^2\right) \cdot 2x$

$\cdot\, g\left(x^2\right) \cdot 2x$

$\cdot\, h'\left(\left(-1\right)^2\right) \cdot 2 \cdot \left(-1\right)$

$\cdot\, g\left(\left(-1\right)^2\right) \cdot 2 \cdot \left(-1\right)$

$\cdot\, h'\left(1\right) \cdot 2 \cdot \left(-1\right)$

5-6

$\cdot\, g(1) \cdot 2 \cdot (-1)$

$\cdot\, h'\left((-1)^2\right) \cdot (-2)$

$\cdot\, g\left((-1)^2\right) \cdot (-2)$

Responses that do not earn the point include:

$\cdot\, h'\left(x^2\right) \cdot 2$

$\cdot\, g\left(x^2\right) \cdot 2$

$\cdot\, h'(x) \cdot 2x$

$\cdot\, g(x) \cdot 2x$

$\cdot\, h'(1) \cdot (-2)$

$\cdot\, g(1) \cdot (-2)$

$\cdot\, h\left(x^2\right) \cdot 2x$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(-1)$. For example, the second point is earned by $g\left(x^2\right)$ or by $g\left((-1)^2\right)$.

The second point may be earned without earning the first point. For example, $h'\left(x^2\right) \cdot 2 = g\left(x^2\right) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(1)$ to 19.7. The function g does not have to be stated explicitly. Responses that earn the first two points include:

$\cdot\, h'\left((-1)^2\right) \cdot 2 \cdot (-1) = 19.7 \cdot 2 \cdot (-1)$

$\cdot\, h'\left((-1)^2\right) \cdot (-2) = 19.7 \cdot (-2)$

Recall, if $x = -1$ is substituted for x, $x = -1$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $g\left(x^2\right) \cdot 2x$ and $g\left((-1)^2\right) \cdot 2 \cdot (-1)$ earn the first two points.

The third point is earned for the correct answer of -39.4, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $h'(x)^2 \cdot 2x$ or $\left(h'(x)\right)^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the

5-6

third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

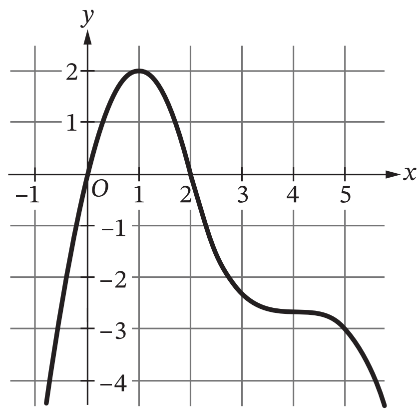
$k'(x) = h'\left(x^2\right) \cdot 2x$

$= g\left(x^2\right) \cdot 2x$

$k'(-1) = g\left((-1)^2\right) \cdot 2 \cdot (-1) = g(1) \cdot (-2) = 19.7 \cdot (-2) = -39.4$

5-6

59. The graph of f', the derivative of the function f, is shown.



Graph of f'

At which of the following values of x does the graph of f have a point of inflection?

(A) 0

(B) 1 ✓

(C) 2

(D) 4

Answer B

Correct. The graph of f will have a point of inflection where the graph of f' changes from increasing to decreasing, or from decreasing to increasing. The only place where this occurs in this graph of f' is at $x = 1$ where f' has a local maximum. At $x = 1$ the graph of f changes from concave up to concave down.

5-6

60. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The number of mosquitoes in a field after a major rainfall is modeled by the function M defined by $M(t) = -t^3 + 12t^2 + 144t$, where t is the number of days after the rainfall ended and $0 \leq t \leq 18$.

(a) Using correct units, interpret the meaning of $M'(15) = -171$ in the context of the problem.

(b) Based on the model, what is the absolute maximum number of mosquitoes in the field over the time interval $0 \leq t \leq 18$? Justify your answer.

(c) For what values of t is the rate of change of the number of mosquitoes in the field increasing?

(d) For $0 \leq t \leq 18$, the number of bats in the field is modeled by the differentiable function B, where B is a function of the number of mosquitoes in the field. Based on the models, write an expression for the rate of change of the number of bats in the field at time $t = a$.

Part A

The point requires a time reference, rate, and units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The student response accurately includes a correct interpretation of $M\prime(15) = -171$

Solution:

At time $t = 15$ days, the number of mosquitoes in the field is decreasing at a rate of 171 mosquitoes per day.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $M'(t)$ with a maximum of one computational error.

5-6

The fourth point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4

The student response accurately includes all four of the criteria below.

- ☐ $M'(t)$
- ☐ sets $M'(t) = 0$
- ☐ identifies $t = 12$ as a candidate
- ☐ answer with justification

Solution:

$M'(t) = -3t^2 + 24t + 144 = -3\left(t^2 - 8t - 48\right) = -3(t-12)(t+4)$

$M'(t) = 0 \;\Rightarrow\; t = 12$

Because $M'(t)$ changes from positive to negative at $t = 12$, M has a relative maximum at $t = 12$.

Because $t = 12$ is the only critical point of M on the interval $0 \le t \le 18$, that is the location of a relative maximum. It is also the location of the absolute maximum of M on the interval $0 \le t \le 18$.

$M(12) = -12^3 + 12\left(12^2\right) + 144(12) = 144(12) = 1728$

The absolute maximum number of mosquitoes in the field over the time interval $0 \le t \le 18$ is $M(12) = 1728$.

Part C

The third point may be earned based on a consistent answer using an incorrect $M''(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ considers $M''(t)$
- ☐ expression for $M''(t)$

5-6

- [ ] answer

Solution:

$M''(t) = -6t + 24 = -6(t-4)$

$M''(t) = -6(t-4) > 0 \;\Rightarrow\; t < 4$

The rate of change of the number of mosquitoes in the field is increasing for $0 \le t \le 4$.

Part D

The point is earned with $B'(M(a)) \cdot M'(a)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The student response accurately includes a correct expression.

Solution:

The number of bats in the field at time $t = a$ is given by $B(M(a))$.

Therefore, the rate of change of the number of bats in the field at time $t = a$ is given by
$B'(M(a)) \cdot M'(a) = B'\left(-a^3 + 12a^2 + 144a\right) \cdot \left(-3a^2 + 24a + 144\right)$.

5-6

61. Let f be the function given by $f(x) = x^2 \ln x$ for $x > 0$. On which of the following intervals is the graph of f concave down?

(A) $\left(0, e^{-3/2}\right)$ only ✓

(B) $\left(0, e^{-1/2}\right)$

(C) $\left(e^{-3/2}, \infty\right)$

(D) $\left(e^{-1/2}, \infty\right)$ only

Answer A

Correct. The graph of f will be concave up on intervals where $f''(x) < 0$.
$f'(x) = 2x \ln x + x$
$f''(x) = 2 \ln x + 2 + 1 = 2 \ln x + 3$
$f''(x) < 0 \Rightarrow 2 \ln x + 3 < 0 \Rightarrow \ln x < -\frac{3}{2} \Rightarrow x < e^{-3/2}$
The graph is concave down only on the interval $\left(0, e^{-3/2}\right)$.

62. If $x + 3y^{\frac{1}{3}} = y$, what is $\frac{dy}{dx}$ at the point $(2, 8)$?

(A) $\frac{1}{3}$

(B) $\frac{3}{4}$

(C) $\frac{5}{4}$

(D) $\frac{4}{3}$ ✓

Answer D

Correct. The chain rule is the basis for implicit differentiation.
$1 + y^{-\frac{2}{3}} \frac{dy}{dx} = \frac{dy}{dx}$
The point $(2, 8)$ is on the curve since $x = 2$ and $y = 8$ satisfy the equation. At this point,
$1 + \frac{1}{4} \frac{dy}{dx} = \frac{dy}{dx} \Rightarrow \frac{3}{4} \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{4}{3}$.

5-6

63. Let f be the function with derivative given by $f'(x) = \sin\left(e^{\frac{x}{4}}\right)$ on the open interval $(5, 8)$. Of the following, which is the largest subinterval of $(5, 8)$ on which the graph of f is concave up?

(A) $(5, 7.448)$

(B) $(5, 8)$

(C) $(6.201, 8)$

(D) $(7.352, 8)$

Answer C

Correct. The graph of f is concave up on intervals where the graph of the derivative, f', is increasing. The graph of f' has a local minimum at $x = 6.201$ and is increasing for $6.201 \leq x \leq 8$.

64. Let f be the function with first derivative given by $f'(x) = x^3 \sin x$. Which of the following is the x-coordinate of a point of inflection of the graph of f ?

(A) 1.571

(B) 1.815

(C) 2.456

(D) 3.142

Answer C

Correct. A point of inflection of the graph of f will occur at a local extrema in the graph of the derivative, f'. Graphing $f'(x) = x^3 \sin x$ on the interval $0 \leq x \leq 4$ reveals only one local extrema (a relative maximum) on this interval. It occurs at $x = 2.456$.

5-6

65. The second derivative of the function g is given by $g''(x) = x^5 - 2.2x^4 - 6.61x^3 + 8.602x^2$. At which values of x in the interval $-3 < x < 4$ does the graph of g have a point of inflection where the concavity of the graph changes from concave up to concave down?

(A) $x = 1.1$ only ✓

(B) $x = -2.3$ and $x = 3.4$ only

(C) $x = -2.3$, $x = 1.1$, and $x = 3.4$ only

(D) $x = -2.3$, $x = 0$, $x = 1.1$, and $x = 3.4$

Answer A

Correct. The graph will have a point of inflection at $x = c$ where the concavity of the graph changes from concave up to concave down if $g''(c) = 0$ and if the sign of g'' changes from positive to negative at c. Use the calculator to graph g'' on the interval $-3 \le x \le 4$ and locate all zeros where the graph goes from positive to negative. There is only one such zero, which occurs at $x = 1.1$.

66. At what values of x does the graph of $y = e^{-x} + 2xe^{-x} + x^2e^{-x}$ have a point of inflection?

(A) $x = -1$ only

(B) $x = -1$ and $x = 1$

(C) $x = -3 - \sqrt{2}$ and $x = -3 + \sqrt{2}$

(D) $x = 1 - \sqrt{2}$ and $x = 1 + \sqrt{2}$ ✓

Answer D

Correct. A point of inflection will occur where the second derivative changes sign.
$y' = -e^{-x}\left(1 + 2x + x^2\right) + e^{-x}\left(2 + 2x\right) = e^{-x}\left(-1 - 2x - x^2 + 2 + 2x\right) = e^{-x}\left(-x^2 + 1\right)$
$y'' = -e^{-x}\left(-x^2 + 1\right) + e^{-x}\left(-2x\right) = e^{-x}\left(x^2 - 2x - 1\right)$
$y'' = 0$ when $x^2 - 2x - 1 = 0 \Rightarrow x = \frac{2 \pm \sqrt{4+4}}{2} = \frac{2 \pm 2\sqrt{2}}{2} \Rightarrow x = 1 \pm \sqrt{2}$.
Since the graph of $f(x) = x^2 - 2x - 1$ is a parabola, y'' will change sign at both $x = 1 - \sqrt{2}$ and $x = 1 + \sqrt{2}$. Therefore, the graph of y will have a point of inflection at these two values of x.

5-6

67. The second derivative of the function f is given by $f''(x) = x^2 \cos\left(\frac{x^2+2x}{6}\right)$. At what values of x in the interval $(-4, 3)$ does the graph of f have a point of inflection?

(A) 2.229 only

(B) 0 and 2.229

(C) -2.357 and 0.987

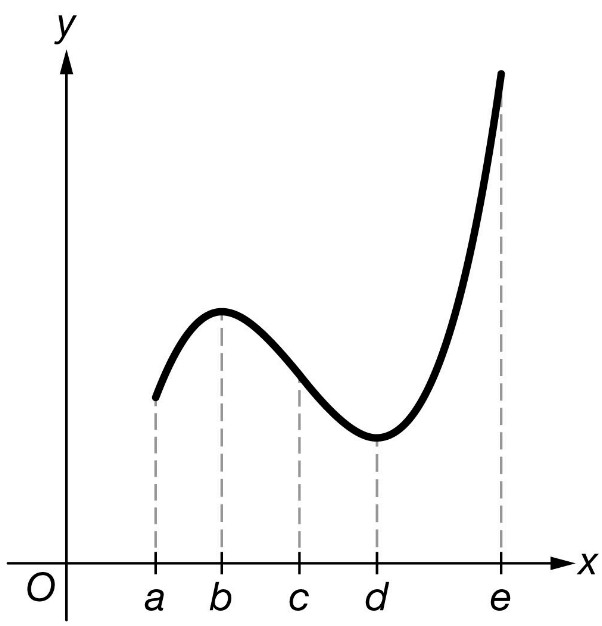
(D) -3.259, 0, and 1.603

Answer A

Correct. A point of inflection will occur where the second derivative changes sign. The calculator is used to graph f'' on the interval $[-4, 3]$, to find the zeros of f'', and to observe whether f'' changes sign at each of these zeros. There are two zeros, but only at $x = 2.229$ does f'' change sign.

5-6

68.



Graph of f'

The graph of f', the derivative of the function f, is shown in the figure above. On what open intervals is the graph of f concave up?

(A) $b < x < d$

(B) $c < x < e$

(C) $d < x < e$ only

(D) $a < x < b$ and $d < x < e$

Answer D

Correct. The graph of f will be concave up on open intervals where f' is increasing. The open intervals on which the graph of f' is increasing are $a < x < b$ and $d < x < e$.

5-6

69. At what values of x does the graph of $y = x^2e^{-2x}$ have a point of inflection?

(A) $x = -2$ and $x = 0$

(B) $x = 0$ and $x = 1$

(C) $x = -2 - \sqrt{2}$ and $x = -2 + \sqrt{2}$

(D) $x = 1 - \frac{\sqrt{2}}{2}$ and $x = 1 + \frac{\sqrt{2}}{2}$ ✓

Answer D

Correct. A point of inflection will occur where the second derivative changes sign.
$y' = 2xe^{-2x} - 2x^2e^{-2x}$
$y'' = 2e^{-2x} - 4xe^{-2x} - 4xe^{-2x} + 4x^2e^{-2x} = 2e^{-2x}\left(2x^2 - 4x + 1\right)$
$y'' = 0$ when $2x^2 - 4x + 1 = 0 \Rightarrow x = \frac{4 \pm \sqrt{16-8}}{4} = 1 \pm \frac{\sqrt{2}}{2}$. Since the graph of $f(x) = 2x^2 - 4x + 1$ is a parabola, y'' will change sign at both $x = 1 - \frac{\sqrt{2}}{2}$ and $x = 1 + \frac{\sqrt{2}}{2}$. Therefore the graph of y will have a point of inflection at these two values of x.

70. The second derivative of the function f is given by $f''(x) = x^2\cos\left(\sqrt{x}\right) - 2x\cos\left(\sqrt{x}\right) + \cos\left(\sqrt{x}\right)$. At what values of x in the interval $(0, 3)$ does the graph of f have a point of inflection?

(A) 2.467 only ✓

(B) 1 and 2.467

(C) 1.443 and 2.734

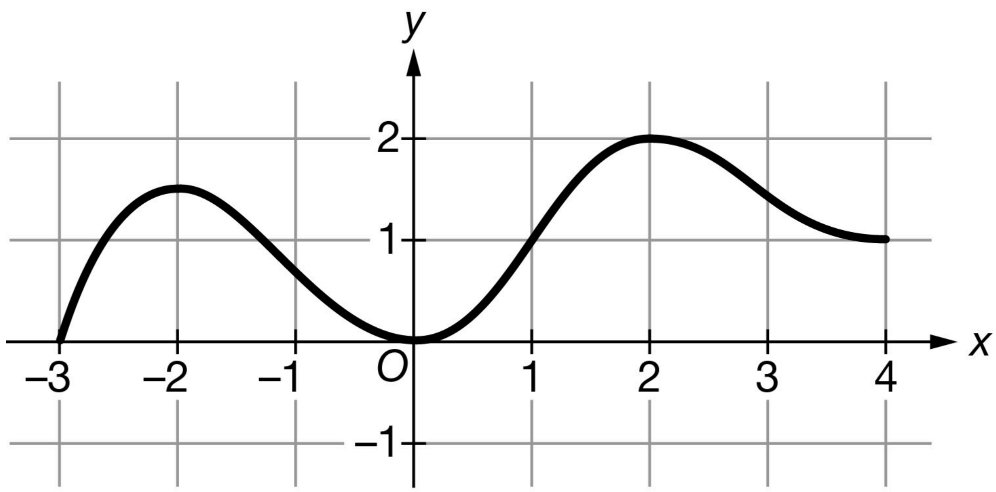
(D) 1 and 1.962

Answer A

Correct. A point of inflection will occur where the second derivative changes sign. The calculator is used to graph f'' on the interval $(0, 3)$, to find the zeros of f'', and to observe whether f'' changes sign at each of these zeros. There are two zeros, but only at $x = 2.467$ does f'' change sign.

5-6

71.



Graph of f'

The graph of f', the derivative of the function f, is shown above. On which of the following open intervals is the graph of f concave down?

(A) $(-2, 0)$ and $(2, 4)$

(B) $(-3, 2)$ and $(0, 2)$

(C) $(-3, -1)$ only

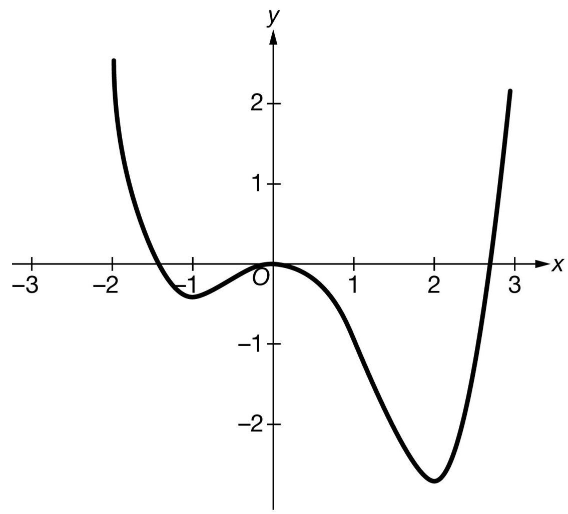
(D) $(0, 4)$

Answer A

Correct. The graph of a function is concave down on an open interval if the function's derivative is decreasing on that interval. Here f' is decreasing on the open intervals $(-2, 0)$ and $(2, 4)$.

5-6

72.



Graph of f'

Let f be the function defined by $f(x) = \frac{x^5}{20} - \frac{x^4}{12} - \frac{x^3}{3}$. The graph of f', the derivative of f, is shown above. On which of the following intervals is the graph of f concave up?

(A) $x < -1$ and $0 < x < 2$

(B) $-1 < x < 0$ and $x > 2$

(C) $x < \frac{2}{3} - \frac{2\sqrt{10}}{3}$ and $x > \frac{2}{3} + \frac{2\sqrt{10}}{3}$

(D) $\frac{2}{3} - \frac{2\sqrt{10}}{3} < x < \frac{2}{3} + \frac{2\sqrt{10}}{3}$

Answer B

Correct. The graph of f is concave up on open intervals where f' is increasing (that is, where $f''(x)$ is positive).

$f'(x) = \frac{x^4}{4} - \frac{x^3}{3} - x^2$

$f''(x) = x^3 - x^2 - 2x = x(x+1)(x-2)$

Use algebraic analysis to determine where $f''(x) > 0$, as follows.

$f''(x) = x(x+1)(x-2)$

$f''(x) = 0$ at $x = -1$, $x = 0$, and $x = 2$.

When $x < -1$,

$(x+1) < 0$, $x < 0$, and $(x-2) < 0$. Therefore, $f''(x) < 0$.

When $-1 < x < 0$,

$(x+1) > 0$, $x < 0$, and $(x-2) < 0$. Therefore, $f''(x) > 0$.

When $0 < x < 2$,

$(x+1) > 0$, $x > 0$, and $(x-2) < 0$. Therefore, $f''(x) < 0$.

When $x > 2$,

$(x+1) > 0$, $x > 0$, and $(x-2) > 0$. Therefore, $f''(x) > 0$.

5-6

73. The first derivative of the function h is given by $h'(x) = x^4 - x^3 + x$. On which of the following intervals is the graph of h concave down?

(A) $(-0.755, 0)$

(B) $(0, 0.5)$ only

(C) $(-0.455, \infty)$

(D) $(-\infty, -0.455)$

Answer D

Correct. The graph of h is concave down on intervals where the second derivative of h is negative. This is where the graph of h' is decreasing. The calculator is used to find that h' is decreasing on the interval $(-\infty, -0.455)$ by observing any of the relative maximum and relative minimum points.
Alternately, $h''(x) = 4x^3 - 3x^2 + 1$, and the calculator is used to find the zero, which occurs at $x = -0.455$. The values of h'' are negative for all $x < -0.455$.

74. The first derivative of the function h is given by $h'(x) = x^5 - 3x^2 + x$. What are all intervals on which the graph of h is concave down?

(A) $(-\infty, 0)$ and $(0.338, 1.307)$

(B) $(-\infty, 0.669)$

(C) $(-\infty, 0.167)$ and $(1, \infty)$

(D) $(0.167, 1)$

Answer D

Correct. The graph of h is concave down on intervals where the second derivative of h is negative. This is where the graph of h' is decreasing. The calculator is used to find that h' is decreasing on the interval $(0.167, 1)$ by observing the relative maximum and relative minimum points.
Alternately, $h''(x) = 5x^4 - 6x + 1$ and the calculator is used to find the zeros, which occur at 0.167 and 1. The values of h'' are negative between 0.167 and 1.

5-6

75. The second derivative of the function g is given by $g''(x) = 0.1x^5 - 0.29x^4 - 0.694x^3 + 1.9136x^2$. At which values of x in the interval $-3 < x < 4$ does the graph of g have a point of inflection where the concavity of the graph changes from concave up to concave down?

(A) $x = 2.3$ only ✓

(B) $x = -2.6$ and $x = 3.2$ only

(C) $x = -2.6$, $x = 2.3$, and $x = 3.2$ only

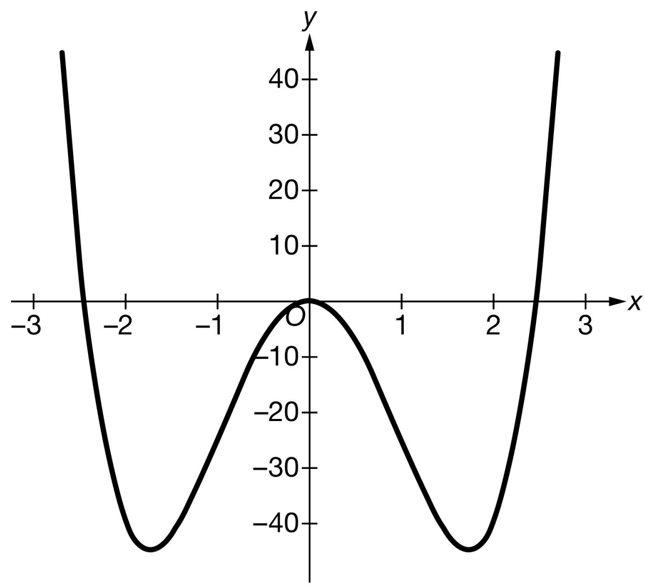
(D) $x = -2.6$, $x = 0$, $x = 2.3$, and $x = 3.2$

Answer A

Correct. The graph will have a point of inflection at $x = c$ where the concavity of the graph changes from concave up to concave down if $g''(c) = 0$ and the sign of g'' changes from positive to negative at c. Use the calculator to graph g'' on the interval $-3 \leq x \leq 4$ and locate all zeros where the graph goes from positive to negative. There is only one such zero, which occurs at $x = 2.3$.

5-6

76.



Graph of f'

Let f be the function defined by $f(x) = x^5 - 10x^3$. The graph of f', the derivative of f, is shown above. On which of the following intervals is the graph of f concave up?

(A) $x < -\sqrt{3}$ and $0 < x < \sqrt{3}$

(B) $-\sqrt{3} < x < 0$ and $x > \sqrt{3}$

(C) $x < -\sqrt{6}$ and $x > \sqrt{6}$

(D) $-\sqrt{6} < x < \sqrt{6}$

Answer B

Correct. The graph of f is concave up on open intervals where f' is increasing (that is, where $f''(x)$ is positive).

$f'(x) = 5x^4 - 30x^2$

$f''(x) = 20x^3 - 60x$

Use algebraic analysis to determine where $f''(x) > 0$, as follows.

$f''(x) = 20x^3 - 60x = 20x\left(x^2 - 3\right) = 20x\left(x - \sqrt{3}\right)\left(x + \sqrt{3}\right)$

$f''(x) = 0$ at $x = -\sqrt{3}$, $x = 0$, and $x = \sqrt{3}$.

When $x < -\sqrt{3}$, then $x < 0$, $x - \sqrt{3} < 0$, and $x + \sqrt{3} < 0$. Therefore, $f''(x) < 0$.

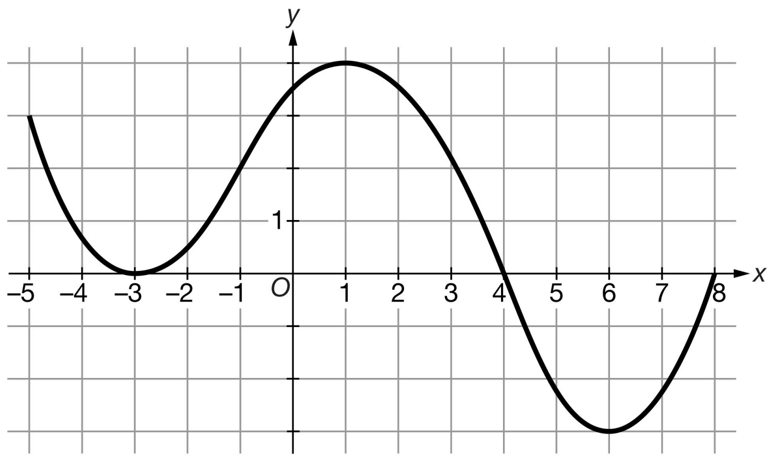
When $-\sqrt{3} < x < 0$, then $x < 0$, $x - \sqrt{3} < 0$, and $x + \sqrt{3} > 0$. Therefore, $f''(x) > 0$.

When $0 < x < \sqrt{3}$, then $x > 0$, $x - \sqrt{3} < 0$, and $x + \sqrt{3} > 0$. Therefore, $f''(x) < 0$.

When $x > \sqrt{3}$, then $x > 0$, $x - \sqrt{3} > 0$, and $x + \sqrt{3} > 0$. Therefore, $f''(x) > 0$.

5-6

77.



Graph of f'

The graph of f', the derivative of the function f, is shown above. On which of the following open intervals is the graph of f concave down?

(A) $(-5,-3)$ and $(1,6)$ ✓

(B) $(-3,1)$ and $(6,8)$

(C) $(-1,4)$

(D) $(4,8)$

Answer A

Correct. The graph of a function is concave down on an open interval if the function's derivative is decreasing on that interval. Here f' is decreasing on the open intervals $(-5,-3)$ and $(1,6)$.

78. For $x>-2$, at any point on the graph of the function f the slope is given by $\sqrt{x^3+8}-\sqrt{3x^2+7}$. How many points of inflection does the graph of f have for $-1<x<5$?

(A) Zero

(B) One

(C) Two ✓

(D) Three

Answer C

Correct. The graph of a function will have a point of inflection where the graph of its derivative has a local maximum or a local minimum. A graph of f' on the interval $[-1,5]$ shows that the derivative has one local maximum and one local minimum on the open interval $(-1,5)$. Therefore, the graph of f has two points of inflection on this interval.

5-6

79. The first derivative of the function f is given by $f'(x) = xe^{\sin x}$. How many points of inflection does the graph of f have on the interval $-3 < x < 6$?

(A) One

(B) Two

(C) Three

(D) Four

Answer B

Correct. The graph of a function will have a point of inflection where the graph of its derivative has a local maximum or a local minimum. The graph of f' on the interval $[-3, 6]$ shows that the derivative of f has one local maximum and one local minimum on the open interval $(-3, 6)$. Therefore, the graph of f has two points of inflection on this interval.

5-6

80. Let f be a function with second derivative given by $f''(x) = (x-1)^2(x-2)(x-3)^3$. How many points of inflection does the graph of f have?

(A) Two

(B) Three

(C) Five

(D) Six

Answer A

Correct. Since f'' exists for all values of x, the graph of f will have a point of inflection at a value of x where the second derivative is 0 and changes sign from negative to positive, or from positive to negative. There are three values of x where $f''(x) = 0$, at $x = 1$, $x = 2$, and $x = 3$.

Near $x = 1$, the graph of $f''(x)$ behaves like a parabola since the exponent of the $(x-1)$ term is 2. Thus $f''(x)$ has the same sign (positive) for values of x near $x = 1$, so the graph of f does not have a point of inflection at $x = 1$.

Near $x = 2$, the graph of $f''(x)$ behaves like a linear function since the exponent of the $(x-2)$ term is 1. Thus $f''(x)$ changes sign (from positive to negative) for values of x near $x = 2$, so the graph of f does have a point of inflection at $x = 2$.

Near $x = 3$, the graph of $f''(x)$ behaves like a cubic function since the exponent of the $(x-3)$ term is 3. Thus $f''(x)$ changes sign (from negative to positive) for values of x near $x = 3$, so the graph of f does have a point of inflection at $x = 3$.

Therefore the graph of f has two points of inflection.